# Summary Report, v5: A 100-Metric Library Calibrated on Answers of Known Quality

2026

## 1 Setup

### 1.1 Shared setup (identical across versions)

- **Suite.** 31 datasets: 11 entity matching (EM), 6 schema matching (SM), 11 table question answering and understanding (TQA), 3 financial documents (FIN). Official test or dev splits only; 150 items per dataset by seeded stratified sampling, 100 flagged for answering (OfficeQA Pro V2: 90; Alaska schema: 93); 3,083 answered items per model. Entity-matching pools are 50/50 match / non-match (Papadakis–Christen keeps its real 1.8 % ratio); schema matching is one question per source column with a candidate list; BIRD is graded on the executed result of the gold SQL.
- **Answers.** Ten models (Claude Opus 5, Claude Sonnet 5, GPT-5.5, GPT-5.4-mini, Gemini 3.8 Flash, Grok 4, DeepSeek-chat, DeepSeek-R1, gpt-oss-120b, Llama 3.3 70B); one prompt per item ending in `FINAL ANSWER: <answer>`; temperature 0 where accepted (Claude 5 and GPT-5.5 ran at provider defaults, recorded per row); output budget 700 tokens with mechanical re-asks at 2,500 and 8,000 for replies cut off before a final line; every answer cached with tokens, cost and finish reason. The same cache feeds every version, so leaderboards are directly comparable.
- **Scoring primitives.** Extraction of the final line, Unicode / case / punctuation normalisation, number parsing with sign, percent and scale words, alias lists per gold (best alias wins), list golds, a hedge flag. Native metric per dataset: numeric tolerance (1 %) for numeric golds, exact match for text and boolean, ROUGE-L for free-form.
- **Coverage rule.** A model is in the study only if it covered $\geq 90$ % of the suite; a dataset counts only items every model answered and only with $\geq 20$ such items.
- **Leaderboard machinery.** Bradley–Terry strengths from item-level pairwise wins (MLE, ridge 0.01, SE from the observed information) as the headline pooling; Borda, Kemeny (with cost), Copeland, mean-$z$, reciprocal-rank fusion and family-balanced $z$ as checks; 1,000 item bootstraps of the Borda and Bradley–Terry ranks; Friedman with Iman–Davenport, Nemenyi CD, Holm-corrected pairwise Wilcoxon across datasets; paired permutation and Wilcoxon per item within datasets (Holm); Kendall's $W$ among datasets; majority-class baselines for boolean sets; post-stratified official-split estimates with a weighted bootstrap and positive-class F1 at the official match ratio; floor / ceiling / self-stability diagnostics per dataset.
- **Sampled designs.** Uniform-lattice, Dirichlet-snapped and Latin-hypercube samples on the weight simplex at step 0.05 (20 seeds each) with centred $L_2$ discrepancy and nearest-neighbour distances, compared with the exhaustive optimum where one exists.
- **Reporting.** Each version's `run_all.sh` regenerates its `REPORT.md` from the cached answers; every number in this document is read from those output files.

### 1.2 What v5 searches

- **Metric library.** 100 per-answer metrics in twelve families (exact-match variants, token overlap, character and sequence similarity, numeric tolerance and graded numeric error, set metrics for list golds, structure and task-specific scores, length, commitment / hedging, semantic placeholders), computed for every (model, item) pair and for every anchor.
- **Anchors as the quality signal.** 31,380 synthetic answers generated from the golds by 20 typed operators with a known utility each (identity, verbose, tagged, unit variant = 1; paraphrase 0.95; near misses 0.75 / 0.45 / 0.15; superset 0.6; typo 0.6; partial 0.5; shuffled 0.4; hedge 0.3; list dump 0.25; digit typo 0.2; sign flip and magnitude 0.05; flip, no answer, wrong random = 0). v5 is the only version whose objective is correctness against this signal rather than agreement among the ranked models.
- **Selection.** Drop constant metrics; cluster $|\text{Spearman}| \geq 0.95$ into representatives; keep candidates above an anchor-agreement floor; forward selection maximising leave-one-dataset-out Spearman of a non-negative least-squares fit to utility.
- **Weights.** Exhaustive grid at step 0.02 over the selected weights (sum to 1) $\times$ five hedge penalties $\lambda$ (score multiplied by $1 - \lambda$ when the answer hedges): 6,630 points. Objective $J = 0.30\,\rho + 0.15\,\text{AUC} + 0.20\,(1 - \text{JS}/\ln 2) + 0.15\,\tau_{\text{native}} + 0.20\,(1 - \text{wrong mean})$, where JS is the Jensen–Shannon divergence between datasets' score distributions at matched operator (cross-dataset consistency) and $\tau_{\text{native}}$ the agreement of the composite's model ranking with the native-metric ranking. The ensemble is the mean of the 100 best weightings, so each answer carries a score and a spread.
- **Validation.** Leave-one-dataset-out re-selection and re-fitting; utility-scale perturbation ($\pm 0.15$, five draws); gaming probes (mean score per operator vs utility against single-metric baselines); sampled designs looked up in the exhaustive grid.
- **Leaderboards.** Exact match, the native metric, the ensemble (headline) and the five best single weightings, each pooled by Bradley–Terry with the full significance battery.

## 2 Results

### 2.1 Three metrics carry the library's information

Of 100 metrics, 45 are distinct at $|\text{Spearman}| \geq 0.95$ and forward selection stops at three: token F1, numeric decay and prefix ratio (Figures 1, 2). Their NNLS fit reaches leave-one-dataset-out $\rho$ 0.875 against 0.814 for the native typed metric and 0.831 for exact match.

### 2.2 The calibrated ensemble

The grid is decay-dominated (Figures 4, 5): the top 100 put 0.77 on numeric decay with a hedge penalty of 0.62. The ensemble reaches $\rho$ 0.904 and AUC 0.996 on the anchors, above every single-metric baseline (Table 8, Figure 6); the probes show where the gain comes from: near misses, unit variants and partial answers get graded credit instead of zero (Figure 7, Table 36). The sampled designs land within 0.0015 of the exhaustive best (Table 9).

### 2.3 Validation

Leaving any dataset out re-selects the same three metrics in 90 % of folds and changes held-out $\rho$ by 0.001 on average (Figure 8); perturbing the utility scale by $\pm 0.15$ moves the weights by 0.022 in

**Table 1:** v5 audit summary.

| | |
|---|---|
| **Library** | 100 metrics, 10 constant on this data, 45 cluster representatives, 28 candidates above the floor |
| **Selected** | tok_f1, num_decay_2p5, prefix_ratio; LODO $\rho$ of the selected set 0.875 vs native typed metric alone 0.814 and exact match alone 0.831 |
| **Grid** | 6,630 weightings (step 0.02 × 5 $\lambda$); $J$ top-1 0.935, top-100 mean 0.933, grid median 0.923; range 0.876–0.935 |
| **Ensemble weights** | tok_f1 0.17, num_decay_2p5 0.77, prefix_ratio 0.06, $\lambda = 0.62$; decay weight spans 0.54–0.96 across the top 100 |
| **Ensemble terms** | $\rho$ 0.904, AUC 0.996, JS 0.053, $\tau$ native 0.898, wrong mean 0.032 |
| **Baselines** | token F1 $\rho$ 0.859 / $J$ 0.907; numeric tolerance $\rho$ 0.806 / $J$ 0.909; ensemble $\rho$ 0.904 / $J$ 0.934 |
| **Sampled designs** | all three within 0.0015 of the exhaustive best (0.9348) |
| **Leave-one-dataset-out** | mean held-out $\rho$ 0.893 (re-fitted) vs 0.894 (full); same metric set selected in 90 % of folds |
| **Utility perturbation** | mean $L_1$ shift of the ensemble weights 0.022; selection changed in 0 % of draws |
| **Probes (MAE to utility)** | ensemble 0.116, token F1 0.144, native 0.155, exact match 0.199 |
| **Leaderboard (ensemble, BT)** | first Gemini 3.8 Flash ($P$(first) 0.95), then Claude Opus 5, GPT-5.5; last Llama 3.3 70B; Friedman $p$ = 2.1e-12; $W = 0.26$ |
| **Other rules** | native metric: Gemini 3.8 Flash first; exact match: Claude Opus 5 first; $\tau$ ensemble vs native 0.78, vs exact match 0.87 |

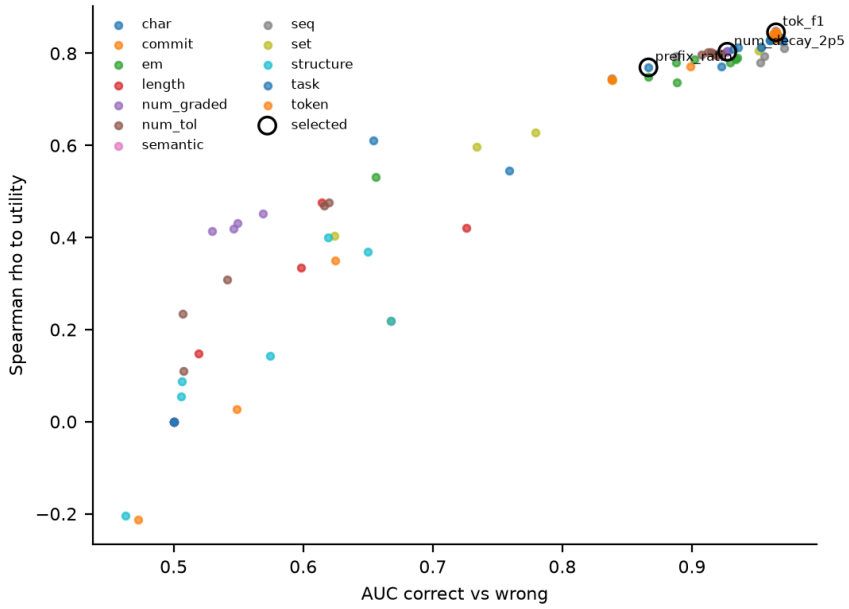



**Figure 1:** The 100-metric library on the anchors: AUC against Spearman $\rho$, coloured by family; the three selected metrics are circled.

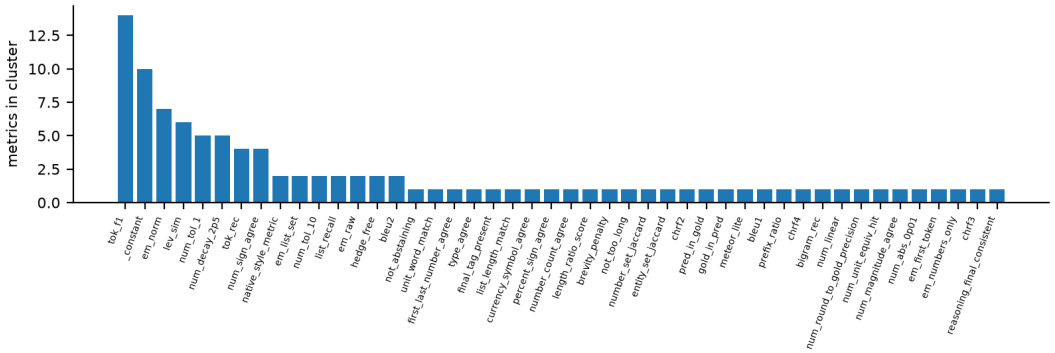



**Figure 2:** Cluster sizes after merging metrics with $|\text{Spearman}| \geq 0.95$ (45 representatives).

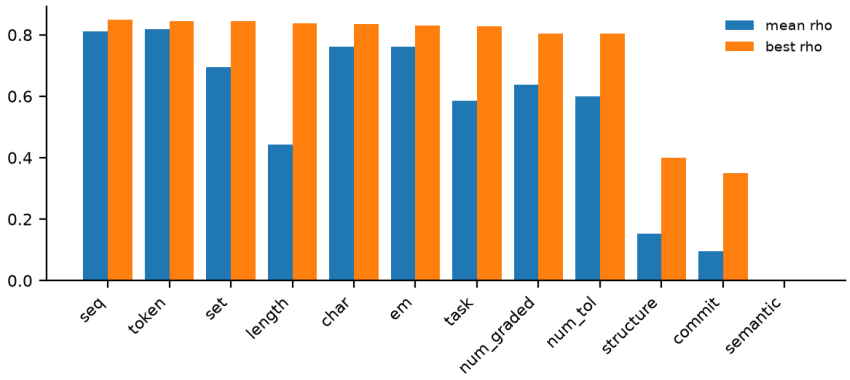



**Figure 3:** Anchor $\rho$ per metric family: mean and best member.

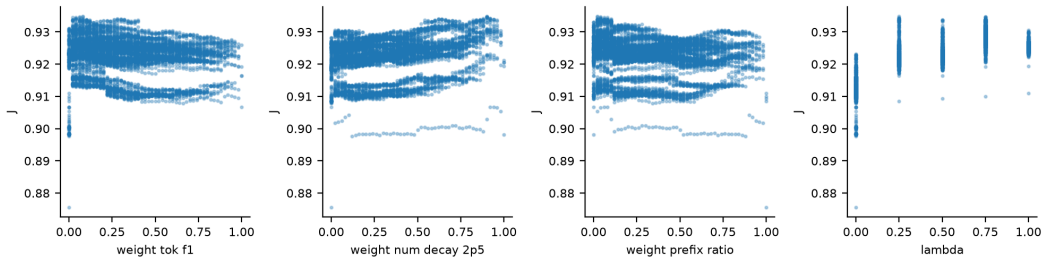



**Figure 4:** Objective $J$ against each selected weight and the hedge penalty $\lambda$ over all 6,630 grid points.

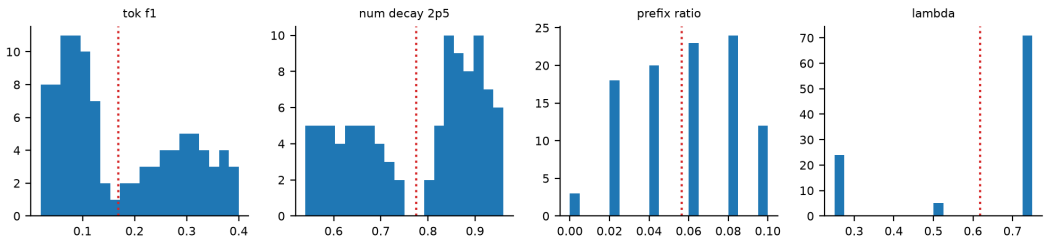



**Figure 5:** Weights and $\lambda$ of the 100 best grid points that form the ensemble (dotted = mean).

**Table 2:** Baselines under the same objective.

| baseline | rho | auc | js | bad | tau_native | J |
|---|---|---|---|---|---|---|
| native_style_metric | 0.826 | 0.935 | 0.031 | 0.014 | 0.992 | 0.925 |
| em_norm | 0.849 | 0.963 | 0.024 | 0.012 | 0.924 | 0.928 |
| tok_f1 | 0.859 | 0.965 | 0.079 | 0.032 | 0.888 | 0.907 |
| rouge_l | 0.862 | 0.965 | 0.082 | 0.032 | 0.892 | 0.907 |
| num_tol_1 | 0.806 | 0.919 | 0.032 | 0.016 | 0.946 | 0.909 |
| uniform_over_selected_no_gate | 0.882 | 0.970 | 0.098 | 0.028 | 0.900 | 0.911 |
| ensemble_top100 | 0.904 | 0.996 | 0.053 | 0.032 | 0.898 | 0.934 |

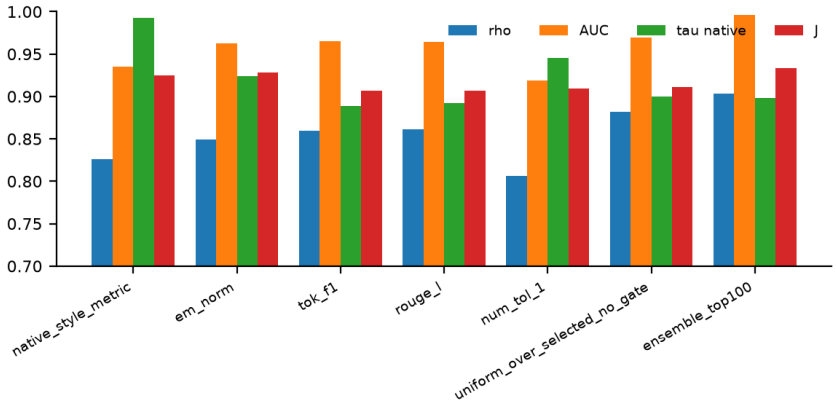



**Figure 6:** Single-metric baselines against the ensemble under the same objective terms.

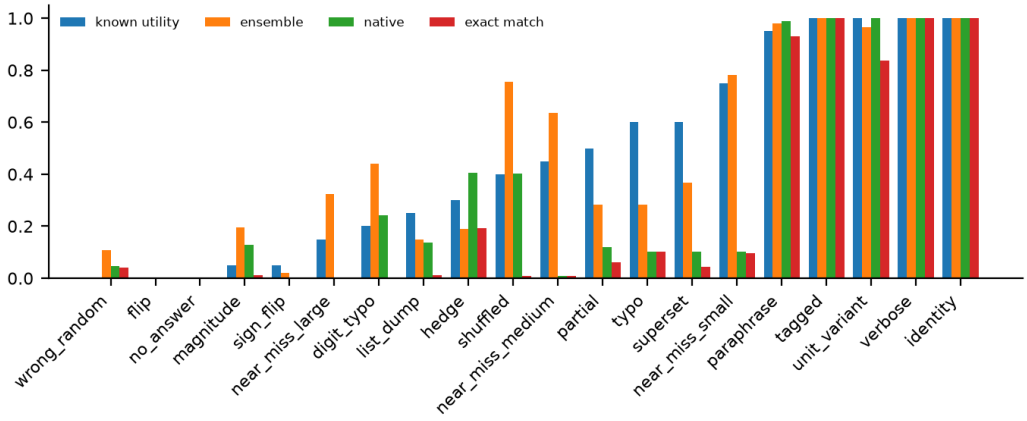



**Figure 7:** Mean score per anchor operator under the ensemble, the native metric and exact match, against known utility.

**Table 3:** Sampled designs vs the exhaustive grid (best $J = 0.9348$ over 6630 points).

| design | best_J_mean | best_J_sd | regret_mean | regret_max | min_pairwise_L1 | discrepancy | zero_share |
|---|---|---|---|---|---|---|---|
| dirichlet_lattice | 0.9342 | 0.0004 | 0.0007 | 0.0014 | 0.0400 | 0.2728 | 0.0615 |
| lhs_simplex | 0.9340 | 0.0003 | 0.0008 | 0.0015 | 0.0400 | 0.2702 | 0.0555 |
| uniform_lattice | 0.9342 | 0.0003 | 0.0007 | 0.0014 | 0.0400 | 0.2639 | 0.1115 |

$L_1$ and never changes the selection. The hand-set utility scale is the main assumption, and these checks say the result is not sensitive to it.

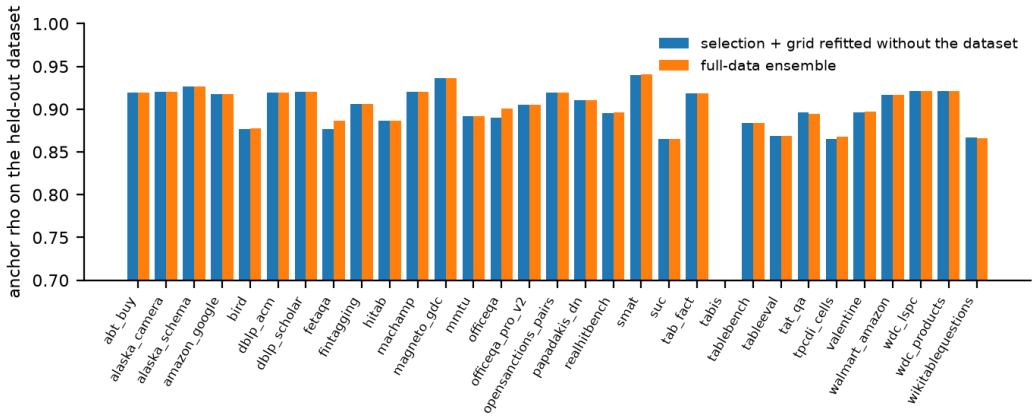



**Figure 8:** Leave-one-dataset-out: anchor $\rho$ on the held-out dataset under the re-fitted and the full-data ensemble.

**Table 4:** Validation summary.

| quantity | value |
|---|---|
| lodo_mean_rho_lodo | 0.8934 |
| lodo_mean_rho_full | 0.8942 |
| lodo_same_selection_share | 0.9032 |
| severity_l1_shift_mean | 0.0220 |
| severity_l1_shift_max | 0.0220 |
| severity_selection_changed_share | 0.0000 |
| probe_mae_ensemble | 0.1158 |
| probe_mae_em | 0.1991 |
| probe_mae_native | 0.1545 |
| probe_mae_tok_f1 | 0.1440 |

## 2.4 Leaderboard

Under the ensemble pooled by Bradley–Terry, Gemini 3.8 Flash is first, Claude Opus 5 and GPT-5.5 next, Llama 3.3 70B last; Borda puts Claude Opus 5 first. The five best single weightings give the same ranking as the ensemble ($\tau = 1.00$); the native metric and exact match move the top three. Dataset agreement stays moderate ($W = 0.26$). The ensemble spread per answer (Figure 11) is largest on the free-form and list datasets, which is where the weighting matters.

**Table 5:** Leaderboard under the headline rule (ensemble).

| rank | model | BT strength | SE | Borda rank | Borda 95% | P(first) BT | mean score |
|---|---|---|---|---|---|---|---|
| 1 | Gemini 3.8 Flash | 0.128 | 0.016 | 3 | [1, 3] | 0.950 | 0.835 |
| 2 | Claude Opus 5 | 0.110 | 0.000 | 1 | [1, 3] | 0.050 | 0.833 |
| 3 | GPT-5.5 | 0.047 | 0.016 | 2 | [1, 3] | 0.000 | 0.824 |
| 4 | Claude Sonnet 5 | 0.035 | 0.016 | 4 | [4, 6] | 0.000 | 0.814 |
| 5 | Grok 4 | 0.026 | 0.016 | 5 | [4, 7] | 0.000 | 0.811 |
| 6 | GPT-5.4-mini | -0.009 | 0.016 | 6 | [4, 8] | 0.000 | 0.805 |
| 7 | DeepSeek-chat | -0.020 | 0.016 | 8 | [5, 8] | 0.000 | 0.798 |
| 8 | DeepSeek-R1 | -0.050 | 0.016 | 7 | [5, 8] | 0.000 | 0.794 |
| 9 | gpt-oss-120b | -0.089 | 0.016 | 9 | [9, 10] | 0.000 | 0.784 |
| 10 | Llama 3.3 70B | -0.177 | 0.016 | 10 | [9, 10] | 0.000 | 0.747 |

**Table 6:** Bradley–Terry rank (Borda rank in brackets) under each reported rule, Friedman $p$ and Kendall's $W$ among datasets.

| rule | Gemini 3.8 Flash | Claude Opus 5 | GPT-5.5 | Claude Sonnet 5 | Grok 4 | GPT-5.4-mini | DeepSeek-chat | DeepSeek-R1 | gpt-oss-120b | Llama 3.3 70B | Friedman p | W |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| exact match | 2 (3) | 1 (1) | 3 (2) | 5 (4) | 4 (5) | 6 (8) | 8 (7) | 7 (6) | 9 (9) | 10 (10) | 2.7e-14 | 0.28 |
| native metric | 1 (3) | 3 (2) | 2 (1) | 5 (4) | 4 (5) | 8 (7) | 7 (8) | 6 (6) | 9 (9) | 10 (10) | 3.8e-14 | 0.28 |
| ensemble | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 2.1e-12 | 0.26 |
| top1 | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 2.9e-12 | 0.26 |
| top2 | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 2.2e-12 | 0.26 |
| top3 | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 2.7e-12 | 0.26 |
| top4 | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 3.6e-12 | 0.25 |
| top5 | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 4.2e-12 | 0.25 |

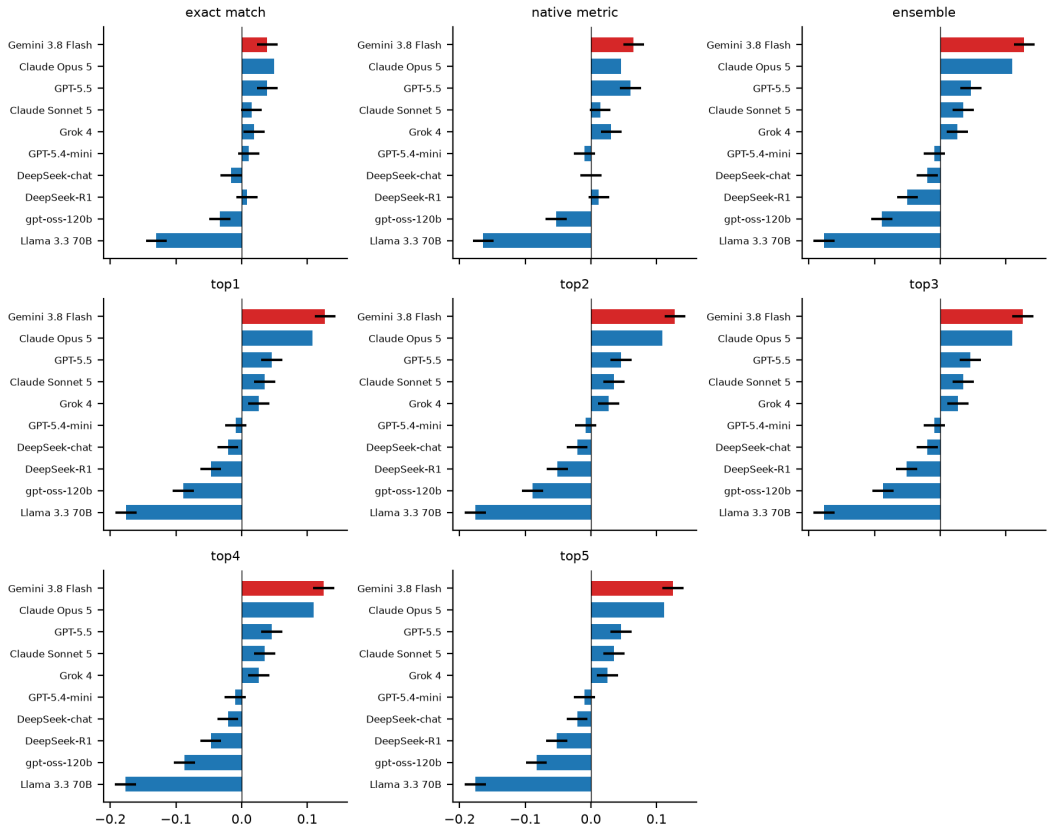



**Figure 9:** Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in the headline order.

## 2.5 Comparability and diagnostics

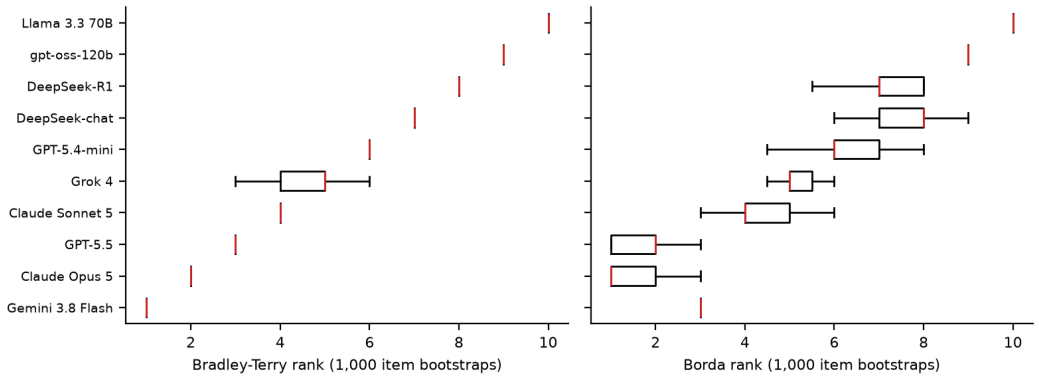



**Figure 10:** Bootstrap distribution of each model's rank under the headline rule (ensemble).

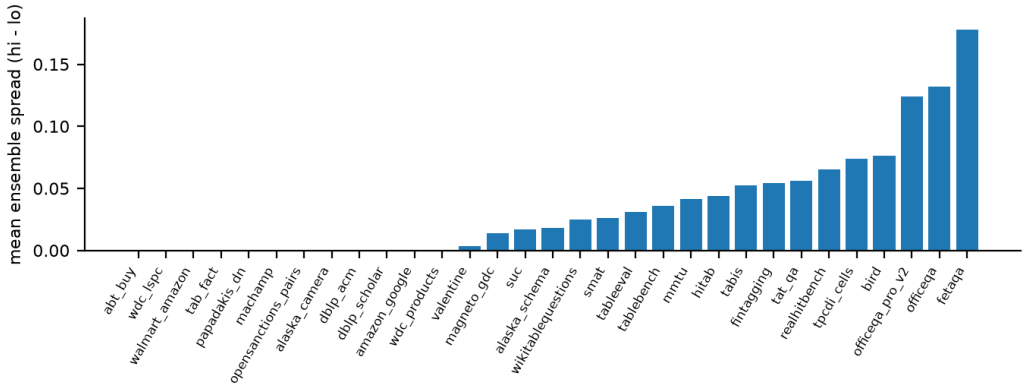



**Figure 11:** Ensemble uncertainty per dataset: mean spread of the per-answer score across the 100 weightings.

**Table 7:** Official-split estimates under the headline rule averaged over the ten models.

| dataset | suite_mean | official_ estimate | suite_f1_ pos | official_precision_ pos | official_recall_ pos | official_f1_ pos | official_positive_ share |
|---|---|---|---|---|---|---|---|
| abt_buy | 0.970 | 0.986 | 0.968 | 0.926 | 0.949 | 0.936 | 0.108 |
| alaska_camera | 0.958 | 0.958 | 0.955 | 1.000 | 0.921 | 0.955 | 0.500 |
| alaska_schema | 0.909 | 0.909 | – | – | – | – | – |
| amazon_google | 0.935 | 0.932 | 0.928 | 0.611 | 0.941 | 0.739 | 0.102 |
| bird | 0.252 | 0.271 | – | – | – | – | – |
| dblp_acm | 0.986 | 0.991 | 0.985 | 0.979 | 0.978 | 0.976 | 0.180 |
| dblp_scholar | 0.949 | 0.966 | 0.948 | 0.900 | 0.924 | 0.910 | 0.186 |
| fetaqa | 0.559 | 0.559 | – | – | – | – | – |
| fintagging | 0.823 | 0.821 | – | – | – | – | – |
| hitab | 0.775 | 0.788 | – | – | – | – | – |
| machamp | 0.887 | 0.890 | 0.892 | 0.644 | 0.853 | 0.733 | 0.175 |
| magneto_gdc | 0.915 | 0.919 | – | – | – | – | – |
| mmtu | 0.862 | 0.810 | – | – | – | – | – |
| officeqa | 0.273 | 0.273 | – | – | – | – | – |
| officeqa_pro_v2 | 0.294 | 0.294 | – | – | – | – | – |
| opensanctions_ pairs | 0.821 | 0.727 | 0.773 | 0.993 | 0.649 | 0.777 | 0.769 |
| papadakis_dn | 0.941 | 0.988 | 0.833 | 0.550 | 0.963 | 0.683 | 0.018 |
| realhitbench | 0.717 | 0.716 | – | – | – | – | – |
| smat | 0.890 | 0.887 | – | – | – | – | – |
| suc | 0.950 | 0.950 | – | – | – | – | – |
| tab_fact | 0.798 | 0.789 | 0.758 | 0.896 | 0.674 | 0.764 | 0.500 |
| tabis | 0.885 | 0.889 | – | – | – | – | – |
| tablebench | 0.854 | 0.860 | – | – | – | – | – |
| tableeval | 0.771 | 0.776 | – | – | – | – | – |
| tat_qa | 0.797 | 0.755 | – | – | – | – | – |
| tpcdi_cells | 0.418 | 0.450 | – | – | – | – | – |
| valentine | 0.981 | 0.974 | – | – | – | – | – |
| walmart_amazon | 0.958 | 0.975 | 0.947 | 0.832 | 0.927 | 0.875 | 0.094 |
| wdc_lspc | 0.982 | 0.983 | 0.983 | 0.965 | 0.979 | 0.971 | 0.264 |
| wdc_products | 0.977 | 0.985 | 0.979 | 0.980 | 0.967 | 0.973 | 0.280 |
| wikitablequestions | 0.850 | 0.849 | – | – | – | – | – |

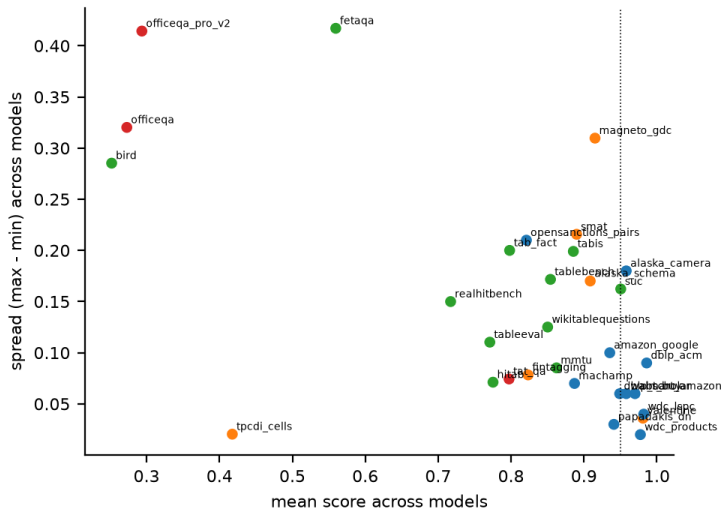



**Figure 12:** Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

# A All figures

Every figure that can be drawn from the saved outputs; source data in `data/`.

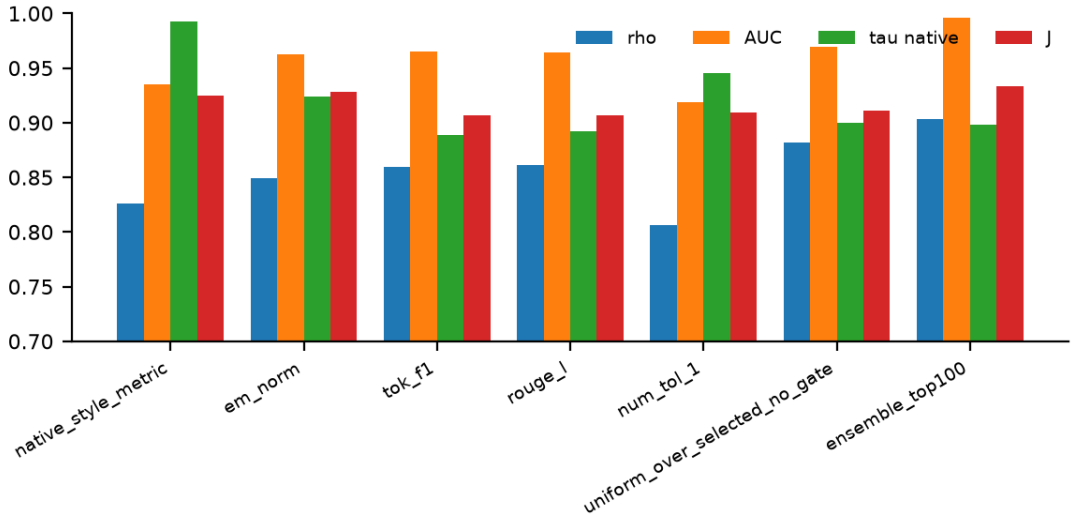



**Figure 13:** Single-metric baselines against the ensemble under the same objective terms.

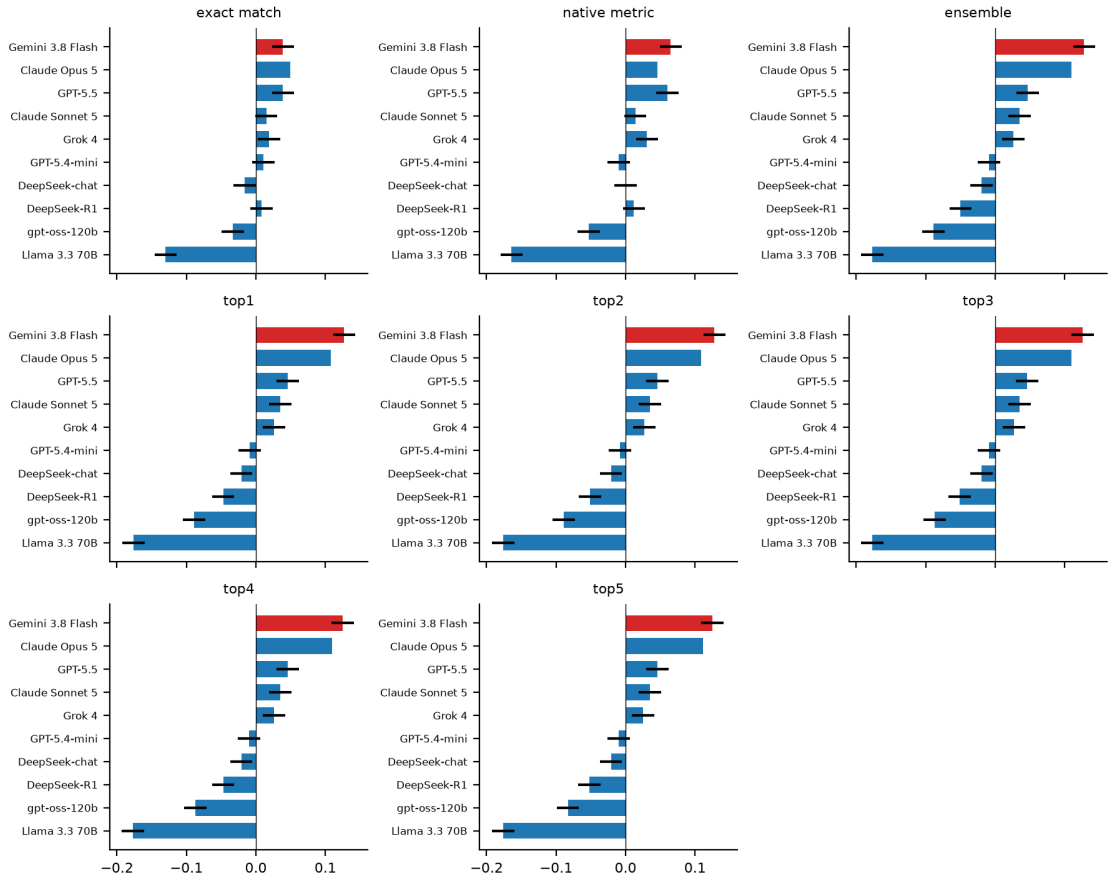



**Figure 14:** Bradley–Terry strengths (± SE vs reference) under every reported rule; models fixed in the headline order.

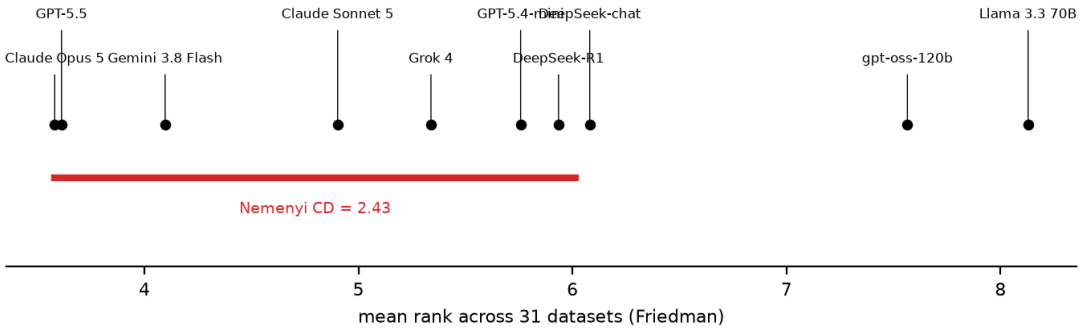



**Figure 15:** Critical-difference view of the Friedman test under the headline rule: mean rank per model and the Nemenyi critical difference at $\alpha = .05$.

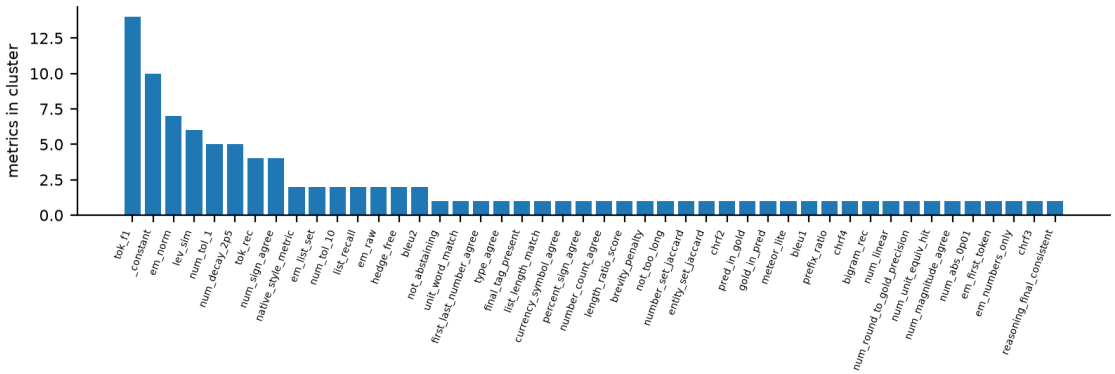



**Figure 16:** Cluster sizes after merging metrics with $|\text{Spearman}| \geq 0.95$ (45 representatives).

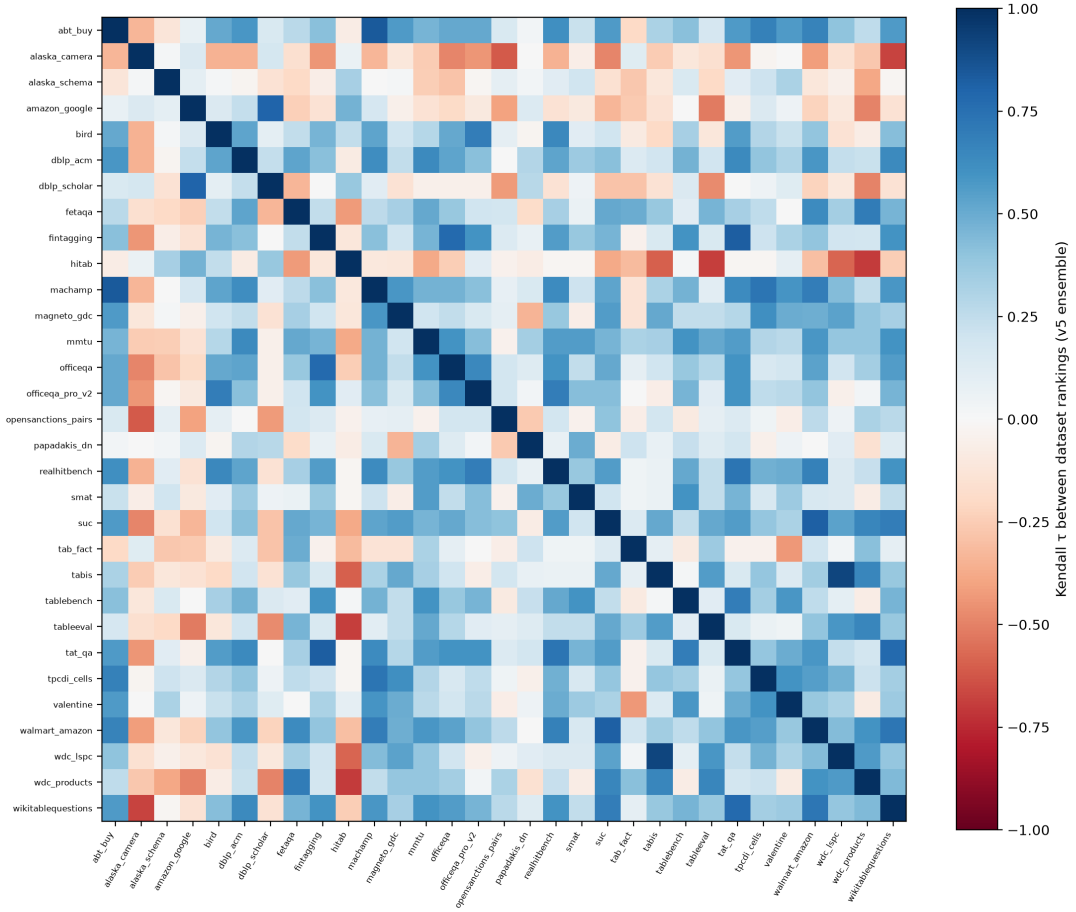



**Figure 17:** Kendall $\tau$ between the model rankings of each pair of datasets under the headline rule.

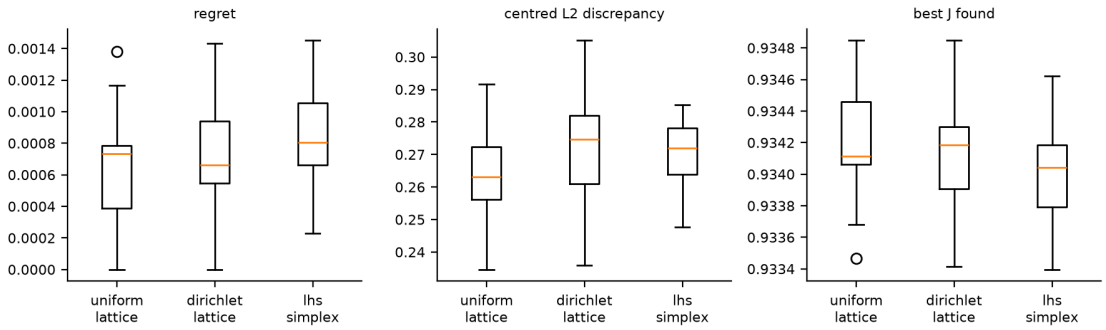



**Figure 18:** Sampled designs over 20 seeds against the exhaustive grid.

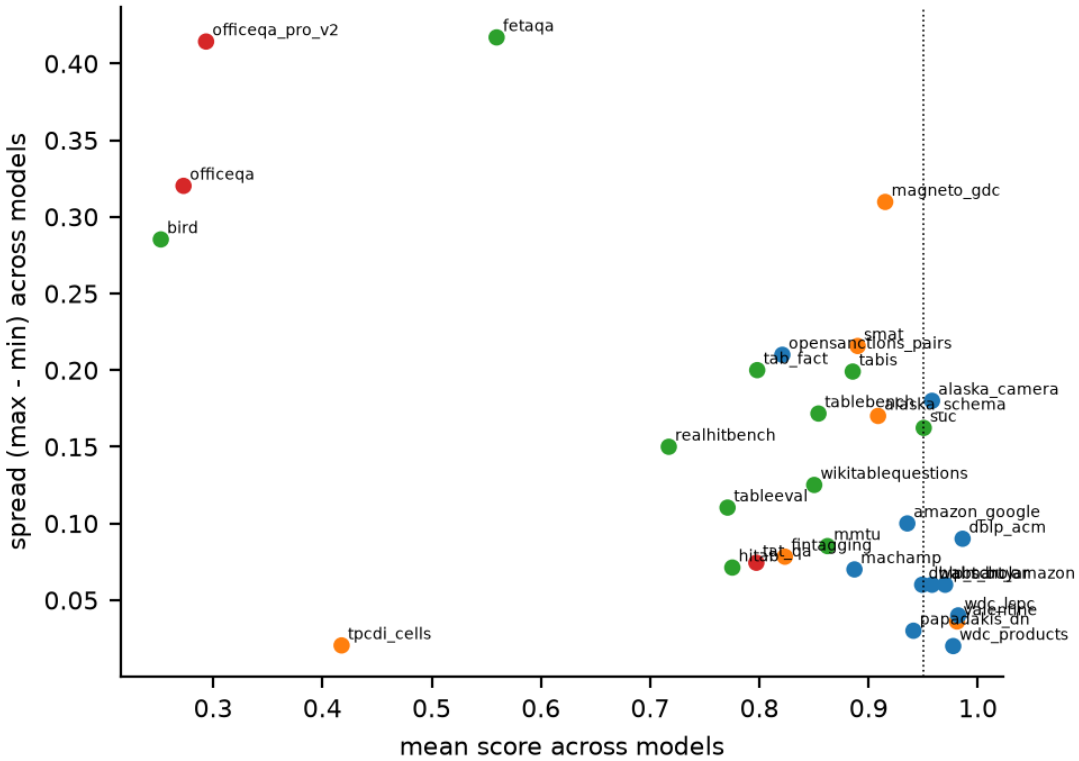



**Figure 19:** Dataset diagnostics: mean score against spread across models; dotted line marks the ceiling flag.

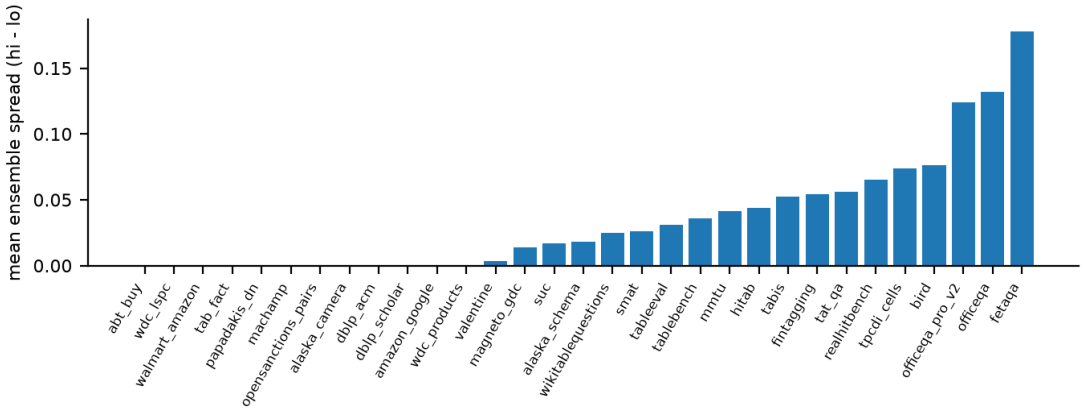



**Figure 20:** Ensemble uncertainty per dataset: mean spread of the per-answer score across the 100 weightings.

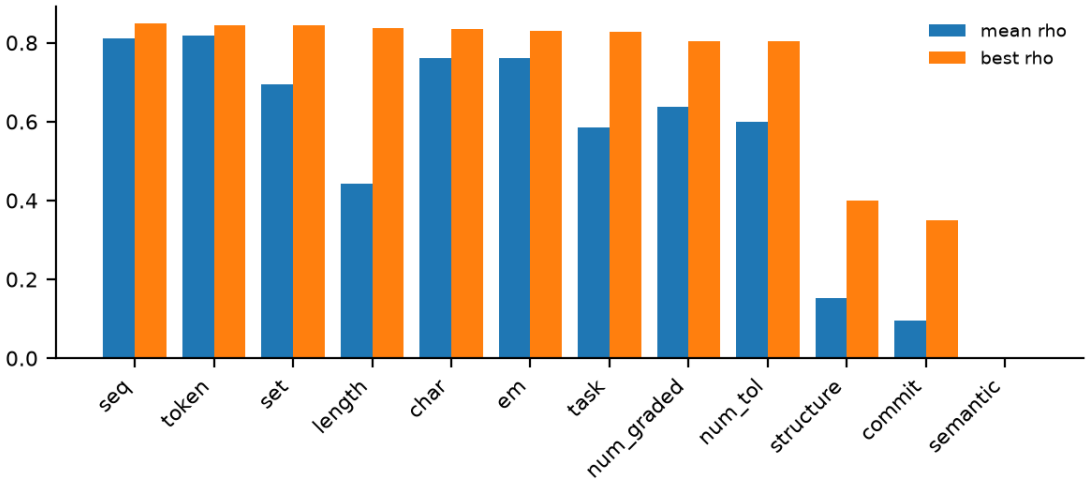



**Figure 21:** Anchor $\rho$ per metric family: mean and best member.

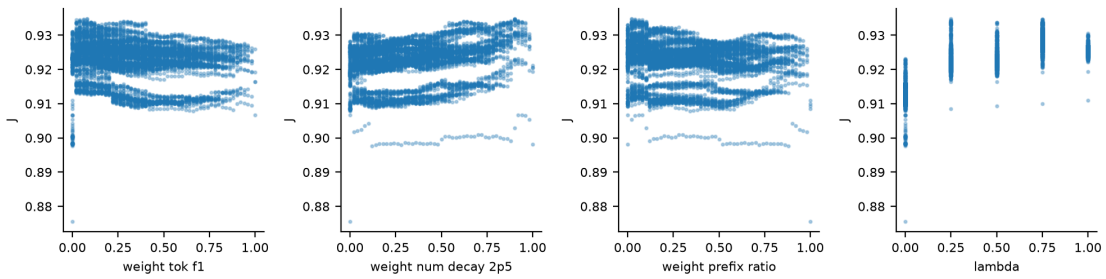



**Figure 22:** Objective $J$ against each selected weight and the hedge penalty $\lambda$ over all 6,630 grid points.

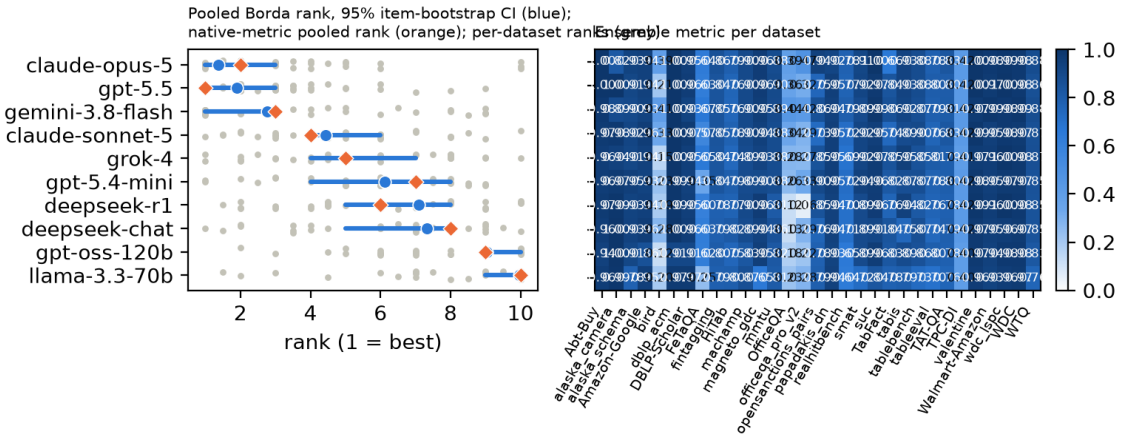



**Figure 23:** Ensemble leaderboard with bootstrap intervals and the native-metric ranking beside.

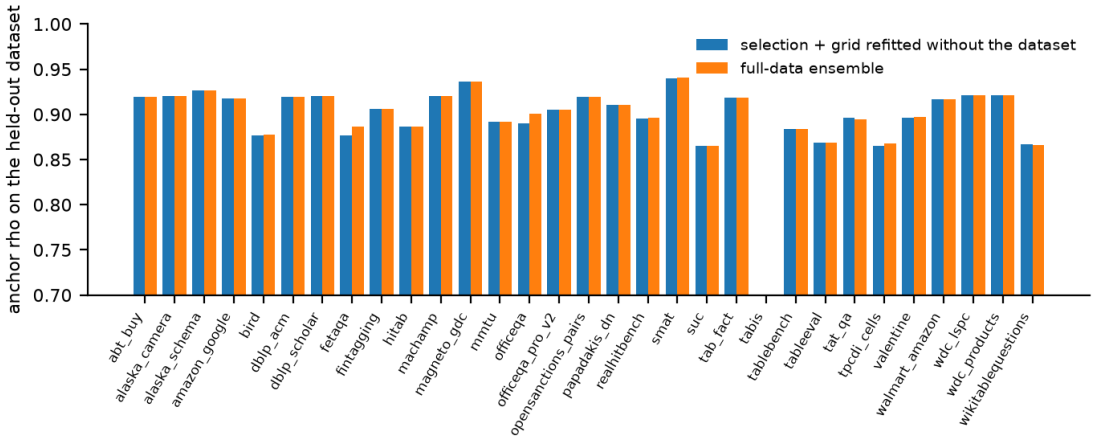



**Figure 24:** Leave-one-dataset-out: anchor $\rho$ on the held-out dataset under the re-fitted and the full-data ensemble.

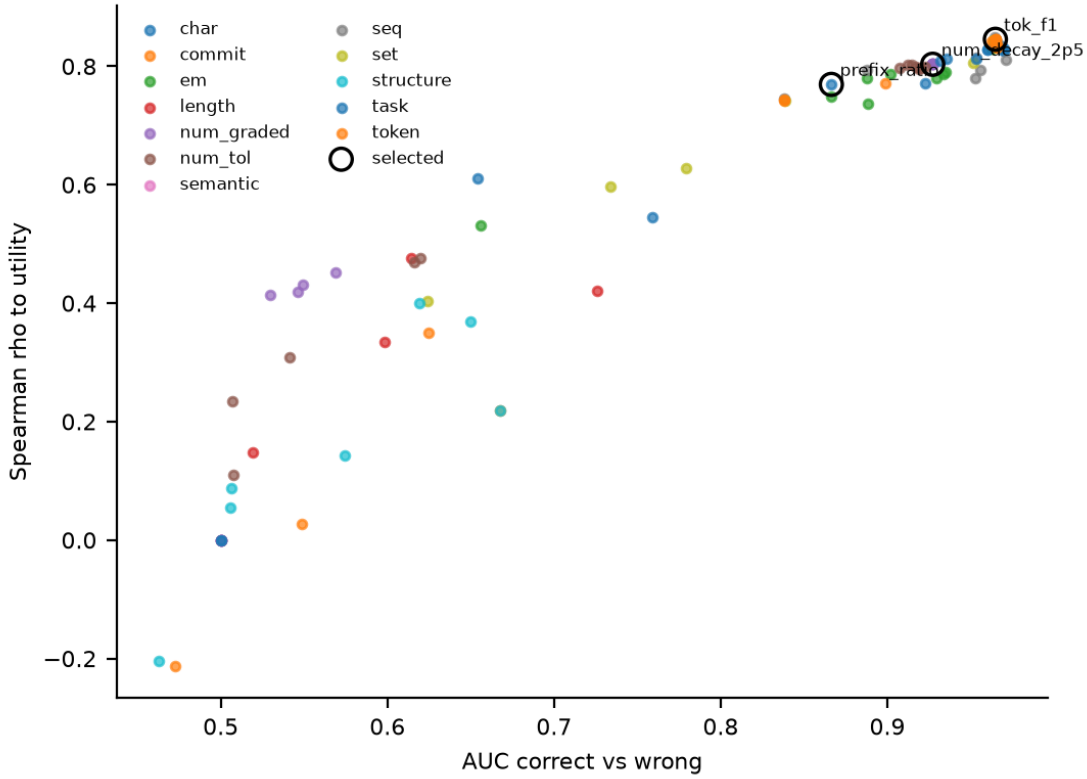



**Figure 25:** The 100-metric library on the anchors: AUC against Spearman $\rho$, coloured by family; the three selected metrics are circled.

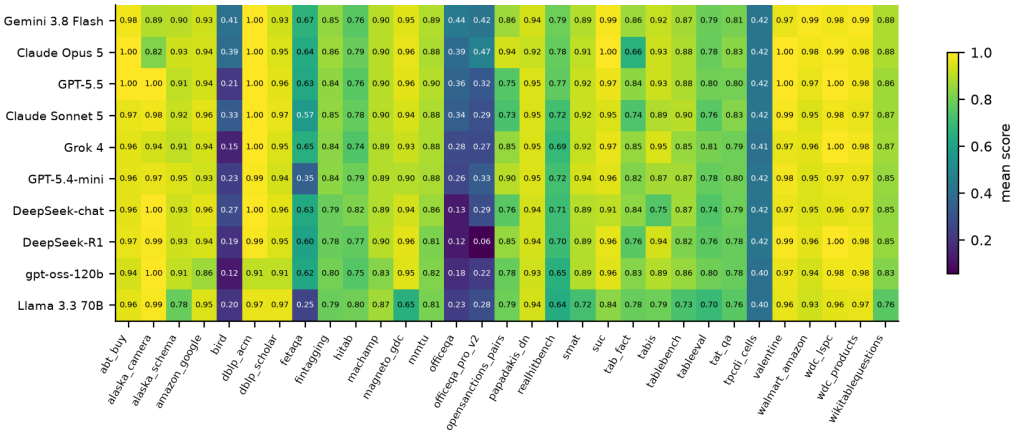



**Figure 26:** Mean score per model and dataset under the headline rule.

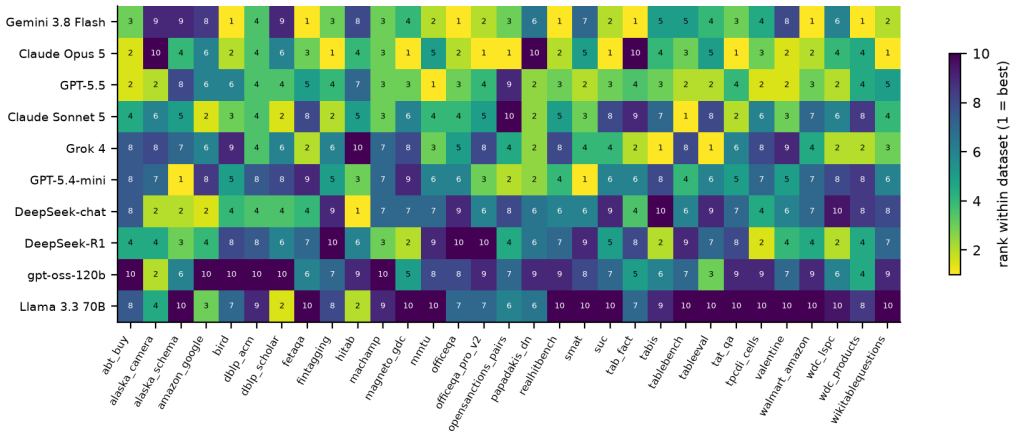



**Figure 27:** Rank per model and dataset under the headline rule.

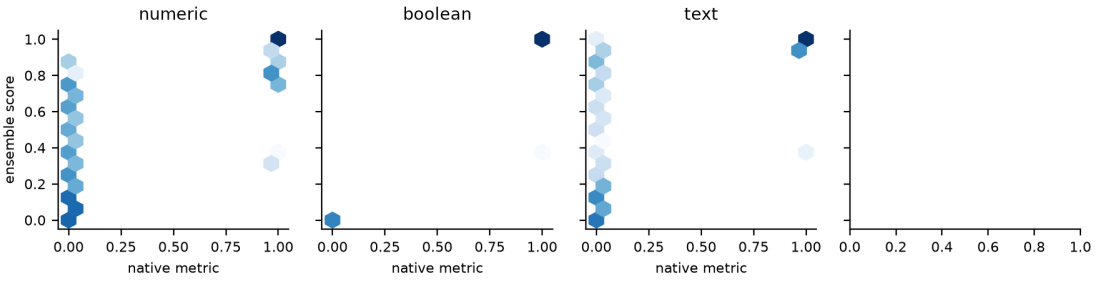



**Figure 28:** Per-answer native metric against the ensemble score, by answer type.

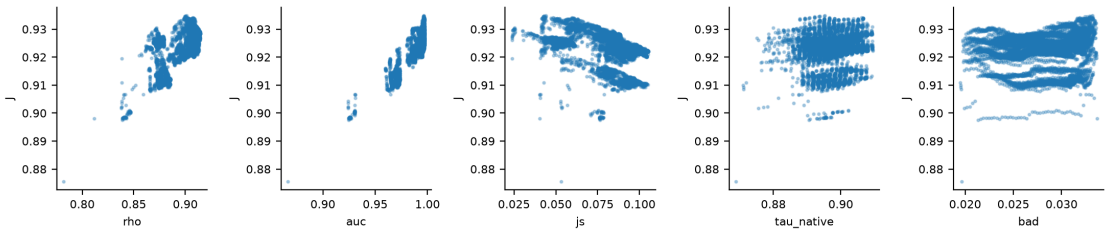



**Figure 29:** Objective $J$ against each of its five terms over the grid.

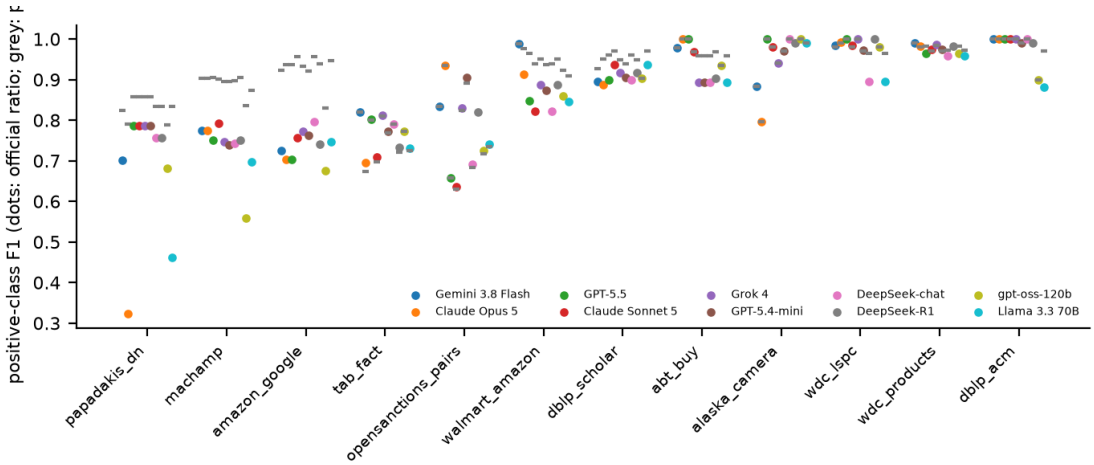



**Figure 30:** Positive-class F1 on the balanced pool (grey) and at the official match ratio (coloured) per model and matching dataset.

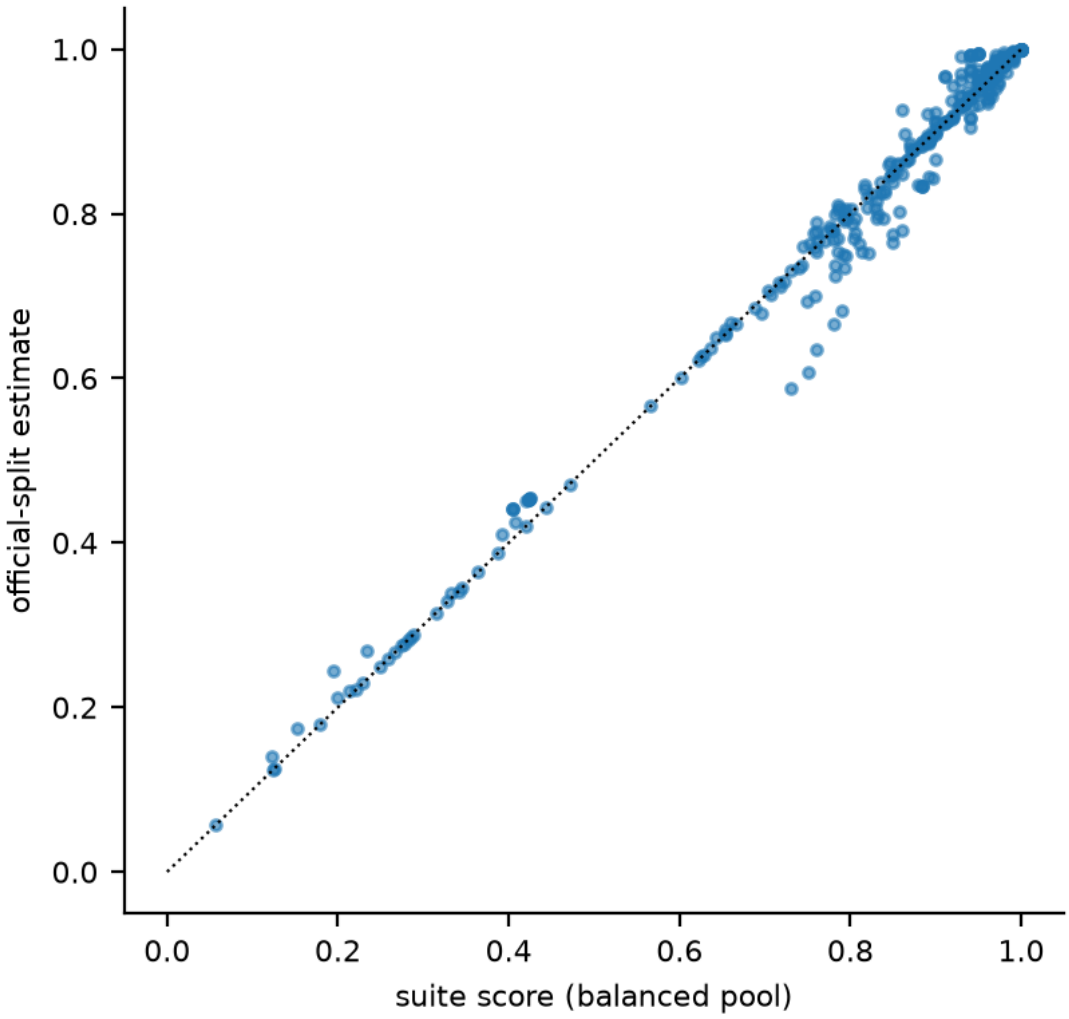



**Figure 31:** Suite score against post-stratified official-split estimate for every (dataset, model) pair under the headline rule.

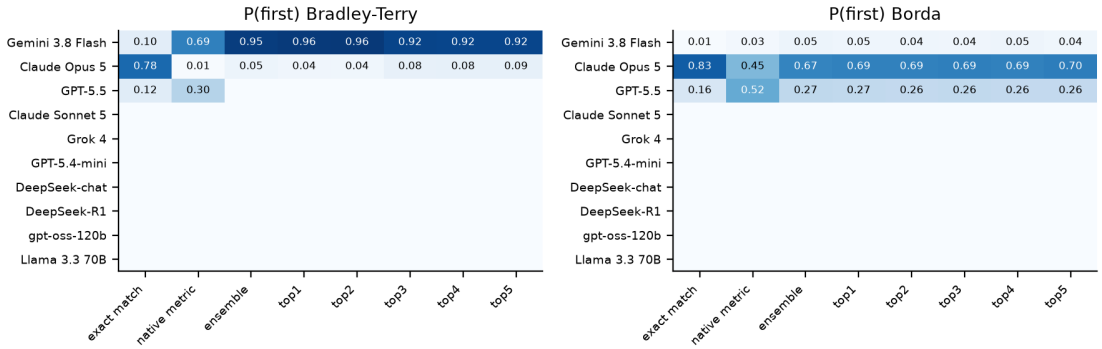



**Figure 32:** Bootstrap probability of first place under every rule, by pooling method.

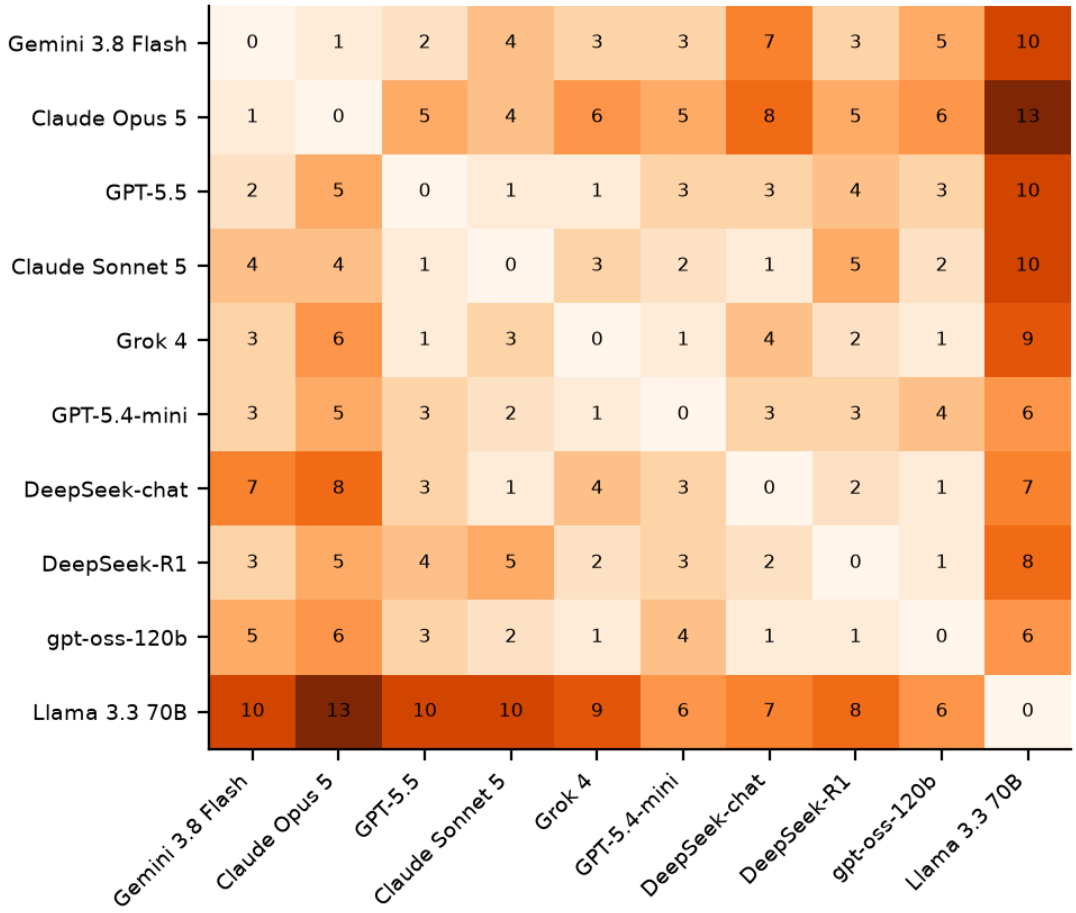



**Figure 33:** Number of datasets (of 31) in which the per-item paired test separates each pair of models under the headline rule (Holm within dataset).

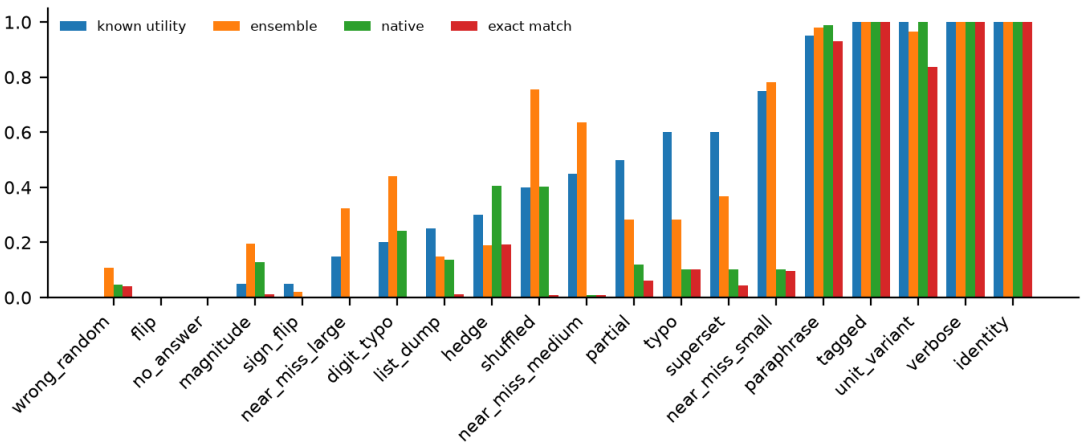



**Figure 34:** Mean score per anchor operator under the ensemble, the native metric and exact match, against known utility.

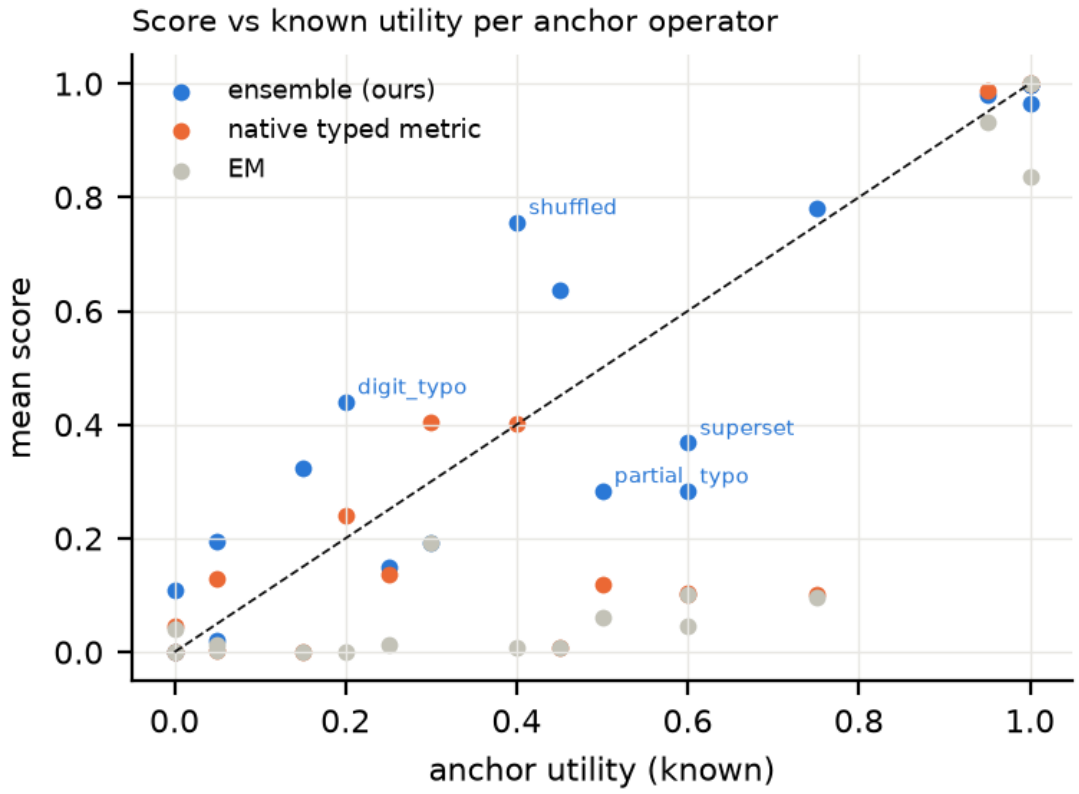



**Figure 35:** Probes: mean score per anchor operator against known utility for the ensemble and the single-metric baselines.

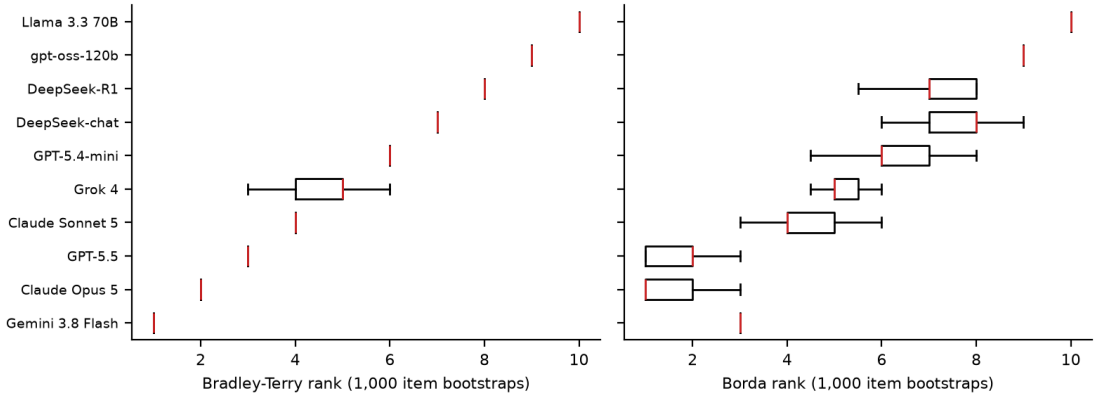



**Figure 36:** Bootstrap distribution of each model's rank under the headline rule (ensemble).

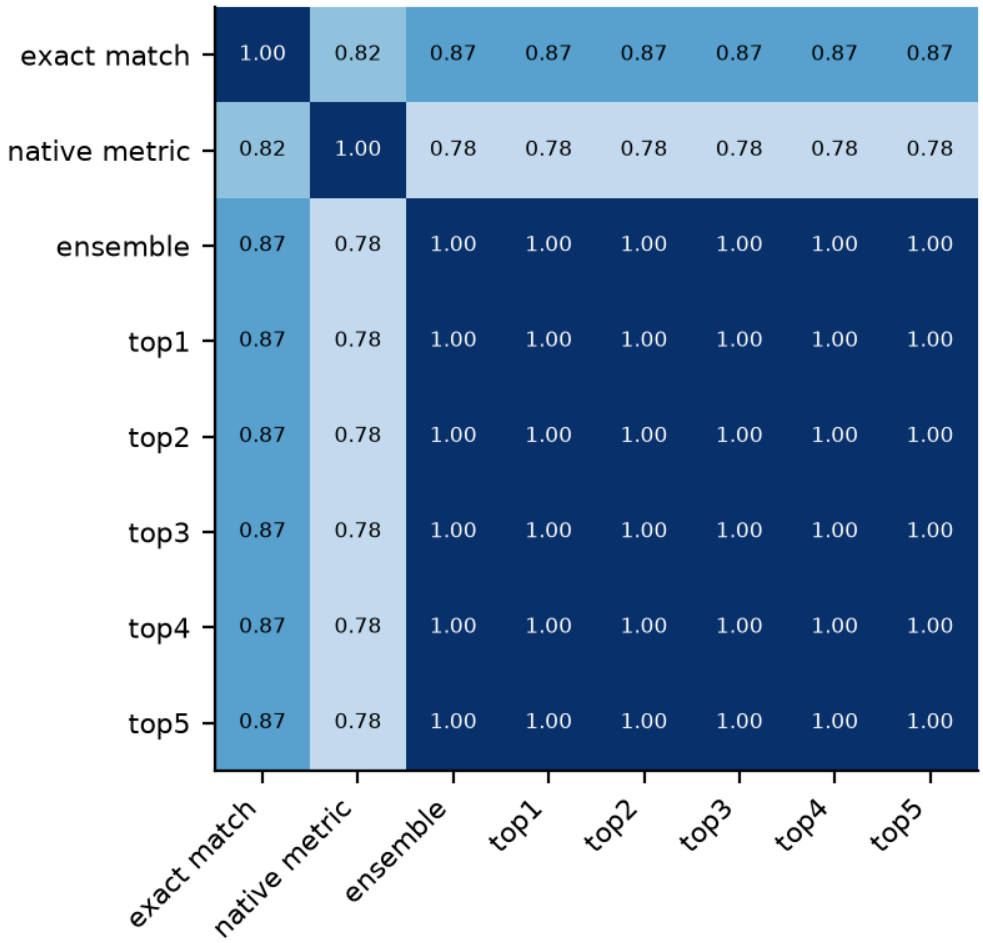



**Figure 37:** Kendall $\tau$ between the Bradley–Terry rankings of each pair of rules.

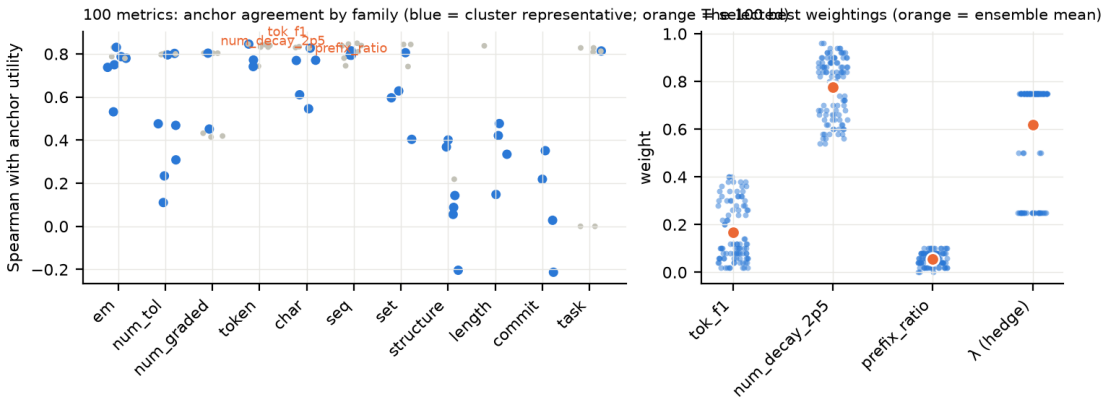



**Figure 38:** Metric selection: anchor agreement of every library metric with the selected set marked.

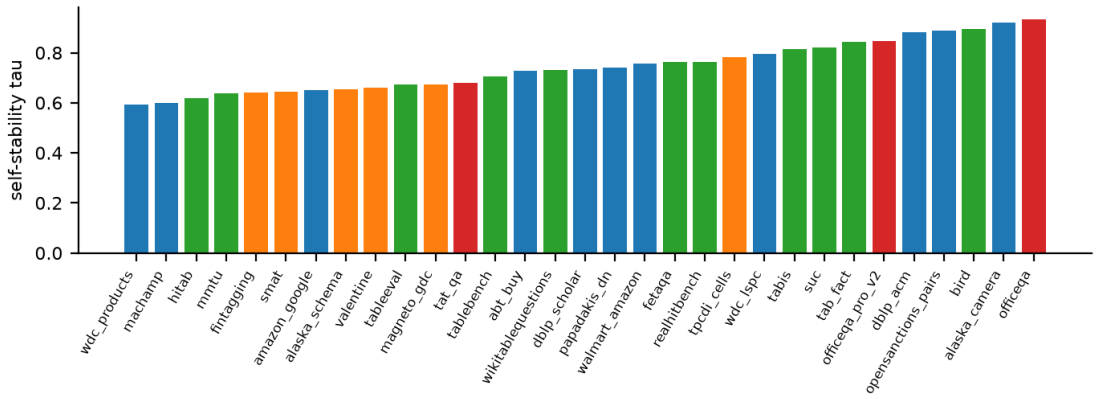



**Figure 39:** Self-stability of each dataset's model ranking (Kendall $\tau$ between bootstrap halves).

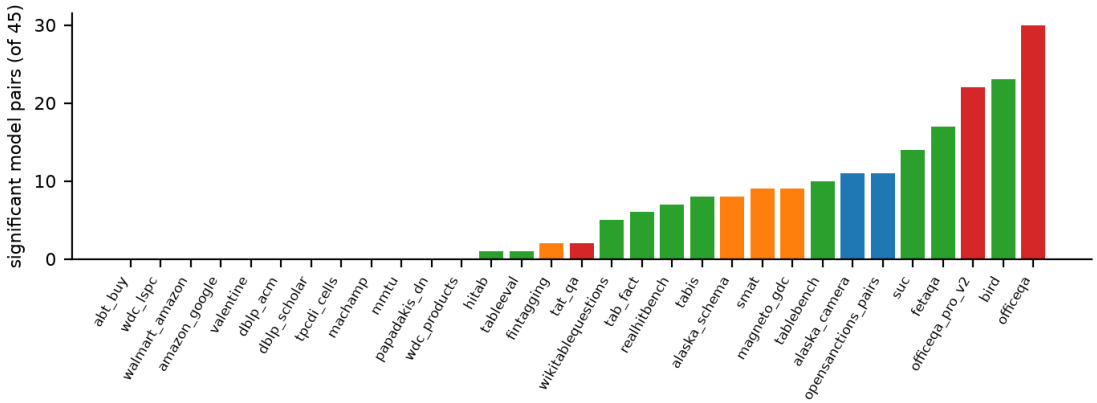



**Figure 40:** How many of the 45 model pairs each dataset separates at the item level under the headline rule (colour = family).

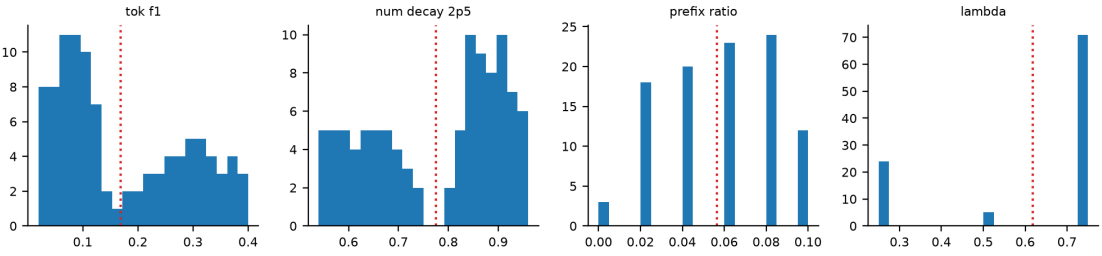



**Figure 41:** Weights and $\lambda$ of the 100 best grid points that form the ensemble (dotted = mean).

# B All tables

Every saved table, one file each under `tables/`; include any in the main text with `\input`.

**Table 8:** Baselines under the same objective.

| baseline | rho | auc | js | bad | tau_native | J |
|---|---|---|---|---|---|---|
| native_style_metric | 0.826 | 0.935 | 0.031 | 0.014 | 0.992 | 0.925 |
| em_norm | 0.849 | 0.963 | 0.024 | 0.012 | 0.924 | 0.928 |
| tok_f1 | 0.859 | 0.965 | 0.079 | 0.032 | 0.888 | 0.907 |
| rouge_l | 0.862 | 0.965 | 0.082 | 0.032 | 0.892 | 0.907 |
| num_tol_1 | 0.806 | 0.919 | 0.032 | 0.016 | 0.946 | 0.909 |
| uniform_over_selected_no_gate | 0.882 | 0.970 | 0.098 | 0.028 | 0.900 | 0.911 |
| ensemble_top100 | 0.904 | 0.996 | 0.053 | 0.032 | 0.898 | 0.934 |

**Table 9:** Sampled designs vs the exhaustive grid (best $J = 0.9348$ over 6630 points).

| design | best_J_mean | best_J_sd | regret_mean | regret_max | min_pairwise_L1 | discrepancy | zero_share |
|---|---|---|---|---|---|---|---|
| dirichlet_lattice | 0.9342 | 0.0004 | 0.0007 | 0.0014 | 0.0400 | 0.2728 | 0.0615 |
| lhs_simplex | 0.9340 | 0.0003 | 0.0008 | 0.0015 | 0.0400 | 0.2702 | 0.0555 |
| uniform_lattice | 0.9342 | 0.0003 | 0.0007 | 0.0014 | 0.0400 | 0.2639 | 0.1115 |

**Table 10:** Dataset diagnostics under the headline rule.

| dataset | family | mean_score | spread_ across_models | floor | ceiling | share_ constant_ bootstrap_ rankings | ranking_self_ stability_tau | tau_to_ others_mean | n_items | exclude_ from_ objective |
|---|---|---|---|---|---|---|---|---|---|---|
| abt_buy | EM | 0.970 | 0.060 | no | no | 0.000 | 0.726 | 0.346 | 100 | no |
| alaska_camera | EM | 0.958 | 0.180 | no | no | 0.000 | 0.919 | -0.226 | 100 | no |
| alaska_schema | SM | 0.909 | 0.170 | no | no | 0.000 | 0.653 | -0.024 | 93 | no |
| amazon_google | EM | 0.935 | 0.100 | no | no | 0.000 | 0.650 | -0.043 | 100 | no |
| bird | TQA | 0.252 | 0.285 | no | no | 0.000 | 0.894 | 0.225 | 100 | no |
| dblp_acm | EM | 0.986 | 0.090 | no | no | 0.000 | 0.884 | 0.337 | 100 | no |
| dblp_scholar | EM | 0.949 | 0.060 | no | no | 0.000 | 0.733 | -0.030 | 100 | no |
| fetaqa | TQA | 0.559 | 0.417 | no | no | 0.000 | 0.764 | 0.223 | 100 | no |
| fintagging | SM | 0.823 | 0.078 | no | no | 0.000 | 0.641 | 0.281 | 100 | no |
| hitab | TQA | 0.775 | 0.071 | no | no | 0.000 | 0.618 | -0.128 | 100 | no |
| machamp | EM | 0.887 | 0.070 | no | no | 0.000 | 0.600 | 0.358 | 100 | no |
| magneto_gdc | SM | 0.915 | 0.310 | no | no | 0.000 | 0.675 | 0.232 | 100 | no |
| mmtu | TQA | 0.862 | 0.085 | no | no | 0.000 | 0.639 | 0.324 | 100 | no |
| officeqa | FIN | 0.273 | 0.320 | no | no | 0.000 | 0.935 | 0.286 | 100 | no |
| officeqa_pro_v2 | FIN | 0.294 | 0.414 | no | no | 0.000 | 0.847 | 0.259 | 90 | no |
| opensanctions_pairs | EM | 0.821 | 0.210 | no | no | 0.000 | 0.888 | 0.034 | 100 | no |
| papadakis_dn | EM | 0.941 | 0.030 | no | no | 0.002 | 0.739 | 0.067 | 100 | no |
| realhitbench | TQA | 0.717 | 0.150 | no | no | 0.000 | 0.764 | 0.347 | 100 | no |
| smat | SM | 0.890 | 0.216 | no | no | 0.000 | 0.643 | 0.200 | 100 | no |
| suc | TQA | 0.950 | 0.162 | no | no | 0.000 | 0.820 | 0.314 | 100 | no |
| tab_fact | TQA | 0.798 | 0.200 | no | no | 0.000 | 0.845 | 0.012 | 100 | no |
| tabis | TQA | 0.885 | 0.199 | no | no | 0.000 | 0.814 | 0.181 | 100 | no |
| tablebench | TQA | 0.854 | 0.172 | no | no | 0.000 | 0.705 | 0.273 | 100 | no |
| tableeval | TQA | 0.771 | 0.110 | no | no | 0.000 | 0.674 | 0.154 | 100 | no |
| tat_qa | FIN | 0.797 | 0.074 | no | no | 0.000 | 0.678 | 0.363 | 100 | no |
| tpcdi_cells | SM | 0.418 | 0.020 | no | no | 0.000 | 0.783 | 0.285 | 100 | no |
| valentine | SM | 0.981 | 0.036 | no | yes | 0.000 | 0.661 | 0.245 | 100 | yes |
| walmart_amazon | EM | 0.958 | 0.060 | no | no | 0.000 | 0.755 | 0.348 | 100 | no |
| wdc_lspc | EM | 0.982 | 0.040 | no | yes | 0.004 | 0.796 | 0.218 | 100 | yes |
| wdc_products | EM | 0.977 | 0.020 | no | yes | 0.006 | 0.592 | 0.162 | 100 | yes |
| wikitablequestions | TQA | 0.850 | 0.125 | no | no | 0.000 | 0.732 | 0.342 | 100 | no |

**Table 11:** Ensemble summary.

| quantity | value |
|---|---|
| selected | ["tok_f1", "num_decay_2p5", "prefix_ratio"] |
| selected_idx | [33, 24, 51] |
| hedge_col | 89 |
| alpha | "rho": 0.3, "auc": 0.15, "consistency": 0.2, "native": 0.15, "bad": 0.2 |
| grid_step | 0.020 |
| n_grid | 6630 |
| top_k | 100 |
| mean_weights | "tok_f1": 0.16860000000000006, "num_decay_2p5": 0.7748, "prefix_ratio": 0.05659999999999999 |
| mean_lambda | 0.618 |
| weight_spread | "tok_f1": [0.02, 0.4], "num_decay_2p5": [0.54, 0.96], "prefix_ratio": [0.0, 0.1] |
| J_top1 | 0.935 |
| J_top100 | 0.933 |
| J_median_grid | 0.923 |
| terms_top1 | "rho": 0.9014035670162496, "auc": 0.9964055948226619, "js": 0.04391693277745681, "tau_native": 0.8951941491376547, "bad": 0.033203903955460724 |
| terms_ensemble_mean | "rho": 0.9035525333633605, "auc": 0.996493843375891, "js": 0.05273826176348726, "tau_native": 0.8980107486682813, "bad": 0.03245605941333208 |

**Table 12:** Mean ensemble spread per dataset.

| dataset | spread |
|---|---|
| abt_buy | 0.000 |
| wdc_lspc | 0.000 |
| walmart_amazon | 0.000 |
| tab_fact | 0.000 |
| papadakis_dn | 0.000 |
| machamp | 0.000 |
| opensanctions_pairs | 0.000 |
| alaska_camera | 0.000 |
| dblp_acm | 0.000 |
| dblp_scholar | 0.000 |
| amazon_google | 0.001 |
| wdc_products | 0.001 |
| valentine | 0.004 |
| magneto_gdc | 0.014 |
| suc | 0.017 |
| alaska_schema | 0.018 |
| wikitablequestions | 0.025 |
| smat | 0.026 |
| tableeval | 0.031 |
| tablebench | 0.036 |
| mmtu | 0.042 |
| hitab | 0.044 |
| tabis | 0.052 |
| fintagging | 0.054 |
| tat_qa | 0.056 |
| realhitbench | 0.066 |
| tpcdi_cells | 0.074 |
| bird | 0.076 |
| officeqa_pro_v2 | 0.124 |
| officeqa | 0.132 |
| fetaqa | 0.178 |

**Table 13:** The 25 best weightings with every objective term.

| rank | rho | auc | js | bad | tau_native | J | w_tok_f1 | w_num_decay_2p5 | w_prefix_ratio | lambda |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.901 | 0.996 | 0.044 | 0.033 | 0.895 | 0.935 | 0.080 | 0.900 | 0.020 | 0.250 |
| 2 | 0.900 | 0.997 | 0.043 | 0.033 | 0.895 | 0.935 | 0.080 | 0.900 | 0.020 | 0.750 |
| 3 | 0.900 | 0.996 | 0.043 | 0.033 | 0.894 | 0.935 | 0.060 | 0.900 | 0.040 | 0.750 |
| 4 | 0.901 | 0.996 | 0.044 | 0.033 | 0.894 | 0.935 | 0.060 | 0.900 | 0.040 | 0.250 |
| 5 | 0.899 | 0.996 | 0.042 | 0.032 | 0.893 | 0.934 | 0.020 | 0.900 | 0.080 | 0.750 |
| 6 | 0.900 | 0.996 | 0.042 | 0.033 | 0.892 | 0.934 | 0.040 | 0.900 | 0.060 | 0.750 |
| 7 | 0.901 | 0.996 | 0.044 | 0.033 | 0.892 | 0.934 | 0.060 | 0.920 | 0.020 | 0.250 |
| 8 | 0.900 | 0.996 | 0.043 | 0.033 | 0.892 | 0.934 | 0.040 | 0.920 | 0.040 | 0.250 |
| 9 | 0.900 | 0.996 | 0.044 | 0.033 | 0.892 | 0.934 | 0.040 | 0.900 | 0.060 | 0.250 |
| 10 | 0.902 | 0.996 | 0.048 | 0.033 | 0.897 | 0.934 | 0.080 | 0.880 | 0.040 | 0.250 |
| 11 | 0.899 | 0.996 | 0.042 | 0.033 | 0.891 | 0.934 | 0.040 | 0.920 | 0.040 | 0.750 |
| 12 | 0.907 | 0.997 | 0.060 | 0.032 | 0.906 | 0.934 | 0.280 | 0.640 | 0.080 | 0.750 |
| 13 | 0.908 | 0.997 | 0.061 | 0.032 | 0.907 | 0.934 | 0.280 | 0.620 | 0.100 | 0.750 |
| 14 | 0.908 | 0.997 | 0.061 | 0.032 | 0.907 | 0.934 | 0.300 | 0.620 | 0.080 | 0.750 |
| 15 | 0.899 | 0.996 | 0.043 | 0.033 | 0.891 | 0.934 | 0.060 | 0.920 | 0.020 | 0.750 |
| 16 | 0.901 | 0.996 | 0.047 | 0.033 | 0.895 | 0.934 | 0.080 | 0.880 | 0.040 | 0.750 |
| 17 | 0.902 | 0.996 | 0.049 | 0.033 | 0.897 | 0.934 | 0.100 | 0.880 | 0.020 | 0.250 |
| 18 | 0.899 | 0.996 | 0.042 | 0.033 | 0.889 | 0.934 | 0.040 | 0.940 | 0.020 | 0.750 |
| 19 | 0.907 | 0.997 | 0.060 | 0.032 | 0.907 | 0.934 | 0.300 | 0.640 | 0.060 | 0.750 |
| 20 | 0.906 | 0.997 | 0.059 | 0.032 | 0.907 | 0.934 | 0.300 | 0.660 | 0.040 | 0.750 |
| 21 | 0.901 | 0.996 | 0.043 | 0.033 | 0.888 | 0.934 | 0.040 | 0.940 | 0.020 | 0.250 |
| 22 | 0.907 | 0.997 | 0.061 | 0.032 | 0.907 | 0.934 | 0.320 | 0.620 | 0.060 | 0.750 |
| 23 | 0.908 | 0.997 | 0.060 | 0.032 | 0.904 | 0.934 | 0.260 | 0.640 | 0.100 | 0.750 |
| 24 | 0.906 | 0.997 | 0.060 | 0.032 | 0.907 | 0.934 | 0.320 | 0.640 | 0.040 | 0.750 |
| 25 | 0.903 | 0.996 | 0.051 | 0.033 | 0.898 | 0.934 | 0.080 | 0.860 | 0.060 | 0.250 |

**Table 14:** Full leaderboard under rule exact match; weights: –.

| model | bt_strength | bt_se_vs_ref | rank_bt | bt_rank_lo95 | bt_rank_hi95 | p_first_bt | rank_borda | borda_rank_lo95 | borda_rank_hi95 | p_first_borda | rank_kemeny | rank_copeland | rank_mean_z | rank_rrf | rank_family_balanced_z | mean_score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Claude Opus 5 | 0.049 | 0.000 | 1.000 | 1.000 | 3.000 | 0.780 | 1.000 | 1.000 | 2.000 | 0.826 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.768 |
| Gemini 3.8 Flash | 0.039 | 0.016 | 2.000 | 1.000 | 3.000 | 0.105 | 3.000 | 2.000 | 5.000 | 0.007 | 3.000 | 3.000 | 3.000 | 3.000 | 3.000 | 0.763 |
| GPT-5.5 | 0.039 | 0.016 | 3.000 | 1.000 | 3.000 | 0.115 | 2.000 | 1.000 | 3.000 | 0.164 | 2.000 | 2.000 | 2.000 | 2.000 | 2.000 | 0.763 |
| Grok 4 | 0.019 | 0.016 | 4.000 | 4.000 | 7.000 | 0.000 | 5.000 | 3.000 | 6.000 | 0.000 | 5.000 | 5.000 | 5.000 | 5.000 | 5.000 | 0.753 |
| Claude Sonnet 5 | 0.015 | 0.016 | 5.000 | 4.000 | 7.000 | 0.000 | 4.000 | 3.000 | 6.000 | 0.000 | 4.000 | 4.000 | 4.000 | 4.000 | 4.000 | 0.751 |
| GPT-5.4-mini | 0.011 | 0.016 | 6.000 | 4.000 | 7.000 | 0.000 | 8.000 | 6.000 | 8.000 | 0.000 | 7.000 | 7.000 | 7.000 | 8.000 | 8.000 | 0.749 |
| DeepSeek-R1 | 0.008 | 0.016 | 7.000 | 5.000 | 7.000 | 0.000 | 6.000 | 4.000 | 8.000 | 0.000 | 6.000 | 6.000 | 6.000 | 6.000 | 6.000 | 0.747 |
| DeepSeek-chat | -0.016 | 0.016 | 8.000 | 8.000 | 9.000 | 0.000 | 7.000 | 6.000 | 8.000 | 0.000 | 8.000 | 8.000 | 8.000 | 7.000 | 7.000 | 0.735 |
| gpt-oss-120b | -0.033 | 0.016 | 9.000 | 8.000 | 9.000 | 0.000 | 9.000 | 9.000 | 9.000 | 0.000 | 9.000 | 9.000 | 9.000 | 9.000 | 9.000 | 0.727 |
| Llama 3.3 70B | -0.130 | 0.016 | 10.000 | 10.000 | 10.000 | 0.000 | 10.000 | 10.000 | 10.000 | 0.000 | 10.000 | 10.000 | 10.000 | 10.000 | 10.000 | 0.678 |

**Table 15:** Full leaderboard under rule ensemble; weights: tok_f1 0.17, num_decay_2p5 0.78, prefix_ratio 0.06, lambda 0.62.

| model | bt_strength | bt_se_vs_ref | rank_bt | bt_rank_lo95 | bt_rank_hi95 | p_first_bt | rank_borda | borda_rank_lo95 | borda_rank_hi95 | p_first_borda | rank_kemeny | rank_copeland | rank_mean_z | rank_rrf | rank_family_balanced_z | mean_score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gemini 3.8 Flash | 0.128 | 0.016 | 1.000 | 1.000 | 2.000 | 0.949 | 3.000 | 1.000 | 3.000 | 0.051 | 3.000 | 3.000 | 3.000 | 3.000 | 2.000 | 0.835 |
| Claude Opus 5 | 0.110 | 0.000 | 2.000 | 1.000 | 2.000 | 0.051 | 1.000 | 1.000 | 3.000 | 0.672 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.833 |
| GPT-5.5 | 0.047 | 0.016 | 3.000 | 3.000 | 4.025 | 0.000 | 2.000 | 1.000 | 3.000 | 0.271 | 2.000 | 2.000 | 2.000 | 2.000 | 3.000 | 0.824 |
| Claude Sonnet 5 | 0.035 | 0.016 | 4.000 | 3.000 | 5.000 | 0.000 | 4.000 | 4.000 | 6.000 | 0.000 | 4.000 | 4.000 | 4.000 | 4.000 | 4.000 | 0.814 |
| Grok 4 | 0.026 | 0.016 | 5.000 | 3.000 | 5.000 | 0.000 | 5.000 | 4.000 | 7.000 | 0.000 | 5.000 | 5.000 | 5.000 | 5.000 | 6.000 | 0.811 |
| GPT-5.4-mini | -0.009 | 0.016 | 6.000 | 6.000 | 7.000 | 0.000 | 6.000 | 4.000 | 8.000 | 0.000 | 6.000 | 6.000 | 6.000 | 6.000 | 5.000 | 0.805 |
| DeepSeek-chat | -0.020 | 0.016 | 7.000 | 6.000 | 7.000 | 0.000 | 8.000 | 5.000 | 8.000 | 0.000 | 7.000 | 7.000 | 8.000 | 8.000 | 7.000 | 0.798 |
| DeepSeek-R1 | -0.050 | 0.016 | 8.000 | 8.000 | 8.000 | 0.000 | 7.000 | 5.000 | 8.000 | 0.000 | 8.000 | 8.000 | 7.000 | 7.000 | 8.000 | 0.794 |
| gpt-oss-120b | -0.089 | 0.016 | 9.000 | 9.000 | 9.000 | 0.000 | 9.000 | 9.000 | 10.000 | 0.000 | 9.000 | 9.000 | 9.000 | 9.000 | 9.000 | 0.784 |
| Llama 3.3 70B | -0.177 | 0.016 | 10.000 | 10.000 | 10.000 | 0.000 | 10.000 | 9.000 | 10.000 | 0.000 | 10.000 | 10.000 | 10.000 | 10.000 | 10.000 | 0.747 |

**Table 16:** Leaderboard under the headline rule (ensemble).

| rank | model | BT strength | SE | Borda rank | Borda 95% | P(first) BT | mean score |
|---|---|---|---|---|---|---|---|
| 1 | Gemini 3.8 Flash | 0.128 | 0.016 | 3 | [1, 3] | 0.950 | 0.835 |
| 2 | Claude Opus 5 | 0.110 | 0.000 | 1 | [1, 3] | 0.050 | 0.833 |
| 3 | GPT-5.5 | 0.047 | 0.016 | 2 | [1, 3] | 0.000 | 0.824 |
| 4 | Claude Sonnet 5 | 0.035 | 0.016 | 4 | [4, 6] | 0.000 | 0.814 |
| 5 | Grok 4 | 0.026 | 0.016 | 5 | [4, 7] | 0.000 | 0.811 |
| 6 | GPT-5.4-mini | -0.009 | 0.016 | 6 | [4, 8] | 0.000 | 0.805 |
| 7 | DeepSeek-chat | -0.020 | 0.016 | 8 | [5, 8] | 0.000 | 0.798 |
| 8 | DeepSeek-R1 | -0.050 | 0.016 | 7 | [5, 8] | 0.000 | 0.794 |
| 9 | gpt-oss-120b | -0.089 | 0.016 | 9 | [9, 10] | 0.000 | 0.784 |
| 10 | Llama 3.3 70B | -0.177 | 0.016 | 10 | [9, 10] | 0.000 | 0.747 |

**Table 17:** Full leaderboard under rule native metric; weights: –.

| model | bt_strength | bt_se_vs_ref | rank_bt | bt_rank_lo95 | bt_rank_hi95 | p_first_bt | rank_borda | borda_rank_lo95 | borda_rank_hi95 | p_first_borda | rank_kemeny | rank_copeland | rank_mean_z | rank_rrf | rank_family_balanced_z | mean_score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gemini 3.8 Flash | 0.065 | 0.016 | 1.000 | 1.000 | 2.000 | 0.689 | 3.000 | 1.000 | 4.000 | 0.026 | 3.000 | 3.000 | 3.000 | 3.000 | 3.000 | 0.792 |
| GPT-5.5 | 0.060 | 0.016 | 2.000 | 1.000 | 3.000 | 0.304 | 1.000 | 1.000 | 3.000 | 0.519 | 2.000 | 2.000 | 1.000 | 1.000 | 2.000 | 0.792 |
| Claude Opus 5 | 0.046 | 0.000 | 3.000 | 2.000 | 4.000 | 0.007 | 2.000 | 1.000 | 3.000 | 0.451 | 1.000 | 1.000 | 2.000 | 2.000 | 1.000 | 0.792 |
| Grok 4 | 0.031 | 0.016 | 4.000 | 3.000 | 5.000 | 0.000 | 5.000 | 4.000 | 6.000 | 0.000 | 5.000 | 5.000 | 5.000 | 5.000 | 5.000 | 0.779 |
| Claude Sonnet 5 | 0.014 | 0.016 | 5.000 | 4.000 | 7.000 | 0.000 | 4.000 | 3.000 | 6.000 | 0.000 | 4.000 | 4.000 | 4.000 | 4.000 | 4.000 | 0.776 |
| DeepSeek-R1 | 0.012 | 0.016 | 6.000 | 5.000 | 7.000 | 0.000 | 6.000 | 4.488 | 8.000 | 0.000 | 6.000 | 6.000 | 6.000 | 6.000 | 7.000 | 0.771 |
| DeepSeek-chat | -0.000 | 0.016 | 7.000 | 5.000 | 8.000 | 0.000 | 8.000 | 6.000 | 8.000 | 0.000 | 8.000 | 8.000 | 8.000 | 8.000 | 8.000 | 0.761 |
| GPT-5.4-mini | -0.010 | 0.016 | 8.000 | 7.000 | 8.000 | 0.000 | 7.000 | 5.000 | 8.000 | 0.000 | 7.000 | 7.000 | 7.000 | 7.000 | 6.000 | 0.770 |
| gpt-oss-120b | -0.053 | 0.016 | 9.000 | 9.000 | 9.000 | 0.000 | 9.000 | 9.000 | 9.000 | 0.000 | 9.000 | 9.000 | 9.000 | 9.000 | 9.000 | 0.748 |
| Llama 3.3 70B | -0.164 | 0.016 | 10.000 | 10.000 | 10.000 | 0.000 | 10.000 | 10.000 | 10.000 | 0.000 | 10.000 | 10.000 | 10.000 | 10.000 | 10.000 | 0.699 |

**Table 18:** Full leaderboard under rule top1; weights: tok_f1 0.08, num_decay_2p5 0.90, prefix_ratio 0.02, lambda 0.25.

| model | bt_strength | bt_se_vs_ref | rank_bt | bt_rank_lo95 | bt_rank_hi95 | p_first_bt | rank_borda | borda_rank_lo95 | borda_rank_hi95 | p_first_borda | rank_kemeny | rank_copeland | rank_mean_z | rank_rrf | rank_family_balanced_z | mean_score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gemini 3.8 Flash | 0.127 | 0.016 | 1.000 | 1.000 | 2.000 | 0.959 | 3.000 | 1.000 | 4.000 | 0.046 | 3.000 | 3.000 | 3.000 | 3.000 | 2.000 | 0.840 |
| Claude Opus 5 | 0.108 | 0.000 | 2.000 | 1.000 | 2.000 | 0.041 | 1.000 | 1.000 | 3.000 | 0.686 | 1.000 | 1.000 | 2.000 | 1.000 | 1.000 | 0.838 |
| GPT-5.5 | 0.046 | 0.016 | 3.000 | 3.000 | 5.000 | 0.000 | 2.000 | 1.000 | 3.000 | 0.265 | 2.000 | 2.000 | 1.000 | 2.000 | 3.000 | 0.828 |
| Claude Sonnet 5 | 0.035 | 0.016 | 4.000 | 3.000 | 5.000 | 0.000 | 4.000 | 3.000 | 6.000 | 0.000 | 4.000 | 4.000 | 4.000 | 4.000 | 4.000 | 0.820 |
| Grok 4 | 0.026 | 0.016 | 5.000 | 3.000 | 5.000 | 0.000 | 5.000 | 4.000 | 7.000 | 0.000 | 5.000 | 5.000 | 5.000 | 5.000 | 6.000 | 0.815 |
| GPT-5.4-mini | -0.009 | 0.016 | 6.000 | 6.000 | 7.000 | 0.000 | 6.000 | 4.000 | 8.000 | 0.000 | 6.000 | 6.000 | 6.000 | 6.000 | 5.000 | 0.809 |
| DeepSeek-chat | -0.021 | 0.016 | 7.000 | 6.000 | 7.025 | 0.000 | 8.000 | 6.000 | 8.000 | 0.000 | 8.000 | 8.000 | 8.000 | 8.000 | 7.000 | 0.802 |
| DeepSeek-R1 | -0.047 | 0.016 | 8.000 | 7.000 | 8.000 | 0.000 | 7.000 | 5.000 | 8.000 | 0.000 | 7.000 | 7.000 | 7.000 | 7.000 | 8.000 | 0.799 |
| gpt-oss-120b | -0.089 | 0.016 | 9.000 | 9.000 | 9.000 | 0.000 | 9.000 | 9.000 | 10.000 | 0.000 | 9.000 | 9.000 | 9.000 | 9.000 | 9.000 | 0.788 |
| Llama 3.3 70B | -0.176 | 0.016 | 10.000 | 10.000 | 10.000 | 0.000 | 10.000 | 9.000 | 10.000 | 0.000 | 10.000 | 10.000 | 10.000 | 10.000 | 10.000 | 0.751 |

**Table 19:** Full leaderboard under rule top2; weights: tok_f1 0.08, num_decay_2p5 0.90, prefix_ratio 0.02, lambda 0.75.

| model | bt_strength | bt_se_vs_ref | rank_bt | bt_rank_lo95 | bt_rank_hi95 | p_first_bt | rank_borda | borda_rank_lo95 | borda_rank_hi95 | p_first_borda | rank_kemeny | rank_copeland | rank_mean_z | rank_rrf | rank_family_balanced_z | mean_score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gemini 3.8 Flash | 0.128 | 0.016 | 1.000 | 1.000 | 2.000 | 0.962 | 3.000 | 1.000 | 4.000 | 0.043 | 3.000 | 3.000 | 3.000 | 3.000 | 3.000 | 0.840 |
| Claude Opus 5 | 0.109 | 0.000 | 2.000 | 1.000 | 2.000 | 0.038 | 1.000 | 1.000 | 3.000 | 0.686 | 1.000 | 1.000 | 2.000 | 1.000 | 1.000 | 0.838 |
| GPT-5.5 | 0.046 | 0.016 | 3.000 | 3.000 | 5.000 | 0.000 | 2.000 | 1.000 | 3.000 | 0.260 | 2.000 | 2.000 | 1.000 | 2.000 | 2.000 | 0.828 |
| Claude Sonnet 5 | 0.035 | 0.016 | 4.000 | 3.000 | 5.000 | 0.000 | 4.000 | 3.000 | 6.000 | 0.000 | 4.000 | 4.000 | 4.000 | 4.000 | 4.000 | 0.819 |
| Grok 4 | 0.027 | 0.016 | 5.000 | 3.000 | 5.000 | 0.000 | 5.000 | 4.000 | 7.000 | 0.000 | 5.000 | 5.000 | 5.000 | 5.000 | 6.000 | 0.814 |
| GPT-5.4-mini | -0.008 | 0.016 | 6.000 | 6.000 | 7.000 | 0.000 | 6.000 | 4.000 | 8.000 | 0.000 | 6.000 | 6.000 | 6.000 | 6.000 | 5.000 | 0.809 |
| DeepSeek-chat | -0.021 | 0.016 | 7.000 | 6.000 | 7.000 | 0.000 | 8.000 | 5.000 | 8.000 | 0.000 | 7.000 | 7.000 | 8.000 | 8.000 | 7.000 | 0.801 |
| DeepSeek-R1 | -0.051 | 0.016 | 8.000 | 8.000 | 8.000 | 0.000 | 7.000 | 5.000 | 8.000 | 0.000 | 8.000 | 8.000 | 7.000 | 7.000 | 8.000 | 0.795 |
| gpt-oss-120b | -0.089 | 0.016 | 9.000 | 9.000 | 9.000 | 0.000 | 9.000 | 9.000 | 10.000 | 0.000 | 9.000 | 9.000 | 9.000 | 9.000 | 9.000 | 0.788 |
| Llama 3.3 70B | -0.176 | 0.016 | 10.000 | 10.000 | 10.000 | 0.000 | 10.000 | 9.000 | 10.000 | 0.000 | 10.000 | 10.000 | 10.000 | 10.000 | 10.000 | 0.749 |

**Table 20:** Full leaderboard under rule top3; weights: tok_f1 0.06, num_decay_2p5 0.90, prefix_ratio 0.04, lambda 0.75.

| model | bt_strength | bt_se_vs_ref | rank_bt | bt_rank_lo95 | bt_rank_hi95 | p_first_bt | rank_borda | borda_rank_lo95 | borda_rank_hi95 | p_first_borda | rank_kemeny | rank_copeland | rank_mean_z | rank_rrf | rank_family_balanced_z | mean_score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gemini 3.8 Flash | 0.126 | 0.016 | 1.000 | 1.000 | 2.000 | 0.921 | 3.000 | 1.000 | 4.000 | 0.041 | 3.000 | 3.000 | 3.000 | 3.000 | 3.000 | 0.839 |
| Claude Opus 5 | 0.110 | 0.000 | 2.000 | 1.000 | 2.000 | 0.079 | 1.000 | 1.000 | 3.000 | 0.692 | 1.000 | 1.000 | 2.000 | 1.000 | 1.000 | 0.837 |
| GPT-5.5 | 0.046 | 0.016 | 3.000 | 3.000 | 5.000 | 0.000 | 2.000 | 1.000 | 3.000 | 0.262 | 2.000 | 2.000 | 1.000 | 2.000 | 2.000 | 0.827 |
| Claude Sonnet 5 | 0.035 | 0.016 | 4.000 | 3.000 | 5.000 | 0.000 | 4.000 | 3.000 | 6.000 | 0.000 | 4.000 | 4.000 | 4.000 | 4.000 | 4.000 | 0.818 |
| Grok 4 | 0.027 | 0.016 | 5.000 | 3.000 | 5.000 | 0.000 | 5.000 | 4.000 | 7.000 | 0.000 | 5.000 | 5.000 | 5.000 | 5.000 | 6.000 | 0.814 |
| GPT-5.4-mini | -0.009 | 0.016 | 6.000 | 6.000 | 7.000 | 0.000 | 6.000 | 4.000 | 8.000 | 0.000 | 6.000 | 6.000 | 6.000 | 6.000 | 5.000 | 0.808 |
| DeepSeek-chat | -0.020 | 0.016 | 7.000 | 6.000 | 7.000 | 0.000 | 8.000 | 5.000 | 8.000 | 0.000 | 7.000 | 7.000 | 8.000 | 8.000 | 7.000 | 0.800 |
| DeepSeek-R1 | -0.051 | 0.016 | 8.000 | 8.000 | 8.000 | 0.000 | 7.000 | 5.000 | 8.000 | 0.000 | 8.000 | 8.000 | 7.000 | 7.000 | 8.000 | 0.794 |
| gpt-oss-120b | -0.087 | 0.016 | 9.000 | 9.000 | 9.000 | 0.000 | 9.000 | 9.000 | 10.000 | 0.000 | 9.000 | 9.000 | 9.000 | 9.000 | 9.000 | 0.787 |
| Llama 3.3 70B | -0.177 | 0.016 | 10.000 | 10.000 | 10.000 | 0.000 | 10.000 | 9.000 | 10.000 | 0.000 | 10.000 | 10.000 | 10.000 | 10.000 | 10.000 | 0.749 |

**Table 21:** Full leaderboard under rule top4; weights: tok_f1 0.06, num_decay_2p5 0.90, prefix_ratio 0.04, lambda 0.25.

| model | bt_strength | bt_se_vs_ref | rank_bt | bt_rank_lo95 | bt_rank_hi95 | p_first_bt | rank_borda | borda_rank_lo95 | borda_rank_hi95 | p_first_borda | rank_kemeny | rank_copeland | rank_mean_z | rank_rrf | rank_family_balanced_z | mean_score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gemini 3.8 Flash | 0.125 | 0.016 | 1.000 | 1.000 | 2.000 | 0.917 | 3.000 | 1.000 | 4.000 | 0.047 | 3.000 | 3.000 | 3.000 | 3.000 | 3.000 | 0.840 |
| Claude Opus 5 | 0.110 | 0.000 | 2.000 | 1.000 | 2.000 | 0.083 | 1.000 | 1.000 | 3.000 | 0.686 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.838 |
| GPT-5.5 | 0.046 | 0.016 | 3.000 | 3.000 | 5.000 | 0.000 | 2.000 | 1.000 | 3.000 | 0.261 | 2.000 | 2.000 | 2.000 | 2.000 | 2.000 | 0.828 |
| Claude Sonnet 5 | 0.035 | 0.016 | 4.000 | 3.000 | 5.000 | 0.000 | 4.000 | 3.000 | 6.000 | 0.000 | 4.000 | 4.000 | 4.000 | 4.000 | 4.000 | 0.819 |
| Grok 4 | 0.026 | 0.016 | 5.000 | 3.000 | 5.000 | 0.000 | 5.000 | 4.000 | 7.000 | 0.000 | 5.000 | 5.000 | 5.000 | 5.000 | 6.000 | 0.814 |
| GPT-5.4-mini | -0.010 | 0.016 | 6.000 | 6.000 | 7.000 | 0.000 | 6.000 | 4.000 | 8.000 | 0.000 | 6.000 | 6.000 | 6.000 | 6.000 | 5.000 | 0.808 |
| DeepSeek-chat | -0.021 | 0.016 | 7.000 | 6.000 | 7.000 | 0.000 | 8.000 | 6.000 | 8.000 | 0.000 | 8.000 | 8.000 | 8.000 | 8.000 | 7.000 | 0.801 |
| DeepSeek-R1 | -0.047 | 0.016 | 8.000 | 8.000 | 8.000 | 0.000 | 7.000 | 5.000 | 8.000 | 0.000 | 7.000 | 7.000 | 7.000 | 7.000 | 8.000 | 0.798 |
| gpt-oss-120b | -0.087 | 0.016 | 9.000 | 9.000 | 9.000 | 0.000 | 9.000 | 9.000 | 10.000 | 0.000 | 9.000 | 9.000 | 9.000 | 9.000 | 9.000 | 0.787 |
| Llama 3.3 70B | -0.177 | 0.016 | 10.000 | 10.000 | 10.000 | 0.000 | 10.000 | 9.000 | 10.000 | 0.000 | 10.000 | 10.000 | 10.000 | 10.000 | 10.000 | 0.750 |

**Table 22:** Full leaderboard under rule top5; weights: tok_f1 0.02, num_decay_2p5 0.90, prefix_ratio 0.08, lambda 0.75.

| model | bt_strength | bt_se_vs_ref | rank_bt | bt_rank_lo95 | bt_rank_hi95 | p_first_bt | rank_borda | borda_rank_lo95 | borda_rank_hi95 | p_first_borda | rank_kemeny | rank_copeland | rank_mean_z | rank_rrf | rank_family_balanced_z | mean_score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gemini 3.8 Flash | 0.125 | 0.016 | 1.000 | 1.000 | 2.000 | 0.915 | 3.000 | 1.000 | 4.000 | 0.036 | 3.000 | 3.000 | 3.000 | 3.000 | 3.000 | 0.838 |
| Claude Opus 5 | 0.111 | 0.000 | 2.000 | 1.000 | 2.000 | 0.085 | 1.000 | 1.000 | 3.000 | 0.702 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.836 |
| GPT-5.5 | 0.046 | 0.016 | 3.000 | 3.000 | 4.000 | 0.000 | 2.000 | 1.000 | 3.000 | 0.258 | 2.000 | 2.000 | 2.000 | 2.000 | 2.000 | 0.826 |
| Claude Sonnet 5 | 0.035 | 0.016 | 4.000 | 3.000 | 5.000 | 0.000 | 4.000 | 3.000 | 6.000 | 0.000 | 4.000 | 4.000 | 4.000 | 4.000 | 4.000 | 0.817 |
| Grok 4 | 0.025 | 0.016 | 5.000 | 3.000 | 5.000 | 0.000 | 5.000 | 4.000 | 7.000 | 0.000 | 5.000 | 5.000 | 5.000 | 5.000 | 6.000 | 0.812 |
| GPT-5.4-mini | -0.010 | 0.016 | 6.000 | 6.000 | 7.000 | 0.000 | 6.000 | 4.000 | 8.000 | 0.000 | 6.000 | 6.000 | 6.000 | 6.000 | 5.000 | 0.807 |
| DeepSeek-chat | -0.021 | 0.016 | 7.000 | 6.000 | 7.000 | 0.000 | 8.000 | 6.000 | 8.000 | 0.000 | 7.000 | 7.000 | 8.000 | 8.000 | 7.000 | 0.799 |
| DeepSeek-R1 | -0.052 | 0.016 | 8.000 | 8.000 | 8.000 | 0.000 | 7.000 | 5.000 | 8.000 | 0.000 | 8.000 | 8.000 | 7.000 | 7.000 | 8.000 | 0.792 |
| gpt-oss-120b | -0.083 | 0.016 | 9.000 | 9.000 | 9.000 | 0.000 | 9.000 | 9.000 | 10.000 | 0.000 | 9.000 | 9.000 | 9.000 | 9.000 | 9.000 | 0.786 |
| Llama 3.3 70B | -0.176 | 0.016 | 10.000 | 10.000 | 10.000 | 0.000 | 10.000 | 9.000 | 10.000 | 0.000 | 10.000 | 10.000 | 10.000 | 10.000 | 10.000 | 0.748 |

**Table 23:** Leave-one-dataset-out.

| held_out | selected_lodo | rho_ heldout_ lodo | rho_ heldout_ full | js_all_ lodo | same_ selection |
|---|---|---|---|---|---|
| abt_buy | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.919 | 0.919 | 0.057 | yes |
| alaska_camera | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.921 | 0.921 | 0.057 | yes |
| alaska_schema | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.926 | 0.926 | 0.057 | yes |
| amazon_google | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.918 | 0.918 | 0.057 | yes |
| bird | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.877 | 0.877 | 0.055 | yes |
| dblp_acm | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.919 | 0.919 | 0.057 | yes |
| dblp_scholar | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.920 | 0.920 | 0.057 | yes |
| fetaqa | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.877 | 0.886 | 0.066 | yes |
| fintagging | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.906 | 0.906 | 0.068 | yes |
| hitab | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.887 | 0.887 | 0.057 | yes |
| machamp | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.921 | 0.921 | 0.057 | yes |
| magneto_gdc | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.936 | 0.936 | 0.057 | yes |
| mmtu | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.892 | 0.892 | 0.057 | yes |
| officeqa | ['tok_f1', 'num_tol_10', 'prefix_ratio'] | 0.890 | 0.901 | 0.038 | no |
| officeqa_pro_ v2 | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.905 | 0.905 | 0.056 | yes |
| opensanctions_ pairs | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.919 | 0.919 | 0.057 | yes |
| papadakis_dn | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.910 | 0.910 | 0.057 | yes |
| realhitbench | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.896 | 0.896 | 0.057 | yes |
| smat | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.940 | 0.941 | 0.058 | yes |
| suc | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.865 | 0.865 | 0.057 | yes |
| tab_fact | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.918 | 0.918 | 0.057 | yes |
| tabis | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.598 | 0.598 | 0.066 | yes |
| tablebench | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.884 | 0.884 | 0.057 | yes |
| tableeval | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.869 | 0.869 | 0.057 | yes |
| tat_qa | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.896 | 0.895 | 0.057 | yes |
| tpcdi_cells | ['tok_f1', 'num_decay_2p5', 'prefix_ratio', 'em_raw'] | 0.865 | 0.867 | 0.096 | no |
| valentine | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.897 | 0.897 | 0.057 | yes |
| walmart_ amazon | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.917 | 0.917 | 0.057 | yes |
| wdc_lspc | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.921 | 0.921 | 0.057 | yes |
| wdc_products | ['tok_f1', 'num_decay_2p5', 'prefix_ratio'] | 0.921 | 0.921 | 0.057 | yes |
| wikitablequestions | ['tok_f1', 'num_decay_2p5', 'prefix_ratio', 'em_raw'] | 0.867 | 0.866 | 0.095 | no |

**Table 24:** Every library metric: anchor $\rho$, AUC, gap, family, cluster.

| metric | rho | auc | gap | family | representative | cluster |
|---|---|---|---|---|---|---|
| rouge_l | 0.849 | 0.965 | 0.800 | seq | no | tok_f1 |
| rouge1 | 0.846 | 0.965 | 0.802 | seq | no | tok_f1 |
| tok_f1 | 0.846 | 0.965 | 0.802 | token | yes | tok_f1 |
| stem_f1 | 0.846 | 0.965 | 0.802 | token | no | tok_f1 |
| tok_f0p5 | 0.845 | 0.965 | 0.828 | token | no | tok_f1 |
| jaccard | 0.844 | 0.964 | 0.836 | set | no | tok_f1 |
| dice | 0.844 | 0.964 | 0.794 | set | no | tok_f1 |
| content_f1 | 0.843 | 0.963 | 0.786 | token | no | tok_f1 |
| tok_f2 | 0.841 | 0.965 | 0.749 | token | no | tok_f1 |
| tok_prec | 0.839 | 0.965 | 0.840 | token | no | tok_f1 |
| ter_score | 0.839 | 0.966 | 0.912 | seq | no | em_norm |
| bigram_f1 | 0.838 | 0.966 | 0.871 | token | no | tok_f1 |
| rouge2 | 0.838 | 0.966 | 0.871 | seq | no | tok_f1 |
| extra_token_score | 0.838 | 0.964 | 0.838 | length | no | tok_f1 |
| jaro_winkler | 0.834 | 0.968 | 0.490 | char | no | lev_sim |
| jaro_sim | 0.833 | 0.968 | 0.503 | char | no | lev_sim |
| trigram_f1 | 0.832 | 0.966 | 0.897 | token | no | em_norm |
| bigram_prec | 0.832 | 0.966 | 0.890 | token | no | tok_f1 |
| em_no_space | 0.831 | 0.963 | 0.925 | em | no | em_norm |
| em_norm | 0.831 | 0.963 | 0.925 | em | yes | em_norm |
| lcs_substr_ratio | 0.829 | 0.970 | 0.747 | char | no | lev_sim |
| bool_label_agree | 0.828 | 0.960 | 0.921 | task | no | em_norm |
| attribute_match_score | 0.828 | 0.960 | 0.921 | task | no | em_norm |
| lev_sim | 0.827 | 0.970 | 0.725 | char | yes | lev_sim |
| damerau_sim | 0.826 | 0.970 | 0.724 | char | no | lev_sim |
| bleu1 | 0.814 | 0.953 | 0.561 | seq | yes | bleu1 |
| native_style_metric | 0.814 | 0.935 | 0.866 | task | yes | native_style_metric |
| year_aware_em | 0.812 | 0.953 | 0.907 | task | no | em_norm |
| wer_score | 0.810 | 0.971 | 0.827 | seq | no | lev_sim |
| numeric_or_em | 0.809 | 0.932 | 0.864 | task | no | native_style_metric |
| pred_in_gold | 0.806 | 0.952 | 0.903 | set | yes | pred_in_gold |
| num_decay_10 | 0.803 | 0.927 | 0.825 | num_graded | no | num_decay_2p5 |
| num_decay_5 | 0.803 | 0.927 | 0.812 | num_graded | no | num_decay_2p5 |
| num_decay_0p5 | 0.803 | 0.927 | 0.674 | num_graded | no | num_decay_2p5 |
| num_decay_1 | 0.803 | 0.927 | 0.728 | num_graded | no | num_decay_2p5 |
| num_decay_2p5 | 0.803 | 0.927 | 0.787 | num_graded | yes | num_decay_2p5 |
| num_tol_5 | 0.803 | 0.915 | 0.829 | num_tol | no | num_tol_1 |
| num_tol_10 | 0.802 | 0.913 | 0.825 | num_tol | yes | num_tol_10 |
| num_tol_0p1 | 0.800 | 0.923 | 0.846 | num_tol | no | num_tol_1 |
| num_tol_2 | 0.798 | 0.917 | 0.835 | num_tol | no | num_tol_1 |
| num_tol_20 | 0.798 | 0.907 | 0.814 | num_tol | no | num_tol_10 |
| num_tol_0p5 | 0.797 | 0.920 | 0.841 | num_tol | no | num_tol_1 |
| num_tol_1 | 0.795 | 0.919 | 0.838 | num_tol | yes | num_tol_1 |
| meteor_lite | 0.794 | 0.888 | 0.467 | seq | yes | meteor_lite |
| bleu2 | 0.794 | 0.956 | 0.392 | seq | yes | bleu2 |
| em_list_ordered | 0.790 | 0.935 | 0.870 | em | no | em_list_set |
| em_casefold | 0.788 | 0.902 | 0.804 | em | no | em_raw |
| em_list_set | 0.787 | 0.934 | 0.868 | em | yes | em_list_set |
| bleu4 | 0.780 | 0.953 | 0.278 | seq | no | bleu2 |
| em_raw | 0.780 | 0.888 | 0.775 | em | yes | em_raw |
| list_precision | 0.780 | 0.929 | 0.834 | em | no | list_recall |
| bigram_rec | 0.772 | 0.899 | 0.792 | token | yes | bigram_rec |
| chrf2 | 0.771 | 0.922 | 0.630 | char | yes | chrf2 |
| prefix_ratio | 0.769 | 0.866 | 0.700 | char | yes | prefix_ratio |
| em_first_token | 0.750 | 0.866 | 0.732 | em | yes | em_first_token |
| rouge_l_recall | 0.745 | 0.838 | 0.664 | seq | no | tok_rec |
| content_rec | 0.743 | 0.838 | 0.665 | token | no | tok_rec |
| overlap_coef | 0.741 | 0.839 | 0.666 | set | no | tok_rec |
| tok_rec | 0.741 | 0.838 | 0.666 | token | yes | tok_rec |
| list_recall | 0.738 | 0.888 | 0.761 | em | yes | list_recall |
| gold_in_pred | 0.628 | 0.779 | 0.558 | set | yes | gold_in_pred |
| chrf4 | 0.610 | 0.654 | 0.297 | char | yes | chrf4 |
| number_set_jaccard | 0.596 | 0.734 | 0.423 | set | yes | number_set_jaccard |
| chrf3 | 0.546 | 0.759 | 0.438 | char | yes | chrf3 |
| em_numbers_only | 0.531 | 0.656 | 0.312 | em | yes | em_numbers_only |
| not_abstaining | 0.477 | 0.614 | 0.228 | length | yes | not_abstaining |
| num_abs_0p01 | 0.476 | 0.620 | 0.240 | num_tol | yes | num_abs_0p01 |
| num_round_to_gold_precision | 0.468 | 0.616 | 0.232 | num_tol | yes | num_round_to_gold_precision |
| num_linear | 0.451 | 0.569 | 0.167 | num_graded | yes | num_linear |
| num_log_ratio | 0.432 | 0.549 | 0.155 | num_graded | no | num_sign_agree |
| length_ratio_score | 0.421 | 0.726 | 0.305 | length | yes | length_ratio_score |
| num_smape | 0.419 | 0.546 | 0.134 | num_graded | no | num_sign_agree |
| num_inv_rel | 0.414 | 0.530 | 0.103 | num_graded | no | num_sign_agree |
| entity_set_jaccard | 0.403 | 0.624 | 0.234 | set | yes | entity_set_jaccard |
| final_tag_present | 0.401 | 0.619 | 0.238 | structure | yes | final_tag_present |
| type_agree | 0.368 | 0.650 | 0.300 | structure | yes | type_agree |
| number_count_agree | 0.351 | 0.624 | 0.249 | commit | yes | number_count_agree |
| not_too_long | 0.334 | 0.598 | 0.196 | length | yes | not_too_long |
| num_magnitude_agree | 0.308 | 0.541 | 0.082 | num_tol | yes | num_magnitude_agree |
| num_sign_agree | 0.234 | 0.507 | 0.014 | num_tol | yes | num_sign_agree |
| single_candidate | 0.219 | 0.667 | 0.335 | structure | no | hedge_free |
| hedge_free | 0.219 | 0.667 | 0.335 | commit | yes | hedge_free |
| brevity_penalty | 0.148 | 0.519 | 0.030 | length | yes | brevity_penalty |
| list_length_match | 0.143 | 0.574 | 0.149 | structure | yes | list_length_match |
| num_unit_equiv_hit | 0.110 | 0.507 | 0.015 | num_tol | yes | num_unit_equiv_hit |
| unit_word_match | 0.088 | 0.507 | 0.013 | structure | yes | unit_word_match |
| percent_sign_agree | 0.055 | 0.506 | 0.011 | structure | yes | percent_sign_agree |

| | | | | | | |
|---|---|---|---|---|---|---|
| first__last__number__agree | 0.028 | 0.549 | 0.098 | commit | yes | first__last__number__agree |
| emb__cos__bge | 0.000 | 0.500 | 0.000 | semantic | no | __constant |
| nli__pred__entails__gold | 0.000 | 0.500 | 0.000 | semantic | no | __constant |
| emb__cos__minilm | 0.000 | 0.500 | 0.000 | semantic | no | __constant |
| sql__result__match__placeholder | 0.000 | 0.500 | 0.000 | task | no | __constant |
| program__exec__match | 0.000 | 0.500 | 0.000 | task | no | __constant |
| bertscore__f__placeholder | 0.000 | 0.500 | 0.000 | semantic | no | __constant |
| emb__cos__minilm__thresh | 0.000 | 0.500 | 0.000 | semantic | no | __constant |
| emb__cos__bge__thresh | 0.000 | 0.500 | 0.000 | semantic | no | __constant |
| emb__question__free__placeholder | 0.000 | 0.500 | 0.000 | semantic | no | __constant |
| nli__gold__entails__pred | 0.000 | 0.500 | 0.000 | semantic | no | __constant |
| currency__symbol__agree | -0.204 | 0.463 | -0.074 | structure | yes | currency__symbol__agree |
| reasoning__final__consistent | -0.213 | 0.473 | -0.055 | commit | yes | reasoning__final__consistent |

**Table 25:** Official-split estimates under the headline rule for every dataset and model.

| dataset | model | suite_mean | official_ estimate | official_lo95 | official_hi95 | effective_n | suite_f1_pos | official_f1_ pos | official_f1_ lo95 | official_f1_ hi95 | official_ positive_ share |
|---|---|---|---|---|---|---|---|---|---|---|---|
| abt_buy | Claude Opus 5 | 1.000 | 1.000 | 1.000 | 1.000 | 63.075 | 1.000 | 1.000 | 1.000 | 1.000 | 0.108 |
| abt_buy | Claude Sonnet 5 | 0.970 | 0.993 | 0.985 | 1.000 | 63.075 | 0.968 | 0.968 | 0.925 | 1.000 | 0.108 |
| abt_buy | DeepSeek-chat | 0.960 | 0.976 | 0.933 | 0.998 | 63.075 | 0.958 | 0.893 | 0.735 | 0.990 | 0.108 |
| abt_buy | DeepSeek-R1 | 0.970 | 0.978 | 0.931 | 1.000 | 63.075 | 0.969 | 0.904 | 0.755 | 1.000 | 0.108 |
| abt_buy | Gemini 3.8 Flash | 0.980 | 0.996 | 0.988 | 1.000 | 63.075 | 0.979 | 0.979 | 0.944 | 1.000 | 0.108 |
| abt_buy | GPT-5.4-mini | 0.960 | 0.976 | 0.937 | 0.998 | 63.075 | 0.958 | 0.893 | 0.750 | 0.990 | 0.108 |
| abt_buy | GPT-5.5 | 1.000 | 1.000 | 1.000 | 1.000 | 63.075 | 1.000 | 1.000 | 1.000 | 1.000 | 0.108 |
| abt_buy | gpt-oss-120b | 0.940 | 0.987 | 0.974 | 0.996 | 63.075 | 0.935 | 0.935 | 0.879 | 0.981 | 0.108 |
| abt_buy | Grok 4 | 0.960 | 0.976 | 0.934 | 0.998 | 63.075 | 0.958 | 0.893 | 0.745 | 0.990 | 0.108 |
| abt_buy | Llama 3.3 70B | 0.960 | 0.976 | 0.937 | 0.998 | 63.075 | 0.958 | 0.893 | 0.735 | 0.992 | 0.108 |
| alaska_camera | Claude Opus 5 | 0.820 | 0.820 | 0.740 | 0.890 | 100.000 | 0.795 | 0.795 | 0.687 | 0.881 | 0.500 |
| alaska_camera | Claude Sonnet 5 | 0.980 | 0.980 | 0.950 | 1.000 | 100.000 | 0.981 | 0.981 | 0.951 | 1.000 | 0.500 |
| alaska_camera | DeepSeek-chat | 1.000 | 1.000 | 1.000 | 1.000 | 100.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.500 |
| alaska_camera | DeepSeek-R1 | 0.990 | 0.990 | 0.970 | 1.000 | 100.000 | 0.990 | 0.990 | 0.969 | 1.000 | 0.500 |
| alaska_camera | Gemini 3.8 Flash | 0.890 | 0.890 | 0.830 | 0.950 | 100.000 | 0.884 | 0.884 | 0.805 | 0.943 | 0.500 |
| alaska_camera | GPT-5.4-mini | 0.970 | 0.970 | 0.930 | 1.000 | 100.000 | 0.971 | 0.971 | 0.929 | 1.000 | 0.500 |
| alaska_camera | GPT-5.5 | 1.000 | 1.000 | 1.000 | 1.000 | 100.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.500 |
| alaska_camera | gpt-oss-120b | 1.000 | 1.000 | 1.000 | 1.000 | 100.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.500 |
| alaska_camera | Grok 4 | 0.940 | 0.940 | 0.890 | 0.980 | 100.000 | 0.940 | 0.940 | 0.886 | 0.982 | 0.500 |
| alaska_camera | Llama 3.3 70B | 0.990 | 0.990 | 0.970 | 1.000 | 100.000 | 0.990 | 0.990 | 0.967 | 1.000 | 0.500 |
| alaska_schema | Claude Opus 5 | 0.932 | 0.932 | 0.880 | 0.973 | 93.000 | – | – | – | – | – |
| alaska_schema | Claude Sonnet 5 | 0.920 | 0.920 | 0.864 | 0.970 | 93.000 | – | – | – | – | – |
| alaska_schema | DeepSeek-chat | 0.934 | 0.934 | 0.880 | 0.981 | 93.000 | – | – | – | – | – |
| alaska_schema | DeepSeek-R1 | 0.932 | 0.932 | 0.881 | 0.979 | 93.000 | – | – | – | – | – |
| alaska_schema | Gemini 3.8 Flash | 0.901 | 0.901 | 0.841 | 0.953 | 93.000 | – | – | – | – | – |
| alaska_schema | GPT-5.4-mini | 0.951 | 0.951 | 0.901 | 0.982 | 93.000 | – | – | – | – | – |
| alaska_schema | GPT-5.5 | 0.911 | 0.911 | 0.854 | 0.960 | 93.000 | – | – | – | – | – |
| alaska_schema | gpt-oss-120b | 0.914 | 0.914 | 0.858 | 0.969 | 93.000 | – | – | – | – | – |
| alaska_schema | Grok 4 | 0.911 | 0.911 | 0.849 | 0.962 | 93.000 | – | – | – | – | – |
| alaska_schema | Llama 3.3 70B | 0.780 | 0.780 | 0.706 | 0.854 | 93.000 | – | – | – | – | – |
| amazon_google | Claude Opus 5 | 0.940 | 0.916 | 0.841 | 0.981 | 67.152 | 0.936 | 0.704 | 0.526 | 0.907 | 0.102 |
| amazon_google | Claude Sonnet 5 | 0.960 | 0.935 | 0.865 | 0.985 | 67.152 | 0.957 | 0.758 | 0.587 | 0.933 | 0.102 |
| amazon_google | DeepSeek-chat | 0.960 | 0.949 | 0.886 | 0.998 | 67.152 | 0.957 | 0.796 | 0.628 | 0.992 | 0.102 |
| amazon_google | DeepSeek-R1 | 0.944 | 0.931 | 0.866 | 0.983 | 67.152 | 0.939 | 0.740 | 0.576 | 0.924 | 0.102 |
| amazon_google | Gemini 3.8 Flash | 0.930 | 0.928 | 0.863 | 0.979 | 67.152 | 0.923 | 0.725 | 0.561 | 0.926 | 0.102 |
| amazon_google | GPT-5.4-mini | 0.930 | 0.942 | 0.876 | 0.990 | 67.152 | 0.921 | 0.762 | 0.590 | 0.962 | 0.102 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| amazon_google | GPT-5.5 | 0.940 | 0.916 | 0.840 | 0.981 | 67.152 | 0.936 | 0.704 | 0.529 | 0.917 | 0.102 |
| amazon_google | gpt-oss-120b | 0.860 | 0.926 | 0.865 | 0.975 | 67.152 | 0.829 | 0.676 | 0.502 | 0.855 | 0.102 |
| amazon_google | Grok 4 | 0.940 | 0.944 | 0.886 | 0.994 | 67.152 | 0.933 | 0.773 | 0.606 | 0.968 | 0.102 |
| amazon_google | Llama 3.3 70B | 0.950 | 0.932 | 0.867 | 0.985 | 67.152 | 0.946 | 0.747 | 0.583 | 0.932 | 0.102 |
| bird | Claude Opus 5 | 0.393 | 0.410 | 0.315 | 0.508 | 74.082 | – | – | – | – | – |
| bird | Claude Sonnet 5 | 0.332 | 0.338 | 0.253 | 0.434 | 74.082 | – | – | – | – | – |
| bird | DeepSeek-chat | 0.275 | 0.276 | 0.198 | 0.360 | 74.082 | – | – | – | – | – |
| bird | DeepSeek-R1 | 0.194 | 0.244 | 0.163 | 0.329 | 74.082 | – | – | – | – | – |
| bird | Gemini 3.8 Flash | 0.407 | 0.424 | 0.333 | 0.518 | 74.082 | – | – | – | – | – |
| bird | GPT-5.4-mini | 0.233 | 0.268 | 0.177 | 0.365 | 74.082 | – | – | – | – | – |
| bird | GPT-5.5 | 0.213 | 0.220 | 0.135 | 0.311 | 74.082 | – | – | – | – | – |
| bird | gpt-oss-120b | 0.122 | 0.140 | 0.068 | 0.220 | 74.082 | – | – | – | – | – |
| bird | Grok 4 | 0.152 | 0.174 | 0.102 | 0.256 | 74.082 | – | – | – | – | – |
| bird | Llama 3.3 70B | 0.199 | 0.211 | 0.139 | 0.291 | 74.082 | – | – | – | – | – |
| dblp_acm | Claude Opus 5 | 1.000 | 1.000 | 1.000 | 1.000 | 72.166 | 1.000 | 1.000 | 1.000 | 1.000 | 0.180 |
| dblp_acm | Claude Sonnet 5 | 1.000 | 1.000 | 1.000 | 1.000 | 72.166 | 1.000 | 1.000 | 1.000 | 1.000 | 0.180 |
| dblp_acm | DeepSeek-chat | 1.000 | 1.000 | 1.000 | 1.000 | 72.166 | 1.000 | 1.000 | 1.000 | 1.000 | 0.180 |
| dblp_acm | DeepSeek-R1 | 0.990 | 0.996 | 0.988 | 1.000 | 72.166 | 0.990 | 0.990 | 0.966 | 1.000 | 0.180 |
| dblp_acm | Gemini 3.8 Flash | 1.000 | 1.000 | 1.000 | 1.000 | 72.166 | 1.000 | 1.000 | 1.000 | 1.000 | 0.180 |
| dblp_acm | GPT-5.4-mini | 0.990 | 0.996 | 0.987 | 1.000 | 72.166 | 0.990 | 0.990 | 0.964 | 1.000 | 0.180 |
| dblp_acm | GPT-5.5 | 1.000 | 1.000 | 1.000 | 1.000 | 72.166 | 1.000 | 1.000 | 1.000 | 1.000 | 0.180 |
| dblp_acm | gpt-oss-120b | 0.910 | 0.967 | 0.943 | 0.987 | 72.166 | 0.899 | 0.899 | 0.819 | 0.958 | 0.180 |
| dblp_acm | Grok 4 | 1.000 | 1.000 | 1.000 | 1.000 | 72.166 | 1.000 | 1.000 | 1.000 | 1.000 | 0.180 |
| dblp_acm | Llama 3.3 70B | 0.970 | 0.952 | 0.893 | 1.000 | 72.166 | 0.970 | 0.882 | 0.752 | 1.000 | 0.180 |
| dblp_scholar | Claude Opus 5 | 0.950 | 0.956 | 0.906 | 0.993 | 70.463 | 0.950 | 0.888 | 0.777 | 0.983 | 0.186 |
| dblp_scholar | Claude Sonnet 5 | 0.970 | 0.976 | 0.941 | 1.000 | 70.463 | 0.970 | 0.937 | 0.836 | 1.000 | 0.186 |
| dblp_scholar | DeepSeek-chat | 0.960 | 0.959 | 0.911 | 0.996 | 70.463 | 0.961 | 0.898 | 0.793 | 0.991 | 0.186 |
| dblp_scholar | DeepSeek-R1 | 0.950 | 0.969 | 0.930 | 0.993 | 70.463 | 0.949 | 0.917 | 0.811 | 0.990 | 0.186 |
| dblp_scholar | Gemini 3.8 Flash | 0.930 | 0.961 | 0.920 | 0.990 | 70.463 | 0.928 | 0.895 | 0.795 | 0.972 | 0.186 |
| dblp_scholar | GPT-5.4-mini | 0.940 | 0.965 | 0.922 | 0.992 | 70.463 | 0.939 | 0.906 | 0.788 | 0.980 | 0.186 |
| dblp_scholar | GPT-5.5 | 0.960 | 0.959 | 0.907 | 0.996 | 70.463 | 0.961 | 0.898 | 0.778 | 0.991 | 0.186 |
| dblp_scholar | gpt-oss-120b | 0.910 | 0.967 | 0.943 | 0.987 | 70.463 | 0.903 | 0.903 | 0.838 | 0.962 | 0.186 |
| dblp_scholar | Grok 4 | 0.950 | 0.969 | 0.930 | 0.996 | 70.463 | 0.949 | 0.917 | 0.822 | 0.990 | 0.186 |
| dblp_scholar | Llama 3.3 70B | 0.970 | 0.976 | 0.938 | 1.000 | 70.463 | 0.970 | 0.937 | 0.851 | 1.000 | 0.186 |
| fetaqa | Claude Opus 5 | 0.637 | 0.637 | 0.563 | 0.701 | 100.000 | – | – | – | – | – |
| fetaqa | Claude Sonnet 5 | 0.566 | 0.566 | 0.493 | 0.636 | 100.000 | – | – | – | – | – |
| fetaqa | DeepSeek-chat | 0.628 | 0.628 | 0.565 | 0.699 | 100.000 | – | – | – | – | – |
| fetaqa | DeepSeek-R1 | 0.602 | 0.602 | 0.530 | 0.678 | 100.000 | – | – | – | – | – |
| fetaqa | Gemini 3.8 Flash | 0.666 | 0.666 | 0.597 | 0.734 | 100.000 | – | – | – | – | – |
| fetaqa | GPT-5.4-mini | 0.345 | 0.345 | 0.274 | 0.416 | 100.000 | – | – | – | – | – |
| fetaqa | GPT-5.5 | 0.626 | 0.626 | 0.556 | 0.694 | 100.000 | – | – | – | – | – |
| fetaqa | gpt-oss-120b | 0.622 | 0.622 | 0.549 | 0.692 | 100.000 | – | – | – | – | – |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| fetaqa | Grok 4 | 0.654 | 0.654 | 0.577 | 0.724 | 100.000 | – | – | – | – | – |
| fetaqa | Llama 3.3 70B | 0.249 | 0.249 | 0.189 | 0.314 | 100.000 | – | – | – | – | – |
| fintagging | Claude Opus 5 | 0.863 | 0.897 | 0.819 | 0.964 | 51.769 | – | – | – | – | – |
| fintagging | Claude Sonnet 5 | 0.847 | 0.845 | 0.757 | 0.929 | 51.769 | – | – | – | – | – |
| fintagging | DeepSeek-chat | 0.786 | 0.771 | 0.670 | 0.876 | 51.769 | – | – | – | – | – |
| fintagging | DeepSeek-R1 | 0.785 | 0.812 | 0.708 | 0.900 | 51.769 | – | – | – | – | – |
| fintagging | Gemini 3.8 Flash | 0.846 | 0.862 | 0.772 | 0.937 | 51.769 | – | – | – | – | – |
| fintagging | GPT-5.4-mini | 0.838 | 0.794 | 0.688 | 0.892 | 51.769 | – | – | – | – | – |
| fintagging | GPT-5.5 | 0.845 | 0.859 | 0.762 | 0.939 | 51.769 | – | – | – | – | – |
| fintagging | gpt-oss-120b | 0.802 | 0.787 | 0.684 | 0.881 | 51.769 | – | – | – | – | – |
| fintagging | Grok 4 | 0.836 | 0.825 | 0.726 | 0.910 | 51.769 | – | – | – | – | – |
| fintagging | Llama 3.3 70B | 0.786 | 0.753 | 0.654 | 0.851 | 51.769 | – | – | – | – | – |
| hitab | Claude Opus 5 | 0.786 | 0.805 | 0.735 | 0.873 | 94.537 | – | – | – | – | – |
| hitab | Claude Sonnet 5 | 0.777 | 0.781 | 0.705 | 0.853 | 94.537 | – | – | – | – | – |
| hitab | DeepSeek-chat | 0.816 | 0.835 | 0.770 | 0.897 | 94.537 | – | – | – | – | – |
| hitab | DeepSeek-R1 | 0.771 | 0.777 | 0.702 | 0.854 | 94.537 | – | – | – | – | – |
| hitab | Gemini 3.8 Flash | 0.758 | 0.776 | 0.701 | 0.847 | 94.537 | – | – | – | – | – |
| hitab | GPT-5.4-mini | 0.793 | 0.803 | 0.731 | 0.877 | 94.537 | – | – | – | – | – |
| hitab | GPT-5.5 | 0.760 | 0.778 | 0.696 | 0.849 | 94.537 | – | – | – | – | – |
| hitab | gpt-oss-120b | 0.752 | 0.764 | 0.687 | 0.839 | 94.537 | – | – | – | – | – |
| hitab | Grok 4 | 0.745 | 0.760 | 0.676 | 0.834 | 94.537 | – | – | – | – | – |
| hitab | Llama 3.3 70B | 0.795 | 0.805 | 0.728 | 0.873 | 94.537 | – | – | – | – | – |
| machamp | Claude Opus 5 | 0.900 | 0.912 | 0.833 | 0.970 | 32.491 | 0.904 | 0.774 | 0.623 | 0.920 | 0.175 |
| machamp | Claude Sonnet 5 | 0.900 | 0.923 | 0.844 | 0.971 | 32.491 | 0.902 | 0.793 | 0.645 | 0.922 | 0.175 |
| machamp | DeepSeek-chat | 0.890 | 0.892 | 0.801 | 0.959 | 32.491 | 0.897 | 0.742 | 0.572 | 0.897 | 0.175 |
| machamp | DeepSeek-R1 | 0.900 | 0.897 | 0.807 | 0.959 | 32.491 | 0.906 | 0.751 | 0.593 | 0.897 | 0.175 |
| machamp | Gemini 3.8 Flash | 0.900 | 0.912 | 0.827 | 0.967 | 32.491 | 0.904 | 0.774 | 0.614 | 0.914 | 0.175 |
| machamp | GPT-5.4-mini | 0.890 | 0.893 | 0.805 | 0.956 | 32.491 | 0.895 | 0.739 | 0.574 | 0.897 | 0.175 |
| machamp | GPT-5.5 | 0.900 | 0.897 | 0.805 | 0.963 | 32.491 | 0.906 | 0.751 | 0.593 | 0.904 | 0.175 |
| machamp | gpt-oss-120b | 0.830 | 0.794 | 0.642 | 0.902 | 32.491 | 0.835 | 0.558 | 0.372 | 0.774 | 0.175 |
| machamp | Grok 4 | 0.890 | 0.896 | 0.812 | 0.960 | 32.491 | 0.895 | 0.747 | 0.594 | 0.887 | 0.175 |
| machamp | Llama 3.3 70B | 0.870 | 0.880 | 0.783 | 0.951 | 32.491 | 0.874 | 0.698 | 0.541 | 0.852 | 0.175 |
| magneto_gdc | Claude Opus 5 | 0.963 | 0.966 | 0.929 | 0.993 | 93.027 | – | – | – | – | – |
| magneto_gdc | Claude Sonnet 5 | 0.945 | 0.945 | 0.897 | 0.982 | 93.027 | – | – | – | – | – |
| magneto_gdc | DeepSeek-chat | 0.937 | 0.940 | 0.889 | 0.979 | 93.027 | – | – | – | – | – |
| magneto_gdc | DeepSeek-R1 | 0.962 | 0.965 | 0.926 | 0.992 | 93.027 | – | – | – | – | – |
| magneto_gdc | Gemini 3.8 Flash | 0.954 | 0.958 | 0.922 | 0.991 | 93.027 | – | – | – | – | – |
| magneto_gdc | GPT-5.4-mini | 0.896 | 0.900 | 0.842 | 0.952 | 93.027 | – | – | – | – | – |
| magneto_gdc | GPT-5.5 | 0.962 | 0.967 | 0.934 | 0.992 | 93.027 | – | – | – | – | – |
| magneto_gdc | gpt-oss-120b | 0.953 | 0.957 | 0.913 | 0.992 | 93.027 | – | – | – | – | – |
| magneto_gdc | Grok 4 | 0.926 | 0.930 | 0.879 | 0.973 | 93.027 | – | – | – | – | – |
| magneto_gdc | Llama 3.3 70B | 0.654 | 0.659 | 0.563 | 0.739 | 93.027 | – | – | – | – | – |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| mmtu | Claude Opus 5 | 0.884 | 0.833 | 0.745 | 0.913 | 86.538 | – | – | – | – | – |
| mmtu | Claude Sonnet 5 | 0.884 | 0.834 | 0.741 | 0.918 | 86.538 | – | – | – | – | – |
| mmtu | DeepSeek-chat | 0.858 | 0.803 | 0.712 | 0.884 | 86.538 | – | – | – | – | – |
| mmtu | DeepSeek-R1 | 0.814 | 0.753 | 0.647 | 0.846 | 86.538 | – | – | – | – | – |
| mmtu | Gemini 3.8 Flash | 0.892 | 0.845 | 0.748 | 0.922 | 86.538 | – | – | – | – | – |
| mmtu | GPT-5.4-mini | 0.879 | 0.835 | 0.749 | 0.914 | 86.538 | – | – | – | – | – |
| mmtu | GPT-5.5 | 0.896 | 0.843 | 0.754 | 0.924 | 86.538 | – | – | – | – | – |
| mmtu | gpt-oss-120b | 0.822 | 0.753 | 0.658 | 0.842 | 86.538 | – | – | – | – | – |
| mmtu | Grok 4 | 0.885 | 0.834 | 0.745 | 0.910 | 86.538 | – | – | – | – | – |
| mmtu | Llama 3.3 70B | 0.811 | 0.764 | 0.667 | 0.853 | 86.538 | – | – | – | – | – |
| officeqa | Claude Opus 5 | 0.387 | 0.387 | 0.326 | 0.453 | 100.000 | – | – | – | – | – |
| officeqa | Claude Sonnet 5 | 0.341 | 0.341 | 0.278 | 0.401 | 100.000 | – | – | – | – | – |
| officeqa | DeepSeek-chat | 0.125 | 0.125 | 0.082 | 0.176 | 100.000 | – | – | – | – | – |
| officeqa | DeepSeek-R1 | 0.123 | 0.123 | 0.083 | 0.165 | 100.000 | – | – | – | – | – |
| officeqa | Gemini 3.8 Flash | 0.443 | 0.443 | 0.387 | 0.503 | 100.000 | – | – | – | – | – |
| officeqa | GPT-5.4-mini | 0.259 | 0.259 | 0.209 | 0.315 | 100.000 | – | – | – | – | – |
| officeqa | GPT-5.5 | 0.364 | 0.364 | 0.304 | 0.421 | 100.000 | – | – | – | – | – |
| officeqa | gpt-oss-120b | 0.179 | 0.179 | 0.134 | 0.231 | 100.000 | – | – | – | – | – |
| officeqa | Grok 4 | 0.277 | 0.277 | 0.218 | 0.340 | 100.000 | – | – | – | – | – |
| officeqa | Llama 3.3 70B | 0.229 | 0.229 | 0.183 | 0.280 | 100.000 | – | – | – | – | – |
| officeqa_pro_v2 | Claude Opus 5 | 0.471 | 0.471 | 0.400 | 0.541 | 90.000 | – | – | – | – | – |
| officeqa_pro_v2 | Claude Sonnet 5 | 0.289 | 0.289 | 0.225 | 0.359 | 90.000 | – | – | – | – | – |
| officeqa_pro_v2 | DeepSeek-chat | 0.285 | 0.285 | 0.233 | 0.342 | 90.000 | – | – | – | – | – |
| officeqa_pro_v2 | DeepSeek-R1 | 0.057 | 0.057 | 0.030 | 0.089 | 90.000 | – | – | – | – | – |
| officeqa_pro_v2 | Gemini 3.8 Flash | 0.420 | 0.420 | 0.355 | 0.488 | 90.000 | – | – | – | – | – |
| officeqa_pro_v2 | GPT-5.4-mini | 0.328 | 0.328 | 0.271 | 0.388 | 90.000 | – | – | – | – | – |
| officeqa_pro_v2 | GPT-5.5 | 0.315 | 0.315 | 0.240 | 0.396 | 90.000 | – | – | – | – | – |
| officeqa_pro_v2 | gpt-oss-120b | 0.221 | 0.221 | 0.163 | 0.284 | 90.000 | – | – | – | – | – |
| officeqa_pro_v2 | Grok 4 | 0.267 | 0.267 | 0.209 | 0.330 | 90.000 | – | – | – | – | – |
| officeqa_pro_v2 | Llama 3.3 70B | 0.282 | 0.282 | 0.228 | 0.340 | 90.000 | – | – | – | – | – |
| opensanctions_pairs | Claude Opus 5 | 0.940 | 0.906 | 0.827 | 0.969 | 76.249 | 0.935 | 0.935 | 0.874 | 0.979 | 0.769 |
| opensanctions_pairs | Claude Sonnet 5 | 0.730 | 0.587 | 0.478 | 0.704 | 76.249 | 0.630 | 0.636 | 0.496 | 0.761 | 0.769 |
| opensanctions_pairs | DeepSeek-chat | 0.760 | 0.635 | 0.521 | 0.753 | 76.249 | 0.684 | 0.691 | 0.556 | 0.792 | 0.769 |
| opensanctions_pairs | DeepSeek-R1 | 0.850 | 0.765 | 0.655 | 0.860 | 76.249 | 0.819 | 0.819 | 0.716 | 0.905 | 0.769 |
| opensanctions_pairs | Gemini 3.8 Flash | 0.860 | 0.780 | 0.669 | 0.881 | 76.249 | 0.833 | 0.833 | 0.734 | 0.909 | 0.769 |
| opensanctions_pairs | GPT-5.4-mini | 0.900 | 0.865 | 0.782 | 0.942 | 76.249 | 0.891 | 0.905 | 0.837 | 0.963 | 0.769 |
| opensanctions_pairs | GPT-5.5 | 0.750 | 0.608 | 0.497 | 0.724 | 76.249 | 0.658 | 0.658 | 0.520 | 0.776 | 0.769 |
| opensanctions_pairs | gpt-oss-120b | 0.780 | 0.666 | 0.561 | 0.770 | 76.249 | 0.718 | 0.725 | 0.594 | 0.831 | 0.769 |
| opensanctions_pairs | Grok 4 | 0.850 | 0.776 | 0.677 | 0.868 | 76.249 | 0.824 | 0.830 | 0.733 | 0.914 | 0.769 |
| opensanctions_pairs | Llama 3.3 70B | 0.790 | 0.682 | 0.571 | 0.789 | 76.249 | 0.734 | 0.741 | 0.618 | 0.834 | 0.769 |
| papadakis_dn | Claude Opus 5 | 0.920 | 0.955 | 0.892 | 0.996 | 32.601 | 0.789 | 0.322 | 0.130 | 0.840 | 0.018 |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| papadakis_dn | Claude Sonnet 5 | 0.950 | 0.994 | 0.987 | 0.999 | 32.601 | 0.857 | 0.786 | 0.569 | 0.947 | 0.018 |
| papadakis_dn | DeepSeek-chat | 0.940 | 0.993 | 0.985 | 0.998 | 32.601 | 0.833 | 0.757 | 0.520 | 0.929 | 0.018 |
| papadakis_dn | DeepSeek-R1 | 0.940 | 0.993 | 0.985 | 0.998 | 32.601 | 0.833 | 0.757 | 0.533 | 0.927 | 0.018 |
| papadakis_dn | Gemini 3.8 Flash | 0.940 | 0.992 | 0.984 | 0.998 | 32.601 | 0.824 | 0.702 | 0.459 | 0.889 | 0.018 |
| papadakis_dn | GPT-5.4-mini | 0.950 | 0.994 | 0.987 | 0.999 | 32.601 | 0.857 | 0.786 | 0.535 | 0.936 | 0.018 |
| papadakis_dn | GPT-5.5 | 0.950 | 0.994 | 0.986 | 0.999 | 32.601 | 0.857 | 0.786 | 0.573 | 0.942 | 0.018 |
| papadakis_dn | gpt-oss-120b | 0.930 | 0.992 | 0.983 | 0.997 | 32.601 | 0.788 | 0.682 | 0.425 | 0.881 | 0.018 |
| papadakis_dn | Grok 4 | 0.950 | 0.994 | 0.987 | 0.999 | 32.601 | 0.857 | 0.786 | 0.547 | 0.939 | 0.018 |
| papadakis_dn | Llama 3.3 70B | 0.940 | 0.975 | 0.925 | 0.998 | 32.601 | 0.833 | 0.463 | 0.180 | 0.916 | 0.018 |
| realhitbench | Claude Opus 5 | 0.779 | 0.780 | 0.712 | 0.851 | 98.080 | – | – | – | – | – |
| realhitbench | Claude Sonnet 5 | 0.716 | 0.716 | 0.639 | 0.799 | 98.080 | – | – | – | – | – |
| realhitbench | DeepSeek-chat | 0.707 | 0.701 | 0.611 | 0.782 | 98.080 | – | – | – | – | – |
| realhitbench | DeepSeek-R1 | 0.704 | 0.707 | 0.629 | 0.782 | 98.080 | – | – | – | – | – |
| realhitbench | Gemini 3.8 Flash | 0.792 | 0.792 | 0.722 | 0.860 | 98.080 | – | – | – | – | – |
| realhitbench | GPT-5.4-mini | 0.717 | 0.711 | 0.627 | 0.790 | 98.080 | – | – | – | – | – |
| realhitbench | GPT-5.5 | 0.770 | 0.767 | 0.698 | 0.837 | 98.080 | – | – | – | – | – |
| realhitbench | gpt-oss-120b | 0.654 | 0.655 | 0.571 | 0.736 | 98.080 | – | – | – | – | – |
| realhitbench | Grok 4 | 0.689 | 0.686 | 0.601 | 0.774 | 98.080 | – | – | – | – | – |
| realhitbench | Llama 3.3 70B | 0.642 | 0.649 | 0.566 | 0.730 | 98.080 | – | – | – | – | – |
| smat | Claude Opus 5 | 0.911 | 0.909 | 0.847 | 0.957 | 98.100 | – | – | – | – | – |
| smat | Claude Sonnet 5 | 0.919 | 0.917 | 0.863 | 0.964 | 98.100 | – | – | – | – | – |
| smat | DeepSeek-chat | 0.893 | 0.890 | 0.835 | 0.943 | 98.100 | – | – | – | – | – |
| smat | DeepSeek-R1 | 0.891 | 0.886 | 0.825 | 0.941 | 98.100 | – | – | – | – | – |
| smat | Gemini 3.8 Flash | 0.893 | 0.890 | 0.827 | 0.946 | 98.100 | – | – | – | – | – |
| smat | GPT-5.4-mini | 0.939 | 0.942 | 0.896 | 0.982 | 98.100 | – | – | – | – | – |
| smat | GPT-5.5 | 0.920 | 0.918 | 0.856 | 0.963 | 98.100 | – | – | – | – | – |
| smat | gpt-oss-120b | 0.892 | 0.887 | 0.824 | 0.943 | 98.100 | – | – | – | – | – |
| smat | Grok 4 | 0.918 | 0.915 | 0.860 | 0.962 | 98.100 | – | – | – | – | – |
| smat | Llama 3.3 70B | 0.723 | 0.718 | 0.633 | 0.796 | 98.100 | – | – | – | – | – |
| suc | Claude Opus 5 | 0.998 | 0.998 | 0.994 | 1.000 | 98.920 | – | – | – | – | – |
| suc | Claude Sonnet 5 | 0.951 | 0.950 | 0.912 | 0.980 | 98.920 | – | – | – | – | – |
| suc | DeepSeek-chat | 0.909 | 0.910 | 0.861 | 0.952 | 98.920 | – | – | – | – | – |
| suc | DeepSeek-R1 | 0.965 | 0.965 | 0.934 | 0.988 | 98.920 | – | – | – | – | – |
| suc | Gemini 3.8 Flash | 0.988 | 0.989 | 0.963 | 1.000 | 98.920 | – | – | – | – | – |
| suc | GPT-5.4-mini | 0.959 | 0.957 | 0.927 | 0.984 | 98.920 | – | – | – | – | – |
| suc | GPT-5.5 | 0.974 | 0.973 | 0.948 | 0.993 | 98.920 | – | – | – | – | – |
| suc | gpt-oss-120b | 0.958 | 0.957 | 0.920 | 0.986 | 98.920 | – | – | – | – | – |
| suc | Grok 4 | 0.968 | 0.966 | 0.936 | 0.990 | 98.920 | – | – | – | – | – |
| suc | Llama 3.3 70B | 0.835 | 0.839 | 0.780 | 0.889 | 98.920 | – | – | – | – | – |
| tab_fact | Claude Opus 5 | 0.660 | 0.667 | 0.571 | 0.760 | 99.360 | 0.673 | 0.696 | 0.589 | 0.785 | 0.500 |
| tab_fact | Claude Sonnet 5 | 0.740 | 0.733 | 0.646 | 0.818 | 99.360 | 0.698 | 0.710 | 0.585 | 0.809 | 0.500 |

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|---|---|---|---|---|---|---|---|---|---|---|---|
| tab_fact | DeepSeek-chat | 0.840 | 0.826 | 0.749 | 0.902 | 99.360 | 0.789 | 0.789 | 0.676 | 0.874 | 0.500 |
| tab_fact | DeepSeek-R1 | 0.760 | 0.754 | 0.669 | 0.842 | 99.360 | 0.721 | 0.732 | 0.625 | 0.825 | 0.500 |
| tab_fact | Gemini 3.8 Flash | 0.860 | 0.848 | 0.781 | 0.924 | 99.360 | 0.821 | 0.821 | 0.709 | 0.909 | 0.500 |
| tab_fact | GPT-5.4-mini | 0.820 | 0.808 | 0.723 | 0.883 | 99.360 | 0.769 | 0.772 | 0.652 | 0.867 | 0.500 |
| tab_fact | GPT-5.5 | 0.840 | 0.829 | 0.745 | 0.902 | 99.360 | 0.800 | 0.803 | 0.700 | 0.892 | 0.500 |
| tab_fact | gpt-oss-120b | 0.830 | 0.815 | 0.731 | 0.891 | 99.360 | 0.773 | 0.773 | 0.656 | 0.880 | 0.500 |
| tab_fact | Grok 4 | 0.850 | 0.839 | 0.756 | 0.912 | 99.360 | 0.810 | 0.812 | 0.700 | 0.897 | 0.500 |
| tab_fact | Llama 3.3 70B | 0.780 | 0.769 | 0.690 | 0.849 | 99.360 | 0.725 | 0.732 | 0.605 | 0.835 | 0.500 |
| tabis | Claude Opus 5 | 0.928 | 0.933 | 0.886 | 0.963 | 77.069 | – | – | – | – | – |
| tabis | Claude Sonnet 5 | 0.889 | 0.887 | 0.824 | 0.939 | 77.069 | – | – | – | – | – |
| tabis | DeepSeek-chat | 0.749 | 0.694 | 0.594 | 0.797 | 77.069 | – | – | – | – | – |
| tabis | DeepSeek-R1 | 0.938 | 0.950 | 0.925 | 0.968 | 77.069 | – | – | – | – | – |
| tabis | Gemini 3.8 Flash | 0.918 | 0.938 | 0.907 | 0.963 | 77.069 | – | – | – | – | – |
| tabis | GPT-5.4-mini | 0.871 | 0.876 | 0.806 | 0.930 | 77.069 | – | – | – | – | – |
| tabis | GPT-5.5 | 0.928 | 0.944 | 0.916 | 0.965 | 77.069 | – | – | – | – | – |
| tabis | gpt-oss-120b | 0.890 | 0.921 | 0.885 | 0.951 | 77.069 | – | – | – | – | – |
| tabis | Grok 4 | 0.948 | 0.956 | 0.937 | 0.971 | 77.069 | – | – | – | – | – |
| tabis | Llama 3.3 70B | 0.793 | 0.795 | 0.712 | 0.876 | 77.069 | – | – | – | – | – |
| tablebench | Claude Opus 5 | 0.883 | 0.888 | 0.828 | 0.944 | 89.187 | – | – | – | – | – |
| tablebench | Claude Sonnet 5 | 0.902 | 0.907 | 0.858 | 0.951 | 89.187 | – | – | – | – | – |
| tablebench | DeepSeek-chat | 0.869 | 0.885 | 0.823 | 0.934 | 89.187 | – | – | – | – | – |
| tablebench | DeepSeek-R1 | 0.817 | 0.831 | 0.760 | 0.895 | 89.187 | – | – | – | – | – |
| tablebench | Gemini 3.8 Flash | 0.872 | 0.877 | 0.817 | 0.935 | 89.187 | – | – | – | – | – |
| tablebench | GPT-5.4-mini | 0.874 | 0.877 | 0.814 | 0.934 | 89.187 | – | – | – | – | – |
| tablebench | GPT-5.5 | 0.883 | 0.887 | 0.827 | 0.945 | 89.187 | – | – | – | – | – |
| tablebench | gpt-oss-120b | 0.856 | 0.861 | 0.798 | 0.919 | 89.187 | – | – | – | – | – |
| tablebench | Grok 4 | 0.854 | 0.859 | 0.794 | 0.917 | 89.187 | – | – | – | – | – |
| tablebench | Llama 3.3 70B | 0.730 | 0.730 | 0.657 | 0.801 | 89.187 | – | – | – | – | – |
| tableeval | Claude Opus 5 | 0.782 | 0.799 | 0.703 | 0.890 | 69.689 | – | – | – | – | – |
| tableeval | Claude Sonnet 5 | 0.760 | 0.790 | 0.700 | 0.877 | 69.689 | – | – | – | – | – |
| tableeval | DeepSeek-chat | 0.742 | 0.738 | 0.641 | 0.833 | 69.689 | – | – | – | – | – |
| tableeval | DeepSeek-R1 | 0.762 | 0.768 | 0.670 | 0.861 | 69.689 | – | – | – | – | – |
| tableeval | Gemini 3.8 Flash | 0.788 | 0.808 | 0.715 | 0.892 | 69.689 | – | – | – | – | – |
| tableeval | GPT-5.4-mini | 0.776 | 0.785 | 0.689 | 0.872 | 69.689 | – | – | – | – | – |
| tableeval | GPT-5.5 | 0.800 | 0.805 | 0.701 | 0.892 | 69.689 | – | – | – | – | – |
| tableeval | gpt-oss-120b | 0.796 | 0.794 | 0.699 | 0.884 | 69.689 | – | – | – | – | – |
| tableeval | Grok 4 | 0.806 | 0.794 | 0.689 | 0.891 | 69.689 | – | – | – | – | – |
| tableeval | Llama 3.3 70B | 0.695 | 0.679 | 0.577 | 0.774 | 69.689 | – | – | – | – | – |
| tat_qa | Claude Opus 5 | 0.832 | 0.800 | 0.706 | 0.886 | 60.725 | – | – | – | – | – |
| tat_qa | Claude Sonnet 5 | 0.829 | 0.808 | 0.713 | 0.892 | 60.725 | – | – | – | – | – |
| tat_qa | DeepSeek-chat | 0.792 | 0.750 | 0.651 | 0.846 | 60.725 | – | – | – | – | – |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| tat_qa | DeepSeek-R1 | 0.781 | 0.737 | 0.634 | 0.832 | 60.725 | – | – | – | – | – |
| tat_qa | Gemini 3.8 Flash | 0.806 | 0.777 | 0.682 | 0.863 | 60.725 | – | – | – | – | – |
| tat_qa | GPT-5.4-mini | 0.795 | 0.749 | 0.654 | 0.830 | 60.725 | – | – | – | – | – |
| tat_qa | GPT-5.5 | 0.804 | 0.770 | 0.680 | 0.849 | 60.725 | – | – | – | – | – |
| tat_qa | gpt-oss-120b | 0.781 | 0.724 | 0.629 | 0.816 | 60.725 | – | – | – | – | – |
| tat_qa | Grok 4 | 0.793 | 0.734 | 0.630 | 0.826 | 60.725 | – | – | – | – | – |
| tat_qa | Llama 3.3 70B | 0.758 | 0.700 | 0.593 | 0.801 | 60.725 | – | – | – | – | – |
| tpcdi_cells | Claude Opus 5 | 0.423 | 0.453 | 0.372 | 0.540 | 95.091 | – | – | – | – | – |
| tpcdi_cells | Claude Sonnet 5 | 0.423 | 0.453 | 0.365 | 0.549 | 95.091 | – | – | – | – | – |
| tpcdi_cells | DeepSeek-chat | 0.423 | 0.453 | 0.367 | 0.542 | 95.091 | – | – | – | – | – |
| tpcdi_cells | DeepSeek-R1 | 0.425 | 0.455 | 0.374 | 0.540 | 95.091 | – | – | – | – | – |
| tpcdi_cells | Gemini 3.8 Flash | 0.423 | 0.453 | 0.367 | 0.538 | 95.091 | – | – | – | – | – |
| tpcdi_cells | GPT-5.4-mini | 0.421 | 0.450 | 0.357 | 0.529 | 95.091 | – | – | – | – | – |
| tpcdi_cells | GPT-5.5 | 0.425 | 0.455 | 0.365 | 0.539 | 95.091 | – | – | – | – | – |
| tpcdi_cells | gpt-oss-120b | 0.405 | 0.441 | 0.350 | 0.535 | 95.091 | – | – | – | – | – |
| tpcdi_cells | Grok 4 | 0.405 | 0.442 | 0.360 | 0.529 | 95.091 | – | – | – | – | – |
| tpcdi_cells | Llama 3.3 70B | 0.404 | 0.441 | 0.354 | 0.524 | 95.091 | – | – | – | – | – |
| valentine | Claude Opus 5 | 1.000 | 1.000 | 1.000 | 1.000 | 81.736 | – | – | – | – | – |
| valentine | Claude Sonnet 5 | 0.991 | 0.985 | 0.943 | 1.000 | 81.736 | – | – | – | – | – |
| valentine | DeepSeek-chat | 0.974 | 0.957 | 0.909 | 1.000 | 81.736 | – | – | – | – | – |
| valentine | DeepSeek-R1 | 0.991 | 0.987 | 0.959 | 1.000 | 81.736 | – | – | – | – | – |
| valentine | Gemini 3.8 Flash | 0.970 | 0.988 | 0.970 | 1.000 | 81.736 | – | – | – | – | – |
| valentine | GPT-5.4-mini | 0.982 | 0.972 | 0.930 | 1.000 | 81.736 | – | – | – | – | – |
| valentine | GPT-5.5 | 1.000 | 1.000 | 1.000 | 1.000 | 81.736 | – | – | – | – | – |
| valentine | gpt-oss-120b | 0.973 | 0.960 | 0.907 | 1.000 | 81.736 | – | – | – | – | – |
| valentine | Grok 4 | 0.965 | 0.943 | 0.885 | 0.987 | 81.736 | – | – | – | – | – |
| valentine | Llama 3.3 70B | 0.964 | 0.948 | 0.889 | 0.990 | 81.736 | – | – | – | – | – |
| walmart_amazon | Claude Opus 5 | 0.980 | 0.982 | 0.948 | 1.000 | 70.807 | 0.976 | 0.912 | 0.767 | 1.000 | 0.094 |
| walmart_amazon | Claude Sonnet 5 | 0.950 | 0.962 | 0.914 | 0.997 | 70.807 | 0.938 | 0.823 | 0.645 | 0.980 | 0.094 |
| walmart_amazon | DeepSeek-chat | 0.950 | 0.962 | 0.912 | 0.998 | 70.807 | 0.938 | 0.823 | 0.657 | 0.977 | 0.094 |
| walmart_amazon | DeepSeek-R1 | 0.960 | 0.978 | 0.939 | 0.998 | 70.807 | 0.950 | 0.887 | 0.737 | 0.989 | 0.094 |
| walmart_amazon | Gemini 3.8 Flash | 0.990 | 0.998 | 0.993 | 1.000 | 70.807 | 0.988 | 0.988 | 0.959 | 1.000 | 0.094 |
| walmart_amazon | GPT-5.4-mini | 0.950 | 0.975 | 0.938 | 0.998 | 70.807 | 0.937 | 0.874 | 0.722 | 0.987 | 0.094 |
| walmart_amazon | GPT-5.5 | 0.970 | 0.967 | 0.917 | 1.000 | 70.807 | 0.964 | 0.848 | 0.671 | 1.000 | 0.094 |
| walmart_amazon | gpt-oss-120b | 0.940 | 0.973 | 0.933 | 0.995 | 70.807 | 0.923 | 0.860 | 0.698 | 0.974 | 0.094 |
| walmart_amazon | Grok 4 | 0.960 | 0.978 | 0.942 | 0.998 | 70.807 | 0.950 | 0.887 | 0.720 | 0.989 | 0.094 |
| walmart_amazon | Llama 3.3 70B | 0.930 | 0.971 | 0.932 | 0.994 | 70.807 | 0.909 | 0.847 | 0.681 | 0.965 | 0.094 |
| wdc_lspc | Claude Opus 5 | 0.990 | 0.996 | 0.987 | 1.000 | 75.082 | 0.991 | 0.992 | 0.973 | 1.000 | 0.264 |
| wdc_lspc | Claude Sonnet 5 | 0.980 | 0.992 | 0.978 | 1.000 | 75.082 | 0.981 | 0.984 | 0.958 | 1.000 | 0.264 |
| wdc_lspc | DeepSeek-chat | 0.960 | 0.940 | 0.868 | 0.994 | 75.082 | 0.964 | 0.895 | 0.777 | 0.990 | 0.264 |
| wdc_lspc | DeepSeek-R1 | 1.000 | 1.000 | 1.000 | 1.000 | 75.082 | 1.000 | 1.000 | 1.000 | 1.000 | 0.264 |
| wdc_lspc | Gemini 3.8 Flash | 0.980 | 0.992 | 0.979 | 1.000 | 75.082 | 0.981 | 0.984 | 0.956 | 1.000 | 0.264 |

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| wdc_lspc | GPT-5.4-mini | 0.970 | 0.986 | 0.967 | 1.000 | 75.082 | 0.971 | 0.973 | 0.937 | 1.000 | 0.264 |
| wdc_lspc | GPT-5.5 | 1.000 | 1.000 | 1.000 | 1.000 | 75.082 | 1.000 | 1.000 | 1.000 | 1.000 | 0.264 |
| wdc_lspc | gpt-oss-120b | 0.980 | 0.990 | 0.974 | 1.000 | 75.082 | 0.981 | 0.981 | 0.951 | 1.000 | 0.264 |
| wdc_lspc | Grok 4 | 1.000 | 1.000 | 1.000 | 1.000 | 75.082 | 1.000 | 1.000 | 1.000 | 1.000 | 0.264 |
| wdc_lspc | Llama 3.3 70B | 0.960 | 0.940 | 0.872 | 0.994 | 75.082 | 0.964 | 0.895 | 0.781 | 0.990 | 0.264 |
| wdc_products | Claude Opus 5 | 0.980 | 0.990 | 0.974 | 1.000 | 77.046 | 0.981 | 0.982 | 0.953 | 1.000 | 0.280 |
| wdc_products | Claude Sonnet 5 | 0.970 | 0.986 | 0.967 | 1.000 | 77.046 | 0.971 | 0.974 | 0.936 | 1.000 | 0.280 |
| wdc_products | DeepSeek-chat | 0.970 | 0.977 | 0.944 | 1.000 | 77.046 | 0.972 | 0.958 | 0.895 | 1.000 | 0.280 |
| wdc_products | DeepSeek-R1 | 0.980 | 0.990 | 0.974 | 1.000 | 77.046 | 0.981 | 0.982 | 0.953 | 1.000 | 0.280 |
| wdc_products | Gemini 3.8 Flash | 0.990 | 0.995 | 0.983 | 1.000 | 77.046 | 0.991 | 0.991 | 0.968 | 1.000 | 0.280 |
| wdc_products | GPT-5.4-mini | 0.970 | 0.986 | 0.966 | 1.000 | 77.046 | 0.971 | 0.974 | 0.940 | 1.000 | 0.280 |
| wdc_products | GPT-5.5 | 0.980 | 0.979 | 0.940 | 1.000 | 77.046 | 0.981 | 0.964 | 0.902 | 1.000 | 0.280 |
| wdc_products | gpt-oss-120b | 0.980 | 0.979 | 0.945 | 1.000 | 77.046 | 0.981 | 0.964 | 0.898 | 1.000 | 0.280 |
| wdc_products | Grok 4 | 0.984 | 0.992 | 0.977 | 1.000 | 77.046 | 0.985 | 0.985 | 0.959 | 1.000 | 0.280 |
| wdc_products | Llama 3.3 70B | 0.970 | 0.977 | 0.940 | 1.000 | 77.046 | 0.972 | 0.958 | 0.901 | 1.000 | 0.280 |
| wikitablequestions | Claude Opus 5 | 0.884 | 0.883 | 0.823 | 0.937 | 97.686 | – | – | – | – | – |
| wikitablequestions | Claude Sonnet 5 | 0.867 | 0.867 | 0.800 | 0.930 | 97.686 | – | – | – | – | – |
| wikitablequestions | DeepSeek-chat | 0.848 | 0.848 | 0.778 | 0.908 | 97.686 | – | – | – | – | – |
| wikitablequestions | DeepSeek-R1 | 0.850 | 0.849 | 0.787 | 0.910 | 97.686 | – | – | – | – | – |
| wikitablequestions | Gemini 3.8 Flash | 0.883 | 0.883 | 0.822 | 0.938 | 97.686 | – | – | – | – | – |
| wikitablequestions | GPT-5.4-mini | 0.853 | 0.852 | 0.784 | 0.912 | 97.686 | – | – | – | – | – |
| wikitablequestions | GPT-5.5 | 0.864 | 0.863 | 0.798 | 0.922 | 97.686 | – | – | – | – | – |
| wikitablequestions | gpt-oss-120b | 0.826 | 0.824 | 0.754 | 0.893 | 97.686 | – | – | – | – | – |
| wikitablequestions | Grok 4 | 0.868 | 0.866 | 0.803 | 0.924 | 97.686 | – | – | – | – | – |
| wikitablequestions | Llama 3.3 70B | 0.759 | 0.759 | 0.690 | 0.834 | 97.686 | – | – | – | – | – |

**Table 26:** Official-split estimates under the headline rule averaged over the ten models.

| dataset | suite_mean | official_ estimate | suite_f1_ pos | official_precision_ pos | official_recall_ pos | official_f1_ pos | official_positive_ share |
|---|---|---|---|---|---|---|---|
| abt_buy | 0.970 | 0.986 | 0.968 | 0.926 | 0.949 | 0.936 | 0.108 |
| alaska_camera | 0.958 | 0.958 | 0.955 | 1.000 | 0.921 | 0.955 | 0.500 |
| alaska_schema | 0.909 | 0.909 | – | – | – | – | – |
| amazon_google | 0.935 | 0.932 | 0.928 | 0.611 | 0.941 | 0.739 | 0.102 |
| bird | 0.252 | 0.271 | – | – | – | – | – |
| dblp_acm | 0.986 | 0.991 | 0.985 | 0.979 | 0.978 | 0.976 | 0.180 |
| dblp_scholar | 0.949 | 0.966 | 0.948 | 0.900 | 0.924 | 0.910 | 0.186 |
| fetaqa | 0.559 | 0.559 | – | – | – | – | – |
| fintagging | 0.823 | 0.821 | – | – | – | – | – |
| hitab | 0.775 | 0.788 | – | – | – | – | – |
| machamp | 0.887 | 0.890 | 0.892 | 0.644 | 0.853 | 0.733 | 0.175 |
| magneto_gdc | 0.915 | 0.919 | – | – | – | – | – |
| mmtu | 0.862 | 0.810 | – | – | – | – | – |
| officeqa | 0.273 | 0.273 | – | – | – | – | – |
| officeqa_pro_v2 | 0.294 | 0.294 | – | – | – | – | – |
| opensanctions_ pairs | 0.821 | 0.727 | 0.773 | 0.993 | 0.649 | 0.777 | 0.769 |
| papadakis_dn | 0.941 | 0.988 | 0.833 | 0.550 | 0.963 | 0.683 | 0.018 |
| realhitbench | 0.717 | 0.716 | – | – | – | – | – |
| smat | 0.890 | 0.887 | – | – | – | – | – |
| suc | 0.950 | 0.950 | – | – | – | – | – |
| tab_fact | 0.798 | 0.789 | 0.758 | 0.896 | 0.674 | 0.764 | 0.500 |
| tabis | 0.885 | 0.889 | – | – | – | – | – |
| tablebench | 0.854 | 0.860 | – | – | – | – | – |
| tableeval | 0.771 | 0.776 | – | – | – | – | – |
| tat_qa | 0.797 | 0.755 | – | – | – | – | – |
| tpcdi_cells | 0.418 | 0.450 | – | – | – | – | – |
| valentine | 0.981 | 0.974 | – | – | – | – | – |
| walmart_amazon | 0.958 | 0.975 | 0.947 | 0.832 | 0.927 | 0.875 | 0.094 |
| wdc_lspc | 0.982 | 0.983 | 0.983 | 0.965 | 0.979 | 0.971 | 0.264 |
| wdc_products | 0.977 | 0.985 | 0.979 | 0.980 | 0.967 | 0.973 | 0.280 |
| wikitablequestions | 0.850 | 0.849 | – | – | – | – | – |

**Table 27:** Cross-dataset pairwise tests under rule exact match: Friedman $\chi^2 = 83.9$ ($p = 2.7\text{e-}14$), Iman–Davenport $F = 12.9$, Nemenyi CD = 2.43, Kendall's $W = 0.284$.

| a | b | mean rank diff | wins a | wins b | ties | Wilcoxon p | Holm p | Nemenyi | sig .05 |
|---|---|---|---|---|---|---|---|---|---|
| Claude Opus 5 | Claude Sonnet 5 | -1.3387 | 21 | 7 | 3 | 0.0223 | 0.4677 | no | no |
| Claude Opus 5 | DeepSeek-chat | -2.6935 | 22 | 6 | 3 | 0.0047 | 0.1472 | yes | no |
| Claude Opus 5 | DeepSeek-R1 | -1.8871 | 18 | 6 | 7 | 0.0110 | 0.2963 | no | no |
| Claude Opus 5 | Gemini 3.8 Flash | -1.0000 | 19 | 8 | 4 | 0.0874 | 1.0000 | no | no |
| Claude Opus 5 | GPT-5.4-mini | -2.8065 | 25 | 5 | 1 | 0.0052 | 0.1555 | yes | no |
| Claude Opus 5 | GPT-5.5 | -0.3387 | 13 | 8 | 10 | 0.2297 | 1.0000 | no | no |
| Claude Opus 5 | gpt-oss-120b | -4.2258 | 26 | 3 | 2 | 0.0004 | 0.0127 | yes | yes |
| Claude Opus 5 | Grok 4 | -1.8065 | 18 | 9 | 4 | 0.0444 | 0.8880 | no | no |
| Claude Opus 5 | Llama 3.3 70B | -5.0323 | 26 | 5 | 0 | 0.0001 | 0.0041 | yes | yes |
| Claude Sonnet 5 | DeepSeek-chat | -1.3548 | 19 | 5 | 7 | 0.0113 | 0.2963 | no | no |
| Claude Sonnet 5 | DeepSeek-R1 | -0.5484 | 14 | 13 | 4 | 0.5625 | 1.0000 | no | no |
| Claude Sonnet 5 | Gemini 3.8 Flash | 0.3387 | 11 | 14 | 6 | 0.3363 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.4-mini | -1.4677 | 19 | 7 | 5 | 0.1117 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.5 | 1.0000 | 7 | 15 | 9 | 0.0478 | 0.9077 | no | no |
| Claude Sonnet 5 | gpt-oss-120b | -2.8871 | 22 | 7 | 2 | 0.0084 | 0.2339 | yes | no |
| Claude Sonnet 5 | Grok 4 | -0.4677 | 18 | 9 | 4 | 0.4327 | 1.0000 | no | no |
| Claude Sonnet 5 | Llama 3.3 70B | -3.6935 | 25 | 3 | 3 | 0.0001 | 0.0027 | yes | yes |
| DeepSeek-chat | DeepSeek-R1 | 0.8065 | 11 | 15 | 5 | 0.4297 | 1.0000 | no | no |
| DeepSeek-chat | Gemini 3.8 Flash | 1.6935 | 6 | 18 | 7 | 0.0158 | 0.3783 | no | no |
| DeepSeek-chat | GPT-5.4-mini | -0.1129 | 12 | 15 | 4 | 0.2981 | 1.0000 | no | no |
| DeepSeek-chat | GPT-5.5 | 2.3548 | 6 | 19 | 6 | 0.0040 | 0.1295 | no | no |
| DeepSeek-chat | gpt-oss-120b | -1.5323 | 19 | 9 | 3 | 0.0995 | 1.0000 | no | no |
| DeepSeek-chat | Grok 4 | 0.8871 | 12 | 15 | 4 | 0.2352 | 1.0000 | no | no |
| DeepSeek-chat | Llama 3.3 70B | -2.3387 | 22 | 3 | 6 | 0.0001 | 0.0053 | no | yes |
| DeepSeek-R1 | Gemini 3.8 Flash | 0.8871 | 10 | 17 | 4 | 0.0879 | 1.0000 | no | no |
| DeepSeek-R1 | GPT-5.4-mini | -0.9194 | 17 | 12 | 2 | 0.9686 | 1.0000 | no | no |
| DeepSeek-R1 | GPT-5.5 | 1.5484 | 7 | 19 | 5 | 0.0160 | 0.3783 | no | no |
| DeepSeek-R1 | gpt-oss-120b | -2.3387 | 21 | 5 | 5 | 0.0038 | 0.1256 | no | no |
| DeepSeek-R1 | Grok 4 | 0.0806 | 13 | 14 | 4 | 0.7163 | 1.0000 | no | no |
| DeepSeek-R1 | Llama 3.3 70B | -3.1452 | 25 | 3 | 3 | 0.0000 | 0.0020 | yes | yes |
| Gemini 3.8 Flash | GPT-5.4-mini | -1.8065 | 21 | 9 | 1 | 0.0509 | 0.9171 | no | no |
| Gemini 3.8 Flash | GPT-5.5 | 0.6613 | 10 | 15 | 6 | 0.7383 | 1.0000 | no | no |
| Gemini 3.8 Flash | gpt-oss-120b | -3.2258 | 24 | 2 | 5 | 0.0001 | 0.0033 | yes | yes |
| Gemini 3.8 Flash | Grok 4 | -0.8065 | 16 | 9 | 6 | 0.1574 | 1.0000 | no | no |
| Gemini 3.8 Flash | Llama 3.3 70B | -4.0323 | 26 | 4 | 1 | 0.0001 | 0.0025 | yes | yes |
| GPT-5.4-mini | GPT-5.5 | 2.4677 | 5 | 24 | 2 | 0.0052 | 0.1555 | yes | no |
| GPT-5.4-mini | gpt-oss-120b | -1.4194 | 20 | 10 | 1 | 0.0134 | 0.3360 | no | no |
| GPT-5.4-mini | Grok 4 | 1.0000 | 10 | 17 | 4 | 0.4266 | 1.0000 | no | no |
| GPT-5.4-mini | Llama 3.3 70B | -2.2258 | 23 | 4 | 4 | 0.0001 | 0.0041 | no | yes |
| GPT-5.5 | gpt-oss-120b | -3.8871 | 27 | 1 | 3 | 0.0000 | 0.0003 | yes | yes |
| GPT-5.5 | Grok 4 | -1.4677 | 14 | 6 | 11 | 0.0598 | 1.0000 | no | no |
| GPT-5.5 | Llama 3.3 70B | -4.6935 | 27 | 3 | 1 | 0.0000 | 0.0005 | yes | yes |
| gpt-oss-120b | Grok 4 | 2.4194 | 3 | 24 | 4 | 0.0002 | 0.0056 | no | yes |
| gpt-oss-120b | Llama 3.3 70B | -0.8065 | 20 | 8 | 3 | 0.0167 | 0.3783 | no | no |
| Grok 4 | Llama 3.3 70B | -3.2258 | 24 | 4 | 3 | 0.0001 | 0.0027 | yes | yes |

**Table 28:** Cross-dataset pairwise tests under rule ensemble: Friedman $\chi^2 = 74.4$ ($p = 2.1\text{e-}12$), Iman–Davenport $F = 10.9$, Nemenyi CD = 2.43, Kendall's $W = 0.259$.

| a | b | mean rank diff | wins a | wins b | ties | Wilcoxon p | Holm p | Nemenyi | sig .05 |
|---|---|---|---|---|---|---|---|---|---|
| Claude Opus 5 | Claude Sonnet 5 | -1.3226 | 21 | 8 | 2 | 0.0282 | 0.5915 | no | no |
| Claude Opus 5 | DeepSeek-chat | -2.5000 | 23 | 7 | 1 | 0.0048 | 0.1434 | yes | no |
| Claude Opus 5 | DeepSeek-R1 | -2.3548 | 20 | 8 | 3 | 0.0063 | 0.1689 | no | no |
| Claude Opus 5 | Gemini 3.8 Flash | -0.5161 | 18 | 11 | 2 | 0.2987 | 1.0000 | no | no |
| Claude Opus 5 | GPT-5.4-mini | -2.1774 | 25 | 6 | 0 | 0.0054 | 0.1562 | no | no |
| Claude Opus 5 | GPT-5.5 | -0.0323 | 14 | 11 | 6 | 0.3031 | 1.0000 | no | no |
| Claude Opus 5 | gpt-oss-120b | -3.9839 | 26 | 4 | 1 | 0.0003 | 0.0107 | yes | yes |
| Claude Opus 5 | Grok 4 | -1.7581 | 18 | 10 | 3 | 0.0343 | 0.6858 | no | no |
| Claude Opus 5 | Llama 3.3 70B | -4.5484 | 25 | 6 | 0 | 0.0002 | 0.0077 | yes | yes |
| Claude Sonnet 5 | DeepSeek-chat | -1.1774 | 20 | 7 | 4 | 0.0395 | 0.7115 | no | no |
| Claude Sonnet 5 | DeepSeek-R1 | -1.0323 | 16 | 13 | 2 | 0.2724 | 1.0000 | no | no |
| Claude Sonnet 5 | Gemini 3.8 Flash | 0.8065 | 11 | 17 | 3 | 0.1416 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.4-mini | -0.8548 | 19 | 9 | 3 | 0.1313 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.5 | 1.2903 | 9 | 19 | 3 | 0.0268 | 0.5889 | no | no |
| Claude Sonnet 5 | gpt-oss-120b | -2.6613 | 21 | 9 | 1 | 0.0077 | 0.2001 | yes | no |
| Claude Sonnet 5 | Grok 4 | -0.4355 | 18 | 11 | 2 | 0.2854 | 1.0000 | no | no |
| Claude Sonnet 5 | Llama 3.3 70B | -3.2258 | 25 | 4 | 2 | 0.0001 | 0.0032 | yes | yes |
| DeepSeek-chat | DeepSeek-R1 | 0.1452 | 17 | 13 | 1 | 0.5565 | 1.0000 | no | no |
| DeepSeek-chat | Gemini 3.8 Flash | 1.9839 | 7 | 21 | 3 | 0.0057 | 0.1598 | no | no |
| DeepSeek-chat | GPT-5.4-mini | 0.3226 | 10 | 17 | 4 | 0.2242 | 1.0000 | no | no |
| DeepSeek-chat | GPT-5.5 | 2.4677 | 6 | 21 | 4 | 0.0041 | 0.1308 | yes | no |
| DeepSeek-chat | gpt-oss-120b | -1.4839 | 21 | 9 | 1 | 0.0745 | 1.0000 | no | no |
| DeepSeek-chat | Grok 4 | 0.7419 | 12 | 16 | 3 | 0.3081 | 1.0000 | no | no |
| DeepSeek-chat | Llama 3.3 70B | -2.0484 | 22 | 5 | 4 | 0.0009 | 0.0301 | no | yes |
| DeepSeek-R1 | Gemini 3.8 Flash | 1.8387 | 10 | 19 | 2 | 0.0230 | 0.5291 | no | no |
| DeepSeek-R1 | GPT-5.4-mini | 0.1774 | 14 | 16 | 1 | 0.2514 | 1.0000 | no | no |
| DeepSeek-R1 | GPT-5.5 | 2.3226 | 6 | 21 | 4 | 0.0015 | 0.0508 | no | no |
| DeepSeek-R1 | gpt-oss-120b | -1.6290 | 20 | 10 | 1 | 0.0938 | 1.0000 | no | no |
| DeepSeek-R1 | Grok 4 | 0.5968 | 11 | 16 | 4 | 0.1991 | 1.0000 | no | no |
| DeepSeek-R1 | Llama 3.3 70B | -2.1935 | 21 | 8 | 2 | 0.0046 | 0.1434 | no | no |
| Gemini 3.8 Flash | GPT-5.4-mini | -1.6613 | 21 | 9 | 1 | 0.0351 | 0.6858 | no | no |
| Gemini 3.8 Flash | GPT-5.5 | 0.4839 | 13 | 16 | 2 | 0.8986 | 1.0000 | no | no |
| Gemini 3.8 Flash | gpt-oss-120b | -3.4677 | 26 | 4 | 1 | 0.0001 | 0.0032 | yes | yes |
| Gemini 3.8 Flash | Grok 4 | -1.2419 | 21 | 9 | 1 | 0.0862 | 1.0000 | no | no |
| Gemini 3.8 Flash | Llama 3.3 70B | -4.0323 | 26 | 4 | 1 | 0.0001 | 0.0034 | yes | yes |
| GPT-5.4-mini | GPT-5.5 | 2.1452 | 6 | 24 | 1 | 0.0118 | 0.2828 | no | no |
| GPT-5.4-mini | gpt-oss-120b | -1.8065 | 23 | 8 | 0 | 0.0040 | 0.1308 | no | no |
| GPT-5.4-mini | Grok 4 | 0.4194 | 13 | 15 | 3 | 0.8986 | 1.0000 | no | no |
| GPT-5.4-mini | Llama 3.3 70B | -2.3710 | 25 | 4 | 2 | 0.0000 | 0.0017 | no | yes |
| GPT-5.5 | gpt-oss-120b | -3.9516 | 27 | 2 | 2 | 0.0000 | 0.0004 | yes | yes |
| GPT-5.5 | Grok 4 | -1.7258 | 19 | 8 | 4 | 0.0108 | 0.2705 | no | no |
| GPT-5.5 | Llama 3.3 70B | -4.5161 | 27 | 4 | 0 | 0.0000 | 0.0010 | yes | yes |
| gpt-oss-120b | Grok 4 | 2.2258 | 6 | 25 | 0 | 0.0002 | 0.0077 | no | yes |
| gpt-oss-120b | Llama 3.3 70B | -0.5645 | 20 | 11 | 0 | 0.1263 | 1.0000 | no | no |
| Grok 4 | Llama 3.3 70B | -2.7903 | 24 | 6 | 1 | 0.0005 | 0.0195 | yes | yes |

**Table 29:** Cross-dataset pairwise tests under rule native metric: Friedman $\chi^2 = 83.2$ ($p = 3.8\text{e-}14$), Iman–Davenport $F = 12.7$, Nemenyi CD = 2.43, Kendall's $W = 0.283$.

| a | b | mean rank diff | wins a | wins b | ties | Wilcoxon p | Holm p | Nemenyi | sig .05 |
|---|---|---|---|---|---|---|---|---|---|
| Claude Opus 5 | Claude Sonnet 5 | -1.1452 | 20 | 7 | 4 | 0.0253 | 0.5311 | no | no |
| Claude Opus 5 | DeepSeek-chat | -2.5645 | 21 | 7 | 3 | 0.0094 | 0.2433 | yes | no |
| Claude Opus 5 | DeepSeek-R1 | -1.7742 | 18 | 7 | 6 | 0.0210 | 0.4617 | no | no |
| Claude Opus 5 | Gemini 3.8 Flash | -0.5161 | 15 | 11 | 5 | 0.4315 | 1.0000 | no | no |
| Claude Opus 5 | GPT-5.4-mini | -2.3710 | 24 | 5 | 2 | 0.0055 | 0.1543 | no | no |
| Claude Opus 5 | GPT-5.5 | 0.0968 | 12 | 10 | 9 | 0.4973 | 1.0000 | no | no |
| Claude Opus 5 | gpt-oss-120b | -3.9677 | 26 | 4 | 1 | 0.0005 | 0.0183 | yes | yes |
| Claude Opus 5 | Grok 4 | -1.5484 | 18 | 9 | 4 | 0.0652 | 0.9784 | no | no |
| Claude Opus 5 | Llama 3.3 70B | -4.7581 | 26 | 5 | 0 | 0.0001 | 0.0041 | yes | yes |
| Claude Sonnet 5 | DeepSeek-chat | -1.4194 | 20 | 6 | 5 | 0.0253 | 0.5311 | no | no |
| Claude Sonnet 5 | DeepSeek-R1 | -0.6290 | 14 | 13 | 4 | 0.6653 | 1.0000 | no | no |
| Claude Sonnet 5 | Gemini 3.8 Flash | 0.6290 | 11 | 15 | 5 | 0.2391 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.4-mini | -1.2258 | 20 | 7 | 4 | 0.0554 | 0.8871 | no | no |
| Claude Sonnet 5 | GPT-5.5 | 1.2419 | 8 | 17 | 6 | 0.0141 | 0.3377 | no | no |
| Claude Sonnet 5 | gpt-oss-120b | -2.8226 | 22 | 7 | 2 | 0.0042 | 0.1342 | yes | no |
| Claude Sonnet 5 | Grok 4 | -0.4032 | 16 | 10 | 5 | 0.5895 | 1.0000 | no | no |
| Claude Sonnet 5 | Llama 3.3 70B | -3.6129 | 25 | 4 | 2 | 0.0001 | 0.0023 | yes | yes |
| DeepSeek-chat | DeepSeek-R1 | 0.7903 | 11 | 15 | 5 | 0.3973 | 1.0000 | no | no |
| DeepSeek-chat | Gemini 3.8 Flash | 2.0484 | 5 | 20 | 6 | 0.0070 | 0.1881 | no | no |
| DeepSeek-chat | GPT-5.4-mini | 0.1935 | 10 | 15 | 6 | 0.2760 | 1.0000 | no | no |
| DeepSeek-chat | GPT-5.5 | 2.6613 | 4 | 21 | 6 | 0.0006 | 0.0190 | yes | yes |
| DeepSeek-chat | gpt-oss-120b | -1.4032 | 18 | 8 | 5 | 0.0758 | 1.0000 | no | no |
| DeepSeek-chat | Grok 4 | 1.0161 | 11 | 16 | 4 | 0.1436 | 1.0000 | no | no |
| DeepSeek-chat | Llama 3.3 70B | -2.1935 | 23 | 3 | 5 | 0.0001 | 0.0037 | no | yes |
| DeepSeek-R1 | Gemini 3.8 Flash | 1.2581 | 10 | 17 | 4 | 0.0521 | 0.8851 | no | no |
| DeepSeek-R1 | GPT-5.4-mini | -0.5968 | 16 | 13 | 2 | 0.9921 | 1.0000 | no | no |
| DeepSeek-R1 | GPT-5.5 | 1.8710 | 6 | 20 | 5 | 0.0052 | 0.1503 | no | no |
| DeepSeek-R1 | gpt-oss-120b | -2.1935 | 21 | 6 | 4 | 0.0043 | 0.1342 | no | no |
| DeepSeek-R1 | Grok 4 | 0.2258 | 11 | 16 | 4 | 0.3020 | 1.0000 | no | no |
| DeepSeek-R1 | Llama 3.3 70B | -2.9839 | 24 | 3 | 4 | 0.0001 | 0.0029 | yes | yes |
| Gemini 3.8 Flash | GPT-5.4-mini | -1.8548 | 20 | 9 | 2 | 0.0454 | 0.8179 | no | no |
| Gemini 3.8 Flash | GPT-5.5 | 0.6129 | 11 | 15 | 5 | 0.7833 | 1.0000 | no | no |
| Gemini 3.8 Flash | gpt-oss-120b | -3.4516 | 25 | 2 | 4 | 0.0000 | 0.0021 | yes | yes |
| Gemini 3.8 Flash | Grok 4 | -1.0323 | 19 | 10 | 2 | 0.0771 | 1.0000 | no | no |
| Gemini 3.8 Flash | Llama 3.3 70B | -4.2419 | 26 | 4 | 1 | 0.0001 | 0.0024 | yes | yes |
| GPT-5.4-mini | GPT-5.5 | 2.4677 | 5 | 25 | 1 | 0.0048 | 0.1455 | yes | no |
| GPT-5.4-mini | gpt-oss-120b | -1.5968 | 21 | 10 | 0 | 0.0124 | 0.3092 | no | no |
| GPT-5.4-mini | Grok 4 | 0.8226 | 11 | 15 | 5 | 0.5043 | 1.0000 | no | no |
| GPT-5.4-mini | Llama 3.3 70B | -2.3871 | 25 | 3 | 3 | 0.0000 | 0.0019 | no | yes |
| GPT-5.5 | gpt-oss-120b | -4.0645 | 26 | 1 | 4 | 0.0000 | 0.0003 | yes | yes |
| GPT-5.5 | Grok 4 | -1.6452 | 17 | 6 | 8 | 0.0256 | 0.5311 | no | no |
| GPT-5.5 | Llama 3.3 70B | -4.8548 | 27 | 4 | 0 | 0.0000 | 0.0006 | yes | yes |
| gpt-oss-120b | Grok 4 | 2.4194 | 3 | 24 | 4 | 0.0001 | 0.0042 | no | yes |
| gpt-oss-120b | Llama 3.3 70B | -0.7903 | 19 | 8 | 4 | 0.0196 | 0.4506 | no | no |
| Grok 4 | Llama 3.3 70B | -3.2097 | 23 | 4 | 4 | 0.0001 | 0.0031 | yes | yes |

**Table 30:** Cross-dataset pairwise tests under rule top1: Friedman $\chi^2 = 73.7$ ($p = 2.9\text{e-}12$), Iman–Davenport $F = 10.8$, Nemenyi CD = 2.43, Kendall's $W = 0.256$.

| a | b | mean rank diff | wins a | wins b | ties | Wilcoxon p | Holm p | Nemenyi | sig .05 |
|---|---|---|---|---|---|---|---|---|---|
| Claude Opus 5 | Claude Sonnet 5 | -1.3387 | 21 | 8 | 2 | 0.0303 | 0.6371 | no | no |
| Claude Opus 5 | DeepSeek-chat | -2.7097 | 23 | 7 | 1 | 0.0035 | 0.1119 | yes | no |
| Claude Opus 5 | DeepSeek-R1 | -2.2419 | 20 | 8 | 3 | 0.0066 | 0.1792 | no | no |
| Claude Opus 5 | Gemini 3.8 Flash | -0.5645 | 18 | 11 | 2 | 0.3468 | 1.0000 | no | no |
| Claude Opus 5 | GPT-5.4-mini | -2.2258 | 25 | 6 | 0 | 0.0061 | 0.1699 | no | no |
| Claude Opus 5 | GPT-5.5 | -0.1613 | 14 | 10 | 7 | 0.2390 | 1.0000 | no | no |
| Claude Opus 5 | gpt-oss-120b | -3.9355 | 26 | 4 | 1 | 0.0003 | 0.0124 | yes | yes |
| Claude Opus 5 | Grok 4 | -1.8710 | 19 | 9 | 3 | 0.0242 | 0.5566 | no | no |
| Claude Opus 5 | Llama 3.3 70B | -4.6290 | 25 | 6 | 0 | 0.0002 | 0.0068 | yes | yes |
| Claude Sonnet 5 | DeepSeek-chat | -1.3710 | 20 | 7 | 4 | 0.0334 | 0.6679 | no | no |
| Claude Sonnet 5 | DeepSeek-R1 | -0.9032 | 16 | 13 | 2 | 0.3317 | 1.0000 | no | no |
| Claude Sonnet 5 | Gemini 3.8 Flash | 0.7742 | 11 | 17 | 3 | 0.1497 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.4-mini | -0.8871 | 19 | 9 | 3 | 0.1362 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.5 | 1.1774 | 9 | 19 | 3 | 0.0274 | 0.6025 | no | no |
| Claude Sonnet 5 | gpt-oss-120b | -2.5968 | 21 | 9 | 1 | 0.0094 | 0.2447 | yes | no |
| Claude Sonnet 5 | Grok 4 | -0.5323 | 19 | 10 | 2 | 0.2515 | 1.0000 | no | no |
| Claude Sonnet 5 | Llama 3.3 70B | -3.2903 | 25 | 4 | 2 | 0.0001 | 0.0033 | yes | yes |
| DeepSeek-chat | DeepSeek-R1 | 0.4677 | 14 | 16 | 1 | 0.8985 | 1.0000 | no | no |
| DeepSeek-chat | Gemini 3.8 Flash | 2.1452 | 6 | 22 | 3 | 0.0042 | 0.1305 | no | no |
| DeepSeek-chat | GPT-5.4-mini | 0.4839 | 10 | 17 | 4 | 0.2395 | 1.0000 | no | no |
| DeepSeek-chat | GPT-5.5 | 2.5484 | 6 | 21 | 4 | 0.0048 | 0.1378 | yes | no |
| DeepSeek-chat | gpt-oss-120b | -1.2258 | 20 | 10 | 1 | 0.1288 | 1.0000 | no | no |
| DeepSeek-chat | Grok 4 | 0.8387 | 12 | 16 | 3 | 0.2766 | 1.0000 | no | no |
| DeepSeek-chat | Llama 3.3 70B | -1.9194 | 22 | 5 | 4 | 0.0011 | 0.0384 | no | yes |
| DeepSeek-R1 | Gemini 3.8 Flash | 1.6774 | 10 | 19 | 2 | 0.0396 | 0.7515 | no | no |
| DeepSeek-R1 | GPT-5.4-mini | 0.0161 | 14 | 16 | 1 | 0.3365 | 1.0000 | no | no |
| DeepSeek-R1 | GPT-5.5 | 2.0806 | 7 | 20 | 4 | 0.0033 | 0.1078 | no | no |
| DeepSeek-R1 | gpt-oss-120b | -1.6935 | 19 | 11 | 1 | 0.0882 | 1.0000 | no | no |
| DeepSeek-R1 | Grok 4 | 0.3710 | 11 | 16 | 4 | 0.2514 | 1.0000 | no | no |
| DeepSeek-R1 | Llama 3.3 70B | -2.3871 | 23 | 6 | 2 | 0.0023 | 0.0784 | no | no |
| Gemini 3.8 Flash | GPT-5.4-mini | -1.6613 | 21 | 9 | 1 | 0.0396 | 0.7515 | no | no |
| Gemini 3.8 Flash | GPT-5.5 | 0.4032 | 12 | 17 | 2 | 0.9687 | 1.0000 | no | no |
| Gemini 3.8 Flash | gpt-oss-120b | -3.3710 | 26 | 4 | 1 | 0.0001 | 0.0040 | yes | yes |
| Gemini 3.8 Flash | Grok 4 | -1.3065 | 21 | 9 | 1 | 0.0957 | 1.0000 | no | no |
| Gemini 3.8 Flash | Llama 3.3 70B | -4.0645 | 26 | 4 | 1 | 0.0001 | 0.0028 | yes | yes |
| GPT-5.4-mini | GPT-5.5 | 2.0645 | 6 | 24 | 1 | 0.0224 | 0.5376 | no | no |
| GPT-5.4-mini | gpt-oss-120b | -1.7097 | 23 | 8 | 0 | 0.0043 | 0.1305 | no | no |
| GPT-5.4-mini | Grok 4 | 0.3548 | 13 | 15 | 3 | 0.8909 | 1.0000 | no | no |
| GPT-5.4-mini | Llama 3.3 70B | -2.4032 | 25 | 4 | 2 | 0.0000 | 0.0018 | no | yes |
| GPT-5.5 | gpt-oss-120b | -3.7742 | 26 | 3 | 2 | 0.0000 | 0.0010 | yes | yes |
| GPT-5.5 | Grok 4 | -1.7097 | 19 | 8 | 4 | 0.0139 | 0.3468 | no | no |
| GPT-5.5 | Llama 3.3 70B | -4.4677 | 26 | 5 | 0 | 0.0000 | 0.0013 | yes | yes |
| gpt-oss-120b | Grok 4 | 2.0645 | 6 | 25 | 0 | 0.0002 | 0.0084 | no | yes |
| gpt-oss-120b | Llama 3.3 70B | -0.6935 | 20 | 11 | 0 | 0.1470 | 1.0000 | no | no |
| Grok 4 | Llama 3.3 70B | -2.7581 | 24 | 6 | 1 | 0.0006 | 0.0210 | yes | yes |

**Table 31:** Cross-dataset pairwise tests under rule top2: Friedman $\chi^2 = 74.3$ ($p = 2.2\text{e-}12$), Iman–Davenport $F = 10.9$, Nemenyi CD = 2.43, Kendall's $W = 0.258$.

| a | b | mean rank diff | wins a | wins b | ties | Wilcoxon p | Holm p | Nemenyi | sig .05 |
|---|---|---|---|---|---|---|---|---|---|
| Claude Opus 5 | Claude Sonnet 5 | -1.3065 | 21 | 8 | 2 | 0.0289 | 0.5774 | no | no |
| Claude Opus 5 | DeepSeek-chat | -2.6129 | 23 | 7 | 1 | 0.0035 | 0.1154 | yes | no |
| Claude Opus 5 | DeepSeek-R1 | -2.4032 | 20 | 8 | 3 | 0.0059 | 0.1650 | no | no |
| Claude Opus 5 | Gemini 3.8 Flash | -0.5968 | 18 | 11 | 2 | 0.3081 | 1.0000 | no | no |
| Claude Opus 5 | GPT-5.4-mini | -2.2258 | 25 | 6 | 0 | 0.0061 | 0.1650 | no | no |
| Claude Opus 5 | GPT-5.5 | -0.0968 | 14 | 10 | 7 | 0.2313 | 1.0000 | no | no |
| Claude Opus 5 | gpt-oss-120b | -3.9355 | 26 | 4 | 1 | 0.0003 | 0.0124 | yes | yes |
| Claude Opus 5 | Grok 4 | -1.8710 | 19 | 9 | 3 | 0.0230 | 0.5376 | no | no |
| Claude Opus 5 | Llama 3.3 70B | -4.6290 | 25 | 6 | 0 | 0.0002 | 0.0068 | yes | yes |
| Claude Sonnet 5 | DeepSeek-chat | -1.3065 | 20 | 7 | 4 | 0.0351 | 0.6661 | no | no |
| Claude Sonnet 5 | DeepSeek-R1 | -1.0968 | 17 | 12 | 2 | 0.2204 | 1.0000 | no | no |
| Claude Sonnet 5 | Gemini 3.8 Flash | 0.7097 | 11 | 17 | 3 | 0.1497 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.4-mini | -0.9194 | 19 | 9 | 3 | 0.1362 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.5 | 1.2097 | 9 | 19 | 3 | 0.0235 | 0.5376 | no | no |
| Claude Sonnet 5 | gpt-oss-120b | -2.6290 | 21 | 9 | 1 | 0.0094 | 0.2447 | yes | no |
| Claude Sonnet 5 | Grok 4 | -0.5645 | 19 | 10 | 2 | 0.2515 | 1.0000 | no | no |
| Claude Sonnet 5 | Llama 3.3 70B | -3.3226 | 25 | 4 | 2 | 0.0001 | 0.0031 | yes | yes |
| DeepSeek-chat | DeepSeek-R1 | 0.2097 | 16 | 14 | 1 | 0.6731 | 1.0000 | no | no |
| DeepSeek-chat | Gemini 3.8 Flash | 2.0161 | 7 | 21 | 3 | 0.0054 | 0.1567 | no | no |
| DeepSeek-chat | GPT-5.4-mini | 0.3871 | 10 | 17 | 4 | 0.2096 | 1.0000 | no | no |
| DeepSeek-chat | GPT-5.5 | 2.5161 | 6 | 21 | 4 | 0.0048 | 0.1473 | yes | no |
| DeepSeek-chat | gpt-oss-120b | -1.3226 | 20 | 10 | 1 | 0.1288 | 1.0000 | no | no |
| DeepSeek-chat | Grok 4 | 0.7419 | 12 | 16 | 3 | 0.3417 | 1.0000 | no | no |
| DeepSeek-chat | Llama 3.3 70B | -2.0161 | 22 | 5 | 4 | 0.0011 | 0.0373 | no | yes |
| DeepSeek-R1 | Gemini 3.8 Flash | 1.8065 | 10 | 19 | 2 | 0.0254 | 0.5376 | no | no |
| DeepSeek-R1 | GPT-5.4-mini | 0.1774 | 14 | 16 | 1 | 0.2166 | 1.0000 | no | no |
| DeepSeek-R1 | GPT-5.5 | 2.3065 | 6 | 21 | 4 | 0.0010 | 0.0345 | no | yes |
| DeepSeek-R1 | gpt-oss-120b | -1.5323 | 19 | 11 | 1 | 0.1364 | 1.0000 | no | no |
| DeepSeek-R1 | Grok 4 | 0.5323 | 11 | 16 | 4 | 0.1991 | 1.0000 | no | no |
| DeepSeek-R1 | Llama 3.3 70B | -2.2258 | 22 | 7 | 2 | 0.0052 | 0.1567 | no | no |
| Gemini 3.8 Flash | GPT-5.4-mini | -1.6290 | 21 | 9 | 1 | 0.0396 | 0.7120 | no | no |
| Gemini 3.8 Flash | GPT-5.5 | 0.5000 | 12 | 17 | 2 | 0.9531 | 1.0000 | no | no |
| Gemini 3.8 Flash | gpt-oss-120b | -3.3387 | 26 | 4 | 1 | 0.0001 | 0.0037 | yes | yes |
| Gemini 3.8 Flash | Grok 4 | -1.2742 | 21 | 9 | 1 | 0.0845 | 1.0000 | no | no |
| Gemini 3.8 Flash | Llama 3.3 70B | -4.0323 | 26 | 4 | 1 | 0.0001 | 0.0028 | yes | yes |
| GPT-5.4-mini | GPT-5.5 | 2.1290 | 6 | 24 | 1 | 0.0224 | 0.5376 | no | no |
| GPT-5.4-mini | gpt-oss-120b | -1.7097 | 23 | 8 | 0 | 0.0043 | 0.1390 | no | no |
| GPT-5.4-mini | Grok 4 | 0.3548 | 13 | 15 | 3 | 0.8754 | 1.0000 | no | no |
| GPT-5.4-mini | Llama 3.3 70B | -2.4032 | 25 | 4 | 2 | 0.0000 | 0.0017 | no | yes |
| GPT-5.5 | gpt-oss-120b | -3.8387 | 26 | 3 | 2 | 0.0000 | 0.0010 | yes | yes |
| GPT-5.5 | Grok 4 | -1.7742 | 19 | 8 | 4 | 0.0124 | 0.3107 | no | no |
| GPT-5.5 | Llama 3.3 70B | -4.5323 | 27 | 4 | 0 | 0.0000 | 0.0012 | yes | yes |
| gpt-oss-120b | Grok 4 | 2.0645 | 6 | 25 | 0 | 0.0003 | 0.0097 | no | yes |
| gpt-oss-120b | Llama 3.3 70B | -0.6935 | 20 | 11 | 0 | 0.1363 | 1.0000 | no | no |
| Grok 4 | Llama 3.3 70B | -2.7581 | 24 | 6 | 1 | 0.0005 | 0.0169 | yes | yes |

**Table 32:** Cross-dataset pairwise tests under rule top3: Friedman $\chi^2 = 73.8$ ($p = 2.7\text{e-}12$), Iman–Davenport $F = 10.8$, Nemenyi CD = 2.43, Kendall's $W = 0.257$.

| a | b | mean rank diff | wins a | wins b | ties | Wilcoxon p | Holm p | Nemenyi | sig .05 |
|---|---|---|---|---|---|---|---|---|---|
| Claude Opus 5 | Claude Sonnet 5 | -1.3065 | 21 | 8 | 2 | 0.0289 | 0.5774 | no | no |
| Claude Opus 5 | DeepSeek-chat | -2.6129 | 23 | 7 | 1 | 0.0035 | 0.1154 | yes | no |
| Claude Opus 5 | DeepSeek-R1 | -2.4355 | 20 | 8 | 3 | 0.0063 | 0.1814 | yes | no |
| Claude Opus 5 | Gemini 3.8 Flash | -0.5968 | 18 | 11 | 2 | 0.3081 | 1.0000 | no | no |
| Claude Opus 5 | GPT-5.4-mini | -2.2258 | 25 | 6 | 0 | 0.0064 | 0.1814 | no | no |
| Claude Opus 5 | GPT-5.5 | -0.0968 | 14 | 10 | 7 | 0.2313 | 1.0000 | no | no |
| Claude Opus 5 | gpt-oss-120b | -3.9355 | 26 | 4 | 1 | 0.0003 | 0.0124 | yes | yes |
| Claude Opus 5 | Grok 4 | -1.8710 | 19 | 9 | 3 | 0.0218 | 0.5242 | no | no |
| Claude Opus 5 | Llama 3.3 70B | -4.5968 | 25 | 6 | 0 | 0.0002 | 0.0068 | yes | yes |
| Claude Sonnet 5 | DeepSeek-chat | -1.3065 | 20 | 7 | 4 | 0.0351 | 0.6661 | no | no |
| Claude Sonnet 5 | DeepSeek-R1 | -1.1290 | 17 | 12 | 2 | 0.2204 | 1.0000 | no | no |
| Claude Sonnet 5 | Gemini 3.8 Flash | 0.7097 | 11 | 17 | 3 | 0.1497 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.4-mini | -0.9194 | 19 | 9 | 3 | 0.1414 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.5 | 1.2097 | 9 | 19 | 3 | 0.0248 | 0.5320 | no | no |
| Claude Sonnet 5 | gpt-oss-120b | -2.6290 | 21 | 9 | 1 | 0.0094 | 0.2447 | yes | no |
| Claude Sonnet 5 | Grok 4 | -0.5645 | 19 | 10 | 2 | 0.2515 | 1.0000 | no | no |
| Claude Sonnet 5 | Llama 3.3 70B | -3.2903 | 25 | 4 | 2 | 0.0001 | 0.0031 | yes | yes |
| DeepSeek-chat | DeepSeek-R1 | 0.1774 | 16 | 14 | 1 | 0.6168 | 1.0000 | no | no |
| DeepSeek-chat | Gemini 3.8 Flash | 2.0161 | 7 | 21 | 3 | 0.0054 | 0.1612 | no | no |
| DeepSeek-chat | GPT-5.4-mini | 0.3871 | 10 | 17 | 4 | 0.2096 | 1.0000 | no | no |
| DeepSeek-chat | GPT-5.5 | 2.5161 | 6 | 21 | 4 | 0.0048 | 0.1473 | yes | no |
| DeepSeek-chat | gpt-oss-120b | -1.3226 | 20 | 10 | 1 | 0.1338 | 1.0000 | no | no |
| DeepSeek-chat | Grok 4 | 0.7419 | 12 | 16 | 3 | 0.3417 | 1.0000 | no | no |
| DeepSeek-chat | Llama 3.3 70B | -1.9839 | 22 | 5 | 4 | 0.0011 | 0.0384 | no | yes |
| DeepSeek-R1 | Gemini 3.8 Flash | 1.8387 | 10 | 19 | 2 | 0.0242 | 0.5320 | no | no |
| DeepSeek-R1 | GPT-5.4-mini | 0.2097 | 14 | 16 | 1 | 0.2166 | 1.0000 | no | no |
| DeepSeek-R1 | GPT-5.5 | 2.3387 | 6 | 21 | 4 | 0.0011 | 0.0385 | no | yes |
| DeepSeek-R1 | gpt-oss-120b | -1.5000 | 19 | 11 | 1 | 0.1364 | 1.0000 | no | no |
| DeepSeek-R1 | Grok 4 | 0.5645 | 11 | 16 | 4 | 0.1991 | 1.0000 | no | no |
| DeepSeek-R1 | Llama 3.3 70B | -2.1613 | 21 | 8 | 2 | 0.0063 | 0.1814 | no | no |
| Gemini 3.8 Flash | GPT-5.4-mini | -1.6290 | 21 | 9 | 1 | 0.0415 | 0.7465 | no | no |
| Gemini 3.8 Flash | GPT-5.5 | 0.5000 | 12 | 17 | 2 | 0.9531 | 1.0000 | no | no |
| Gemini 3.8 Flash | gpt-oss-120b | -3.3387 | 26 | 4 | 1 | 0.0001 | 0.0040 | yes | yes |
| Gemini 3.8 Flash | Grok 4 | -1.2742 | 21 | 9 | 1 | 0.0845 | 1.0000 | no | no |
| Gemini 3.8 Flash | Llama 3.3 70B | -4.0000 | 26 | 4 | 1 | 0.0001 | 0.0030 | yes | yes |
| GPT-5.4-mini | GPT-5.5 | 2.1290 | 6 | 24 | 1 | 0.0224 | 0.5242 | no | no |
| GPT-5.4-mini | gpt-oss-120b | -1.7097 | 23 | 8 | 0 | 0.0043 | 0.1390 | no | no |
| GPT-5.4-mini | Grok 4 | 0.3548 | 13 | 15 | 3 | 0.8909 | 1.0000 | no | no |
| GPT-5.4-mini | Llama 3.3 70B | -2.3710 | 25 | 4 | 2 | 0.0000 | 0.0017 | no | yes |
| GPT-5.5 | gpt-oss-120b | -3.8387 | 26 | 3 | 2 | 0.0000 | 0.0013 | yes | yes |
| GPT-5.5 | Grok 4 | -1.7742 | 19 | 8 | 4 | 0.0124 | 0.3107 | no | no |
| GPT-5.5 | Llama 3.3 70B | -4.5000 | 27 | 4 | 0 | 0.0000 | 0.0012 | yes | yes |
| gpt-oss-120b | Grok 4 | 2.0645 | 6 | 25 | 0 | 0.0003 | 0.0097 | no | yes |
| gpt-oss-120b | Llama 3.3 70B | -0.6613 | 20 | 11 | 0 | 0.1363 | 1.0000 | no | no |
| Grok 4 | Llama 3.3 70B | -2.7258 | 24 | 6 | 1 | 0.0004 | 0.0157 | yes | yes |

**Table 33:** Cross-dataset pairwise tests under rule top4: Friedman $\chi^2 = 73.2$ ($p = 3.6\text{e-}12$), Iman–Davenport $F = 10.7$, Nemenyi CD = 2.43, Kendall's $W = 0.255$.

| a | b | mean rank diff | wins a | wins b | ties | Wilcoxon p | Holm p | Nemenyi | sig .05 |
|---|---|---|---|---|---|---|---|---|---|
| Claude Opus 5 | Claude Sonnet 5 | -1.3387 | 21 | 8 | 2 | 0.0319 | 0.6693 | no | no |
| Claude Opus 5 | DeepSeek-chat | -2.7097 | 23 | 7 | 1 | 0.0035 | 0.1148 | yes | no |
| Claude Opus 5 | DeepSeek-R1 | -2.2742 | 20 | 8 | 3 | 0.0070 | 0.1901 | no | no |
| Claude Opus 5 | Gemini 3.8 Flash | -0.5645 | 18 | 11 | 2 | 0.3468 | 1.0000 | no | no |
| Claude Opus 5 | GPT-5.4-mini | -2.2258 | 25 | 6 | 0 | 0.0064 | 0.1803 | no | no |
| Claude Opus 5 | GPT-5.5 | -0.1613 | 14 | 10 | 7 | 0.2390 | 1.0000 | no | no |
| Claude Opus 5 | gpt-oss-120b | -3.9355 | 26 | 4 | 1 | 0.0003 | 0.0124 | yes | yes |
| Claude Opus 5 | Grok 4 | -1.8710 | 19 | 9 | 3 | 0.0242 | 0.5566 | no | no |
| Claude Opus 5 | Llama 3.3 70B | -4.5968 | 25 | 6 | 0 | 0.0002 | 0.0068 | yes | yes |
| Claude Sonnet 5 | DeepSeek-chat | -1.3710 | 20 | 7 | 4 | 0.0334 | 0.6693 | no | no |
| Claude Sonnet 5 | DeepSeek-R1 | -0.9355 | 16 | 13 | 2 | 0.3317 | 1.0000 | no | no |
| Claude Sonnet 5 | Gemini 3.8 Flash | 0.7742 | 11 | 17 | 3 | 0.1497 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.4-mini | -0.8871 | 19 | 9 | 3 | 0.1362 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.5 | 1.1774 | 9 | 19 | 3 | 0.0274 | 0.6025 | no | no |
| Claude Sonnet 5 | gpt-oss-120b | -2.5968 | 21 | 9 | 1 | 0.0094 | 0.2447 | yes | no |
| Claude Sonnet 5 | Grok 4 | -0.5323 | 19 | 10 | 2 | 0.2515 | 1.0000 | no | no |
| Claude Sonnet 5 | Llama 3.3 70B | -3.2581 | 25 | 4 | 2 | 0.0001 | 0.0033 | yes | yes |
| DeepSeek-chat | DeepSeek-R1 | 0.4355 | 14 | 16 | 1 | 0.8985 | 1.0000 | no | no |
| DeepSeek-chat | Gemini 3.8 Flash | 2.1452 | 6 | 22 | 3 | 0.0042 | 0.1305 | no | no |
| DeepSeek-chat | GPT-5.4-mini | 0.4839 | 10 | 17 | 4 | 0.2395 | 1.0000 | no | no |
| DeepSeek-chat | GPT-5.5 | 2.5484 | 6 | 21 | 4 | 0.0048 | 0.1378 | yes | no |
| DeepSeek-chat | gpt-oss-120b | -1.2258 | 20 | 10 | 1 | 0.1338 | 1.0000 | no | no |
| DeepSeek-chat | Grok 4 | 0.8387 | 12 | 16 | 3 | 0.2766 | 1.0000 | no | no |
| DeepSeek-chat | Llama 3.3 70B | -1.8871 | 22 | 5 | 4 | 0.0011 | 0.0384 | no | yes |
| DeepSeek-R1 | Gemini 3.8 Flash | 1.7097 | 10 | 19 | 2 | 0.0343 | 0.6693 | no | no |
| DeepSeek-R1 | GPT-5.4-mini | 0.0484 | 14 | 16 | 1 | 0.3268 | 1.0000 | no | no |
| DeepSeek-R1 | GPT-5.5 | 2.1129 | 7 | 20 | 4 | 0.0035 | 0.1148 | no | no |
| DeepSeek-R1 | gpt-oss-120b | -1.6613 | 19 | 11 | 1 | 0.0882 | 1.0000 | no | no |
| DeepSeek-R1 | Grok 4 | 0.4032 | 11 | 16 | 4 | 0.2434 | 1.0000 | no | no |
| DeepSeek-R1 | Llama 3.3 70B | -2.3226 | 22 | 7 | 2 | 0.0030 | 0.1015 | no | no |
| Gemini 3.8 Flash | GPT-5.4-mini | -1.6613 | 21 | 9 | 1 | 0.0396 | 0.7120 | no | no |
| Gemini 3.8 Flash | GPT-5.5 | 0.4032 | 12 | 17 | 2 | 0.9531 | 1.0000 | no | no |
| Gemini 3.8 Flash | gpt-oss-120b | -3.3710 | 26 | 4 | 1 | 0.0001 | 0.0043 | yes | yes |
| Gemini 3.8 Flash | Grok 4 | -1.3065 | 21 | 9 | 1 | 0.0996 | 1.0000 | no | no |
| Gemini 3.8 Flash | Llama 3.3 70B | -4.0323 | 26 | 4 | 1 | 0.0001 | 0.0030 | yes | yes |
| GPT-5.4-mini | GPT-5.5 | 2.0645 | 6 | 24 | 1 | 0.0224 | 0.5376 | no | no |
| GPT-5.4-mini | gpt-oss-120b | -1.7097 | 23 | 8 | 0 | 0.0043 | 0.1305 | no | no |
| GPT-5.4-mini | Grok 4 | 0.3548 | 13 | 15 | 3 | 0.9064 | 1.0000 | no | no |
| GPT-5.4-mini | Llama 3.3 70B | -2.3710 | 25 | 4 | 2 | 0.0000 | 0.0018 | no | yes |
| GPT-5.5 | gpt-oss-120b | -3.7742 | 26 | 3 | 2 | 0.0000 | 0.0013 | yes | yes |
| GPT-5.5 | Grok 4 | -1.7097 | 19 | 8 | 4 | 0.0139 | 0.3468 | no | no |
| GPT-5.5 | Llama 3.3 70B | -4.4355 | 26 | 5 | 0 | 0.0000 | 0.0013 | yes | yes |
| gpt-oss-120b | Grok 4 | 2.0645 | 6 | 25 | 0 | 0.0002 | 0.0084 | no | yes |
| gpt-oss-120b | Llama 3.3 70B | -0.6613 | 20 | 11 | 0 | 0.1363 | 1.0000 | no | no |
| Grok 4 | Llama 3.3 70B | -2.7258 | 24 | 6 | 1 | 0.0006 | 0.0210 | yes | yes |

**Table 34:** Cross-dataset pairwise tests under rule top5: Friedman $\chi^2 = 72.8$ ($p = 4.2\text{e-}12$), Iman–Davenport $F = 10.6$, Nemenyi CD = 2.43, Kendall's $W = 0.253$.

| a | b | mean rank diff | wins a | wins b | ties | Wilcoxon p | Holm p | Nemenyi | sig .05 |
|---|---|---|---|---|---|---|---|---|---|
| Claude Opus 5 | Claude Sonnet 5 | -1.3387 | 21 | 8 | 2 | 0.0275 | 0.5561 | no | no |
| Claude Opus 5 | DeepSeek-chat | -2.6129 | 23 | 7 | 1 | 0.0035 | 0.1154 | yes | no |
| Claude Opus 5 | DeepSeek-R1 | -2.4355 | 21 | 7 | 3 | 0.0043 | 0.1348 | yes | no |
| Claude Opus 5 | Gemini 3.8 Flash | -0.6290 | 18 | 11 | 2 | 0.3081 | 1.0000 | no | no |
| Claude Opus 5 | GPT-5.4-mini | -2.1935 | 25 | 6 | 0 | 0.0064 | 0.1751 | no | no |
| Claude Opus 5 | GPT-5.5 | -0.0968 | 14 | 10 | 7 | 0.2390 | 1.0000 | no | no |
| Claude Opus 5 | gpt-oss-120b | -3.9032 | 25 | 5 | 1 | 0.0004 | 0.0144 | yes | yes |
| Claude Opus 5 | Grok 4 | -1.8710 | 19 | 9 | 3 | 0.0230 | 0.5519 | no | no |
| Claude Opus 5 | Llama 3.3 70B | -4.5968 | 25 | 6 | 0 | 0.0002 | 0.0074 | yes | yes |
| Claude Sonnet 5 | DeepSeek-chat | -1.2742 | 20 | 7 | 4 | 0.0334 | 0.6345 | no | no |
| Claude Sonnet 5 | DeepSeek-R1 | -1.0968 | 16 | 13 | 2 | 0.2355 | 1.0000 | no | no |
| Claude Sonnet 5 | Gemini 3.8 Flash | 0.7097 | 11 | 17 | 3 | 0.1497 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.4-mini | -0.8548 | 19 | 9 | 3 | 0.1468 | 1.0000 | no | no |
| Claude Sonnet 5 | GPT-5.5 | 1.2419 | 9 | 19 | 3 | 0.0248 | 0.5561 | no | no |
| Claude Sonnet 5 | gpt-oss-120b | -2.5645 | 21 | 9 | 1 | 0.0100 | 0.2591 | yes | no |
| Claude Sonnet 5 | Grok 4 | -0.5323 | 19 | 10 | 2 | 0.2515 | 1.0000 | no | no |
| Claude Sonnet 5 | Llama 3.3 70B | -3.2581 | 25 | 4 | 2 | 0.0001 | 0.0032 | yes | yes |
| DeepSeek-chat | DeepSeek-R1 | 0.1774 | 16 | 14 | 1 | 0.6447 | 1.0000 | no | no |
| DeepSeek-chat | Gemini 3.8 Flash | 1.9839 | 7 | 21 | 3 | 0.0054 | 0.1558 | no | no |
| DeepSeek-chat | GPT-5.4-mini | 0.4194 | 10 | 17 | 4 | 0.2096 | 1.0000 | no | no |
| DeepSeek-chat | GPT-5.5 | 2.5161 | 6 | 21 | 4 | 0.0045 | 0.1348 | yes | no |
| DeepSeek-chat | gpt-oss-120b | -1.2903 | 20 | 10 | 1 | 0.1553 | 1.0000 | no | no |
| DeepSeek-chat | Grok 4 | 0.7419 | 12 | 16 | 3 | 0.3034 | 1.0000 | no | no |
| DeepSeek-chat | Llama 3.3 70B | -1.9839 | 22 | 5 | 4 | 0.0010 | 0.0335 | no | yes |
| DeepSeek-R1 | Gemini 3.8 Flash | 1.8065 | 10 | 19 | 2 | 0.0242 | 0.5561 | no | no |
| DeepSeek-R1 | GPT-5.4-mini | 0.2419 | 14 | 16 | 1 | 0.2094 | 1.0000 | no | no |
| DeepSeek-R1 | GPT-5.5 | 2.3387 | 6 | 21 | 4 | 0.0016 | 0.0541 | no | no |
| DeepSeek-R1 | gpt-oss-120b | -1.4677 | 19 | 11 | 1 | 0.1470 | 1.0000 | no | no |
| DeepSeek-R1 | Grok 4 | 0.5645 | 11 | 16 | 4 | 0.2060 | 1.0000 | no | no |
| DeepSeek-R1 | Llama 3.3 70B | -2.1613 | 21 | 8 | 2 | 0.0063 | 0.1751 | no | no |
| Gemini 3.8 Flash | GPT-5.4-mini | -1.5645 | 20 | 10 | 1 | 0.0500 | 0.8992 | no | no |
| Gemini 3.8 Flash | GPT-5.5 | 0.5323 | 12 | 17 | 2 | 0.9375 | 1.0000 | no | no |
| Gemini 3.8 Flash | gpt-oss-120b | -3.2742 | 26 | 4 | 1 | 0.0001 | 0.0043 | yes | yes |
| Gemini 3.8 Flash | Grok 4 | -1.2419 | 21 | 9 | 1 | 0.0957 | 1.0000 | no | no |
| Gemini 3.8 Flash | Llama 3.3 70B | -3.9677 | 26 | 4 | 1 | 0.0001 | 0.0032 | yes | yes |
| GPT-5.4-mini | GPT-5.5 | 2.0968 | 6 | 24 | 1 | 0.0261 | 0.5561 | no | no |
| GPT-5.4-mini | gpt-oss-120b | -1.7097 | 23 | 8 | 0 | 0.0041 | 0.1307 | no | no |
| GPT-5.4-mini | Grok 4 | 0.3226 | 13 | 15 | 3 | 0.8446 | 1.0000 | no | no |
| GPT-5.4-mini | Llama 3.3 70B | -2.4032 | 25 | 4 | 2 | 0.0000 | 0.0017 | no | yes |
| GPT-5.5 | gpt-oss-120b | -3.8065 | 26 | 3 | 2 | 0.0000 | 0.0014 | yes | yes |
| GPT-5.5 | Grok 4 | -1.7742 | 19 | 8 | 4 | 0.0118 | 0.2939 | no | no |
| GPT-5.5 | Llama 3.3 70B | -4.5000 | 27 | 4 | 0 | 0.0000 | 0.0012 | yes | yes |
| gpt-oss-120b | Grok 4 | 2.0323 | 6 | 25 | 0 | 0.0003 | 0.0122 | no | yes |
| gpt-oss-120b | Llama 3.3 70B | -0.6935 | 20 | 11 | 0 | 0.1363 | 1.0000 | no | no |
| Grok 4 | Llama 3.3 70B | -2.7258 | 24 | 6 | 1 | 0.0004 | 0.0157 | yes | yes |

**Table 35:** Mean score per dataset and model under the headline rule (ensemble).

| dataset | Gemini 3.8 Flash | Claude Opus 5 | GPT-5.5 | Claude Sonnet 5 | Grok 4 | GPT-5.4-mini | DeepSeek-chat | DeepSeek-R1 | gpt-oss-120b | Llama 3.3 70B |
|---|---|---|---|---|---|---|---|---|---|---|
| abt_buy | 0.980 | 1.000 | 1.000 | 0.970 | 0.960 | 0.960 | 0.960 | 0.970 | 0.940 | 0.960 |
| alaska_camera | 0.890 | 0.820 | 1.000 | 0.980 | 0.940 | 0.970 | 1.000 | 0.990 | 1.000 | 0.990 |
| alaska_schema | 0.901 | 0.932 | 0.911 | 0.920 | 0.911 | 0.951 | 0.934 | 0.932 | 0.914 | 0.780 |
| amazon_google | 0.930 | 0.940 | 0.940 | 0.960 | 0.940 | 0.930 | 0.960 | 0.944 | 0.860 | 0.950 |
| bird | 0.407 | 0.393 | 0.213 | 0.332 | 0.152 | 0.233 | 0.275 | 0.194 | 0.122 | 0.199 |
| dblp_acm | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0.990 | 1.000 | 0.990 | 0.910 | 0.970 |
| dblp_scholar | 0.930 | 0.950 | 0.960 | 0.970 | 0.950 | 0.940 | 0.960 | 0.950 | 0.910 | 0.970 |
| fetaqa | 0.666 | 0.637 | 0.626 | 0.566 | 0.654 | 0.345 | 0.628 | 0.602 | 0.622 | 0.249 |
| fintagging | 0.846 | 0.863 | 0.845 | 0.847 | 0.836 | 0.838 | 0.786 | 0.785 | 0.802 | 0.786 |
| hitab | 0.758 | 0.786 | 0.760 | 0.777 | 0.745 | 0.793 | 0.816 | 0.771 | 0.752 | 0.795 |
| machamp | 0.900 | 0.900 | 0.900 | 0.900 | 0.890 | 0.890 | 0.890 | 0.900 | 0.830 | 0.870 |
| magneto_gdc | 0.954 | 0.963 | 0.962 | 0.945 | 0.926 | 0.896 | 0.937 | 0.962 | 0.953 | 0.654 |
| mmtu | 0.892 | 0.884 | 0.896 | 0.884 | 0.885 | 0.879 | 0.858 | 0.814 | 0.822 | 0.811 |
| officeqa | 0.443 | 0.387 | 0.364 | 0.341 | 0.277 | 0.259 | 0.125 | 0.123 | 0.179 | 0.229 |
| officeqa_pro_v2 | 0.420 | 0.471 | 0.315 | 0.289 | 0.267 | 0.328 | 0.285 | 0.057 | 0.221 | 0.282 |
| opensanctions_pairs | 0.860 | 0.940 | 0.750 | 0.730 | 0.850 | 0.900 | 0.760 | 0.850 | 0.780 | 0.790 |
| papadakis_dn | 0.940 | 0.920 | 0.950 | 0.950 | 0.950 | 0.950 | 0.940 | 0.940 | 0.930 | 0.940 |
| realhitbench | 0.792 | 0.779 | 0.770 | 0.716 | 0.689 | 0.717 | 0.707 | 0.704 | 0.654 | 0.642 |
| smat | 0.893 | 0.911 | 0.920 | 0.919 | 0.918 | 0.939 | 0.893 | 0.891 | 0.892 | 0.723 |
| suc | 0.988 | 0.998 | 0.974 | 0.951 | 0.968 | 0.959 | 0.909 | 0.965 | 0.958 | 0.835 |
| tab_fact | 0.860 | 0.660 | 0.840 | 0.740 | 0.850 | 0.820 | 0.840 | 0.760 | 0.830 | 0.780 |
| tabis | 0.918 | 0.928 | 0.928 | 0.889 | 0.948 | 0.871 | 0.749 | 0.938 | 0.890 | 0.793 |
| tablebench | 0.872 | 0.883 | 0.883 | 0.902 | 0.854 | 0.874 | 0.869 | 0.817 | 0.856 | 0.730 |
| tableeval | 0.788 | 0.782 | 0.800 | 0.760 | 0.806 | 0.776 | 0.742 | 0.762 | 0.796 | 0.695 |
| tat_qa | 0.806 | 0.832 | 0.804 | 0.829 | 0.793 | 0.795 | 0.792 | 0.781 | 0.781 | 0.758 |
| tpcdi_cells | 0.423 | 0.423 | 0.425 | 0.423 | 0.405 | 0.421 | 0.423 | 0.425 | 0.405 | 0.404 |
| valentine | 0.970 | 1.000 | 1.000 | 0.991 | 0.965 | 0.982 | 0.974 | 0.991 | 0.973 | 0.964 |
| walmart_amazon | 0.990 | 0.980 | 0.970 | 0.950 | 0.960 | 0.950 | 0.950 | 0.960 | 0.940 | 0.930 |
| wdc_lspc | 0.980 | 0.990 | 1.000 | 0.980 | 1.000 | 0.970 | 0.960 | 1.000 | 0.980 | 0.960 |
| wdc_products | 0.990 | 0.980 | 0.980 | 0.970 | 0.984 | 0.970 | 0.970 | 0.980 | 0.980 | 0.970 |
| wikitablequestions | 0.883 | 0.884 | 0.864 | 0.867 | 0.868 | 0.853 | 0.848 | 0.850 | 0.826 | 0.759 |

**Table 36:** Probes: mean score per anchor operator.

| op | utility | n | ensemble | em | native | tok_f1 | abs_err_ensemble |
|---|---|---|---|---|---|---|---|
| identity | 1.000 | 3083 | 0.999 | 1.000 | 1.000 | 1.000 | 0.001 |
| verbose | 1.000 | 3083 | 0.999 | 1.000 | 1.000 | 1.000 | 0.001 |
| unit_variant | 1.000 | 632 | 0.965 | 0.837 | 1.000 | 0.843 | 0.035 |
| tagged | 1.000 | 3083 | 0.999 | 1.000 | 1.000 | 1.000 | 0.001 |
| paraphrase | 0.950 | 3083 | 0.981 | 0.931 | 0.988 | 0.965 | 0.031 |
| near_miss_small | 0.750 | 632 | 0.781 | 0.097 | 0.101 | 0.112 | 0.031 |
| superset | 0.600 | 1213 | 0.368 | 0.045 | 0.103 | 0.650 | 0.232 |
| typo | 0.600 | 1104 | 0.283 | 0.101 | 0.101 | 0.459 | 0.317 |
| partial | 0.500 | 1213 | 0.284 | 0.060 | 0.119 | 0.481 | 0.216 |
| near_miss_medium | 0.450 | 632 | 0.637 | 0.008 | 0.008 | 0.021 | 0.187 |
| shuffled | 0.400 | 109 | 0.756 | 0.009 | 0.403 | 0.997 | 0.356 |
| hedge | 0.300 | 2974 | 0.191 | 0.192 | 0.405 | 0.504 | 0.109 |
| list_dump | 0.250 | 1845 | 0.149 | 0.013 | 0.136 | 0.394 | 0.101 |
| digit_typo | 0.200 | 632 | 0.440 | 0.000 | 0.241 | 0.013 | 0.240 |
| near_miss_large | 0.150 | 632 | 0.323 | 0.000 | 0.000 | 0.013 | 0.173 |
| sign_flip | 0.050 | 632 | 0.021 | 0.003 | 0.003 | 0.017 | 0.029 |
| magnitude | 0.050 | 632 | 0.195 | 0.013 | 0.128 | 0.027 | 0.145 |
| flip | 0.000 | 1238 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| no_answer | 0.000 | 3083 | 0.000 | 0.000 | 0.000 | 0.001 | 0.000 |
| wrong_random | 0.000 | 1845 | 0.108 | 0.041 | 0.046 | 0.105 | 0.108 |

**Table 37:** Bradley–Terry rank (Borda rank in brackets) under each reported rule, Friedman $p$ and Kendall's $W$ among datasets.

| rule | Gemini 3.8 Flash | Claude Opus 5 | GPT-5.5 | Claude Sonnet 5 | Grok 4 | GPT-5.4-mini | DeepSeek-chat | DeepSeek-R1 | gpt-oss-120b | Llama 3.3 70B | Friedman p | W |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| exact match | 2 (3) | 1 (1) | 3 (2) | 5 (4) | 4 (5) | 6 (8) | 8 (7) | 7 (6) | 9 (9) | 10 (10) | 2.7e-14 | 0.28 |
| native metric | 1 (3) | 3 (2) | 2 (1) | 5 (4) | 4 (5) | 8 (7) | 7 (8) | 6 (6) | 9 (9) | 10 (10) | 3.8e-14 | 0.28 |
| ensemble | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 2.1e-12 | 0.26 |
| top1 | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 2.9e-12 | 0.26 |
| top2 | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 2.2e-12 | 0.26 |
| top3 | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 2.7e-12 | 0.26 |
| top4 | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 3.6e-12 | 0.25 |
| top5 | 1 (3) | 2 (1) | 3 (2) | 4 (4) | 5 (5) | 6 (6) | 7 (8) | 8 (7) | 9 (9) | 10 (10) | 4.2e-12 | 0.25 |

**Table 38:** Kendall $\tau$ between the Bradley–Terry rankings of the rules.

| rule | exact match | native metric | ensemble | top1 | top2 | top3 | top4 | top5 |
|---|---|---|---|---|---|---|---|---|
| exact match | 1.000 | 0.822 | 0.867 | 0.867 | 0.867 | 0.867 | 0.867 | 0.867 |
| native metric | 0.822 | 1.000 | 0.778 | 0.778 | 0.778 | 0.778 | 0.778 | 0.778 |
| ensemble | 0.867 | 0.778 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| top1 | 0.867 | 0.778 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| top2 | 0.867 | 0.778 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| top3 | 0.867 | 0.778 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| top4 | 0.867 | 0.778 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| top5 | 0.867 | 0.778 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |

**Table 39:** Metric selection summary.

| quantity | value |
|---|---|
| anchor | synthetic anchors (31380 answers, 20 operators) |
| n_metrics | 100 |
| n_constant | 10 |
| n_representatives | 45 |
| n_candidates | 28 |
| selected | ["tok_f1", "num_decay_2p5", "prefix_ratio"] |
| selected_idx | [33, 24, 51] |
| nnls_weights | "tok_f1": 0.5610768920589378, "num_decay_2p5": 0.325330660908859, "prefix_ratio": 0.1135924470322032 |
| lodo_rho_selected | 0.875 |
| lodo_rho_native_only | 0.814 |
| lodo_rho_em_only | 0.831 |

**Table 40:** Forward-selection trace.

| step | added | lodo_rho | gain | stopped |
|---|---|---|---|---|
| 1 | tok_f1 | 0.846 | 1.846 | – |
| 2 | num_decay_2p5 | 0.869 | 0.023 | – |
| 3 | prefix_ratio | 0.875 | 0.006 | – |
| 4 | em_raw | 0.877 | 0.002 | yes |

**Table 41:** Validation summary.

| quantity | value |
|---|---|
| lodo_mean_rho_lodo | 0.8934 |
| lodo_mean_rho_full | 0.8942 |
| lodo_same_selection_share | 0.9032 |
| severity_l1_shift_mean | 0.0220 |
| severity_l1_shift_max | 0.0220 |
| severity_selection_changed_share | 0.0000 |
| probe_mae_ensemble | 0.1158 |
| probe_mae_em | 0.1991 |
| probe_mae_native | 0.1545 |
| probe_mae_tok_f1 | 0.1440 |